



**Bansilal Ramnath Agarwal Charitable Trust's**

# **Vishwakarma Institute of Technology**

*(An Autonomous Institute affiliated to Savitribai Phule Pune University)*

## **Structure & Syllabus**

# **B.Tech. Computer Science & Engineering (Artificial Intelligence)**

**With Effect from Academic Year 2025-26**

**Prepared by: - Board of Studies in CSE (Artificial Intelligence)**

**Approved by: - Academic Board, Vishwakarma Institute of Technology, Pune**

**Chairman – BOS**

**Chairman – Academic Board**

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### **Vision of the Institution**

"To be globally acclaimed Institute in Technical Education and Research for holistic Socio-economic development".

### **Mission of the Institution**

- To ensure that 100% students are employable and employed in Industry, Higher Studies, become Entrepreneurs, Civil / Defense Services / Govt. Jobs and other areas like Sports and Theatre.
- To strengthen Academic Practices in terms of Curriculum, Pedagogy, Assessment and Faculty Competence.
- Promote Research Culture among Students and Faculty through Projects and Consultancy.
- To make students Socially Responsible Citizen.

### **Vision of the Department**

Excellence in Computer Science and Engineering, specialized in Artificial Intelligence, fostering skilled professionals and innovators for holistic societal Development

### **Mission of the Department**

- To equip aspiring engineers for industry, academia and entrepreneurship by providing quality education in emerging Artificial Intelligence techniques.
- To impart value-added, research-oriented technical education by inculcating life skills, critical thinking and human values.
- To educate and empower learners with professional integrity through innovation, industry engagement and higher studies in Artificial Intelligence for sustainable solutions.
- To nurture ethical and socially responsible professionals by fostering lifelong learning skills for holistic societal development.

**Program Educational Objectives (PEOs)**

| <b>PEO</b> | <b>PEO Focus</b>            | <b>PEO Statements- AI</b>                                                                                                             |
|------------|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| PEO1       | <b>Core competence</b>      | Demonstrate core <b>competence</b> in principles of computing and AI based technologies'.                                             |
| PEO2       | <b>Breadth</b>              | Apply AI principles, methodologies, algorithms, and tools, to effectively <b>design</b> , develop, and implement AI-driven solutions. |
| PEO3       | <b>Professionalism</b>      | Excel as professionalism with the necessary <b>soft skills</b> to work collaboratively in interdisciplinary teams.                    |
| PEO4       | <b>Learning Environment</b> | Aim for <b>continuing education</b> and <b>entrepreneurship</b> in emerging areas of computing and Artificial Intelligence.           |

**List of Program Outcomes [PO]**

| <b>PO</b>  | <b>PO Statements</b>                                                                                                                                                                                                                                                                             |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>PO1</b> | <b>Engineering knowledge:</b> Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.                                                                                                          |
| <b>PO2</b> | <b>Problem analysis:</b> Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.                                                         |
| <b>PO3</b> | <b>Design/development of solutions:</b> Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations. |
| <b>PO4</b> | <b>Conduct investigations of complex problems:</b> Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.                                                        |
| <b>PO5</b> | <b>Modern tool usage:</b> Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.                                                         |

- PO6      The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
- PO7      Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
- PO8      Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
- PO9      Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
- PO10      Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
- PO11      Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

**PO 12**    **Life-long Learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

**Program Specific Objectives (PSO)**

**PSO1:** Demonstrate proficiency in essential concepts of computer science and programming solutions.

**PSO2:** Formulate robust software design, execution, and testing strategies employing a software paradigms and Artificial Intelligence knowledge to solve real word problems.

**PSO3:** Adapt and exhibit expertise in evolving areas of computer science, engineering and technology.



**Course Name Nomenclature as per NEP (For FY and SY)**

BSC: Basic Science Course

MDOE: Multi-Disciplinary Open Elective

ESC: Engineering Science Course

CC: Co-curricular Course

PCC: Program Core Course

HSSM: Humanities Social Science and Management

PEC: Program Elective Course

IKS: Indian Knowledge System

ELC: Experiential Learning Course

FP: Field Project

MD: Multi Disciplinary

INT: Internship

**Nomenclature for Teaching and Examination Assessment Scheme AY 2025-26**

| <b>Sr No.</b> | <b>Category</b> | <b>Head of Teaching/ Assessment</b>                        | <b>Abbreviation used</b> |
|---------------|-----------------|------------------------------------------------------------|--------------------------|
| 1             | Teaching        | Theory                                                     | Th                       |
| 2             | Teaching        | Laboratory                                                 | Lab                      |
| 3             | Teaching        | Tutorial                                                   | Tut                      |
| 4             | Teaching        | Open Elective                                              | OE                       |
| 5             | Teaching        | Multi-Disciplinary                                         | MD                       |
| 6             | Teaching        | Computer Science and Engineering – Artificial Intelligence | CI                       |
| 7             | Assessment      | Laboratory Continuous Assessment                           | CA                       |
| 8             | Assessment      | Mid Semester Assessment                                    | MSA                      |
| 9             | Assessment      | End Semester Assessment                                    | ESE                      |
| 10            | Assessment      | Home Assignment                                            | HA                       |
| 11            | Assessment      | Course Project                                             | CP                       |
| 12            | Assessment      | Group Discussion                                           | GD                       |
| 13            | Assessment      | PowerPoint Presentation                                    | PPT                      |
| 14            | Assessment      | Class Test –1                                              | CT1                      |
| 15            | Assessment      | Class Test –2                                              | CT2                      |
| 16            | Assessment      | Mid Semester Examination                                   | MSE                      |
| 17            | Assessment      | End Semester Examination                                   | ESE                      |
| 18            | Assessment      | Written Examination                                        | WRT                      |
| 19            | Assessment      | Multiple Choice Questions                                  | MCQ                      |
| 20            | Assessment      | Laboratory                                                 | LAB                      |

**Title: Course Structure**

**FF No. 653**

**Branch:** Computer Science & Engineering (Artificial Intelligence) **Year:** S.Y.      **A.Y.:** 2025-26

**Module:** III

| Sr. No | Subject Code | Subject Name Course                    | Teaching Scheme (Hrs/Week) |     |     | Examination Scheme |            |     |    |     |             |         |         |        |        |        |        |     |             | Credits |
|--------|--------------|----------------------------------------|----------------------------|-----|-----|--------------------|------------|-----|----|-----|-------------|---------|---------|--------|--------|--------|--------|-----|-------------|---------|
|        |              |                                        | Th                         | Lab | Tut |                    |            |     |    |     |             |         |         |        |        |        |        |     |             |         |
|        |              |                                        |                            |     |     | ESE TH (W)         | ESE TH (O) | CVV | CP | LAB | GD/ PPT /HA | MSE (W) | MSE (O) | T1 (W) | T1 (O) | T2 (W) | T2 (O) | ESE | PR AC T+ CV |         |
| S01    | CI2011:      | Fundamentals Of Data Structures        | 2                          | 2   |     |                    |            |     | 30 | 10  |             |         |         |        |        |        |        |     | 60          | 3       |
| S02    | CI2012:      | Database Management Systems            | 2                          | 2   |     | 40                 |            | 20  | 30 | 10  |             |         |         |        |        |        |        |     |             | 3       |
| S03    | CI 2013      | Object Oriented                        | 2                          | 2   |     | 40                 |            | 20  | 30 | 10  |             |         |         |        |        |        |        |     |             | 3       |
| S04    | CI2014       | Digital Systems and Microprocessor     | 2                          | 2   |     |                    | 40         | 20  |    | 10  | 30          |         |         |        |        |        |        |     |             | 3       |
| S05    | CIM001       | Discrete Mathematics                   | 2                          |     | 1   |                    |            |     |    |     | 30          |         |         |        | 35     |        | 35     |     |             | 3       |
| S06    | HS2002       | From Campus to                         | 2                          |     |     |                    | 50         |     |    |     |             |         | 50      |        |        |        |        |     |             | 2       |
| S07    | HS2001       | Reasoning And Aptitude Development - 3 |                            |     | 1   |                    |            |     |    |     |             |         |         |        |        |        |        | 100 |             | 1       |
| S08    | CI2006       | Design Thinking - 1                    |                            |     | 1   |                    |            |     |    |     |             |         |         |        |        |        |        | 100 |             | 1       |
| S09    | CI2007       | Engineering Design and Innovation - I  |                            |     |     |                    |            |     |    |     |             |         | 30      |        |        |        |        | 70  |             | 2       |
|        |              | <b>Total</b>                           | 12                         | 8   | 3   | 40                 | 90         | 60  | 90 | 40  | 60          |         | 80      | 35     | 35     | 35     | 35     | 270 | 60          | 21      |

**Title: Course Structure**

**FF No. 653**

**Branch:** Computer Science & Engineering (Artificial Intelligence) **Year:** S.Y.      **A.Y.:** 2025-26

**Module: IV**

| Sr. No | Subject Code | Subject Name Course                   | Teaching Scheme (Hrs/Week) |          |          | Examination Scheme |            |           |           |           |             |           |           |           |           |           |           |            |               | Credits   |
|--------|--------------|---------------------------------------|----------------------------|----------|----------|--------------------|------------|-----------|-----------|-----------|-------------|-----------|-----------|-----------|-----------|-----------|-----------|------------|---------------|-----------|
|        |              |                                       | Th                         | Lab      | Tut      |                    |            |           |           |           |             |           |           |           |           |           |           |            |               |           |
|        |              |                                       |                            |          |          | ESE TH (W)         | ESE TH (O) | CVV       | CP        | LAB       | GD/ PPT /HA | M SE (W ) | MSE (O)   | T1 (W)    | T1 (O)    | T2 (W)    | T2 (O)    | ESE        | PR AC T+ CV V |           |
| S01    | CI2017       | Advance Data Structure And algorithm  | 2                          | 2        |          |                    |            |           | 30        | 10        |             |           |           |           |           |           |           |            | 60            | 3         |
| S02    | CI2018       | Machine Learning                      | 2                          | 2        |          | 40                 |            | 20        | 30        | 10        |             |           |           |           |           |           |           |            |               | 3         |
| S03    | CI2019       | Data Communication and Networking     | 2                          | 2        |          |                    | 40         | 20        |           | 10        | 30          |           |           |           |           |           |           |            |               | 3         |
| S04    | CI2020       | FullStack Development                 | 2                          | 2        |          | 40                 |            | 20        | 30        | 10        |             |           |           |           |           |           |           |            |               | 3         |
| S05    | MM0902       | Industry Automation 5.0               | 2                          |          | 1        |                    |            |           |           |           | 30          |           |           |           | 35        |           | 35        |            |               | 3         |
| S06    | HS2003       | FROM CAMPUS TO CORPORATE - 2          | 2                          |          |          |                    | 50         |           |           |           |             |           | 50        |           |           |           |           |            |               | 2         |
| S07    | HS2004       | Reasoning and Aptitude Development-4  |                            |          | 1        |                    |            |           |           |           |             |           |           |           |           |           |           | 100        |               | 1         |
| S08    | CI2021       | Design Thinking- 2                    |                            |          | 1        |                    |            |           |           |           |             |           |           |           |           |           |           | 100        |               | 1         |
| S09    | CI2022       | Engineering Design and Innovation - 2 |                            |          |          |                    |            |           |           |           |             |           | 30        |           |           |           |           | 70         |               | 2         |
|        |              | <b>Total</b>                          | <b>12</b>                  | <b>8</b> | <b>3</b> | <b>80</b>          | <b>90</b>  | <b>60</b> | <b>90</b> | <b>40</b> | <b>60</b>   |           | <b>80</b> | <b>35</b> | <b>35</b> | <b>35</b> | <b>35</b> | <b>270</b> | <b>60</b>     | <b>21</b> |

**Title: Course Structure**

**FF No. 653**

**Branch:** Computer Science & Engineering (Artificial Intelligence) **Year:** T.Y.

**A.Y.:** 2024-25

**Module:** V

| Sub No. | Sub Code | Subject Name                          | Teaching Scheme (Hrs/Week) |     |     | Examination Scheme |         |        |         |        |             |         |          |     | Credits |  |
|---------|----------|---------------------------------------|----------------------------|-----|-----|--------------------|---------|--------|---------|--------|-------------|---------|----------|-----|---------|--|
|         |          |                                       | Th                         | Lab | Tut | CA                 | MSA     | ESA    |         |        |             |         |          |     | Total   |  |
|         |          |                                       |                            |     |     | LAB (%)            | MSE (%) | HA (%) | LAB (%) | CP (%) | PPT /GD (%) | CVV (%) | ESE (%)  |     |         |  |
| S1      | CI3001   | Computer Network Technology           | 2                          | 2   | 1   | 10                 | -       | -      | 50      | 20     | -           | 20      | -        | 100 | 4       |  |
| S2      | CI3002   | Design and Analysis of Algorithms     | 2                          | 2   | 1   | 10                 | -       | 20     | -       | 20     | -           | 20      | 30 (WRT) | 100 | 4       |  |
| S3      | CI3003   | Artificial Neural Networks            | 2                          | 2   | 1   | 10                 | -       | -      | -       | 20     | 20 (PPT)    | 20      | 30 (WRT) | 100 | 4       |  |
| S4      | CI3004   | Cloud Computing                       | 2                          | 2   | 1   | 10                 | -       | -      | -       | 20     | 20 (PPT)    | 20      | 30 (MCQ) | 100 | 4       |  |
| S5      | CI3005   | Design Thinking – 5                   | -                          | -   | -   | -                  | -       | -      | -       | -      | -           | -       | 100      | 100 | 1       |  |
| S6      | CI3006   | Engineering Design and Innovation – 5 | -                          | 12  | -   | -                  | 30      | -      | -       | -      | -           | -       | 70       | 100 | 6       |  |
| S8      | SH3001   | Reasoning and Aptitude Development    | -                          | -   | -   | -                  | -       | -      | -       | -      | -           | -       | -        | -   | 1       |  |
| S9      | AC*      | Audit Courses*                        | -                          | -   | -   | -                  | -       | -      | -       | -      | -           | -       | -        | -   | 0       |  |
|         |          | Total                                 | 8                          | 20  | 4   | 40                 | 30      | 20     | 50      | 80     | 40          | 80      | 260      | 600 | 24      |  |

**Title: Course Structure**

**FF No. 653**

**Branch:** Computer Science & Engineering (Artificial Intelligence) **Year:** T.Y.      **A.Y.:** 2025-26

**Module: VI**

| Sub No. | Sub Code    | Subject Name                          | Teaching Scheme (Hrs/Week) |     |     | Examination Scheme |         |        |      |        |             |         |          | Total | Credits |
|---------|-------------|---------------------------------------|----------------------------|-----|-----|--------------------|---------|--------|------|--------|-------------|---------|----------|-------|---------|
|         |             |                                       | T h                        | Lab | Tut | CA                 | MSA     | ESA    |      |        |             |         |          |       |         |
|         |             |                                       |                            |     |     | LA B               | MSE (%) | HA (%) | LA B | CP (%) | PPT /GD (%) | CVV (%) | ESE (%)  |       |         |
| S1      | CI3007      | Software Engineering                  | 2                          | 2   | 1   | 10                 | -       | -      | -    | 20     | 20 (GD)     | 20      | 30 (MCQ) | 100   | 4       |
| S2      | CI3008      | Cyber Security and Blockchain         | 2                          | 2   | 1   | 10                 | -       | 20     | -    | 20     |             | 20      | 30 (WRT) | 100   | 4       |
| S3      | CI3009      | Deep Learning                         | 2                          | 2   | 1   | 10                 | -       | -      | 50   | 20     | -           | 20      | -        | 100   | 4       |
| S4      | MOOC Course | Coursera Courses*                     | -                          | -   | -   | -                  | -       | -      | -    | -      | -           | -       | 100      | 100   | 4       |
| S5      | CI3010      | Design Thinking – 6                   | -                          | -   | -   | -                  | -       | -      | -    | -      | -           | -       | 100      | 100   | 1       |
| S6      | CI3011      | Engineering Design and Innovation – 6 | -                          | 12  | -   | -                  | 30      | -      | -    | -      | -           | -       | 70       | 100   | 6       |
| S7      | SH3001      | Reasoning and Aptitude Development    | -                          | -   | -   | -                  | -       | -      | -    | -      | -           | -       | -        | -     | 1       |
|         |             | Total                                 | 6                          | 18  | 3   | 30                 | 30      | 20     | 50   | 60     | 20          | 60      | 330      | 600   | 24      |

**Coursera Courses\***

| Subject Code | Subject Name                      | Subject Code | Subject Name                            |
|--------------|-----------------------------------|--------------|-----------------------------------------|
| MD3101       | IBM Full Stack Software Developer | MD3122       | Google Digital Marketing and E-Commerce |
| MD3103       | IBM Back-End Developer            | MD3130       | IBM Mainframe Developer                 |
| MD3118       | IBM Data Science                  | MD3141       | AWS Cloud Technology Consultant         |
| MD3142       | Google UX Design                  |              |                                         |

**Title: Course Structure**

**FF No. 653**

**Branch: Computer Science & Engineering (AI) Year: B.Tech**

**A.Y.: 2025-26**

**Module: VII (Course Work-Course Work)**

| Subject No.   | Subject Code | Subject Name      | Teaching Scheme (Hrs/Week) |     |      | Examination Scheme |         |         |         |         |        |          |       | Credits |
|---------------|--------------|-------------------|----------------------------|-----|------|--------------------|---------|---------|---------|---------|--------|----------|-------|---------|
|               |              |                   | Th                         | Lab | Tut. | CA                 |         | MSA     |         | ESA     |        |          | Total |         |
|               |              |                   |                            |     |      | HA (%)             | LAB (%) | MSE (%) | PPT (%) | ESE (%) | GD (%) | Viva (%) |       |         |
| OE1           | Linked in*   | Linked in *       | 2                          |     |      |                    |         |         |         | 100     |        |          | 100   | 2       |
| OE2           | CI4002       | Swayam            | 2                          | -   | -    | 10                 | -       | 30      | -       | 30      | -      | 30       | 100   | 2       |
| OE2           | CI4001       | Generative AI     | 2                          | -   | -    | 10                 | -       | 30      | -       | 30      | -      | 30       | 100   | 2       |
| Major Project | CI4008       | Major Project     | 0                          | 20  | -    | -                  | -       | 30      | -       | 70      | -      | -        | 100   | 9       |
| -             | CI4007       | Design Thinking-7 | -                          | -   | -    | -                  | -       | -       | -       | 100     | -      | -        | 100   | 1       |
|               |              | Total             | 4                          | 20  | -    | 20                 | -       | 90      | -       | 330     | -      | 60       | 500   | 16      |

**Linked in Courses\***

| Subject Code | Subject Name                                    | Subject Code | Subject Name                                  |
|--------------|-------------------------------------------------|--------------|-----------------------------------------------|
| MD4274       | Large Language Models Skill Development         | MD4282       | Natural Language Processing Skill Development |
| MD4275       | Mastering Microsoft Power BI                    | MD4283       | Prompt Engineering Skills                     |
| MD4276       | Generative AI Skills for Developers             | MD4284       | Essentials in Generative AI                   |
| MD4277       | Career in Data Analysis                         | MD4285       | Python in Finance                             |
| MD4278       | Concepts of Data Visualization and Storytelling | MD4286       | Understanding Quantum Computing               |
| MD4279       | AWS Certified Solutions Architect               | MD4287       | Foundational Maths for Machine                |
| MD4280       | IT Security Specialist                          | MD4281       | Technical Program Management                  |

**Title: Course Structure**

**FF No. 653**

**Branch:** Computer Science & Engineering (AI) **Year:** B.Tech **A.Y.:** 2025-26 **Module:** VIII (Course Work-Course Work)

| Subject No.   | Subject Code | Subject Name                         | Teaching Scheme (Hrs/Week) |     |      | Examination Scheme |         |         |         |         |        |          |       | Credits |
|---------------|--------------|--------------------------------------|----------------------------|-----|------|--------------------|---------|---------|---------|---------|--------|----------|-------|---------|
|               |              |                                      | Th                         | Lab | Tut. | CA                 |         | MSA     |         | ESA     |        |          | Total |         |
|               |              |                                      |                            |     |      | HA (%)             | LAB (%) | MSE (%) | PPT (%) | ESE (%) | GD (%) | Viva (%) |       |         |
| OE1           |              | LinkedIn Learning*                   | 1                          |     |      |                    |         |         |         | 100     |        |          | 100   | 2       |
| OE2           | CI4014       | Swayam Course Reinforcement Learning | 1                          | -   | -    | 10                 | -       | 30      | -       | 30      | -      | 30       | 100   | 2       |
| OE3           | CI4005       | Natural Language Processing          | 2                          | -   | -    | 10                 | -       | 30      | -       | 30      | -      | 30       | 100   | 2       |
| Major Project | CI4016       | Major Project                        | 0                          | 20  | -    | -                  | -       | 30      | -       | 70      | -      | -        | 100   | 10      |
|               |              | Total                                | 4                          | 20  | -    | 20                 | -       | 90      | -       | 230     | -      | 60       | 400   | 16      |

**LinkedIn Learning Courses\***

| Subject Code | Subject Name                                    | Subject Code | Subject Name                                  |
|--------------|-------------------------------------------------|--------------|-----------------------------------------------|
| MD4274       | Large Language Models Skill Development         | MD4282       | Natural Language Processing Skill Development |
| MD4275       | Mastering Microsoft Power BI                    | MD4283       | Prompt Engineering Skills                     |
| MD4276       | Generative AI Skills for Developers             | MD4284       | Essentials in Generative AI                   |
| MD4277       | Career in Data Analysis                         | MD4285       | Python in Finance                             |
| MD4278       | Concepts of Data Visualization and Storytelling | MD4286       | Understanding Quantum Computing               |
| MD4279       | AWS Certified Solutions Architect               | MD4287       | Foundational Maths for Machine                |
| MD4280       | IT Security Specialist                          | MD4281       | Technical Program Management                  |



**Title: Course Structure**

**FF No. 653**

**Branch: Computer Science & Engineering (AI) Year: B.Tech**

**A.Y.: 2025-26**

**Module: VII (Internship - Internship)**

| Subject No. | Subject Code | Subject Name        | Teaching Scheme (Hrs/Week) |     |      | Examination Scheme |         |         |         |         |        |          |       | Credits |
|-------------|--------------|---------------------|----------------------------|-----|------|--------------------|---------|---------|---------|---------|--------|----------|-------|---------|
|             |              |                     | Th                         | Lab | Tut. | CA                 |         | MSA     |         | ESA     |        |          | Total |         |
|             |              |                     |                            |     |      | HA (%)             | LAB (%) | MSE (%) | PPT (%) | ESE (%) | GD (%) | Viva (%) |       |         |
| S1          | CI4010       | Industry Internship | -                          | 32  | -    | -                  | -       | 30      | -       | 70      | -      | -        | 100   | 15      |
| S1          | CI4011       | Project Internship  | -                          | 32  | -    | -                  | -       | 30      | -       | 70      | -      | -        | 100   | 15      |
| S1          | CI4012       | Research Internship | -                          | 32  | -    | -                  | -       | 30      | -       | 70      | -      | -        | 100   | 15      |
| S1          | CI4013       | Global Internship   | -                          | 32  | -    | -                  | -       | 30      | -       | 70      | -      | -        | 100   | 15      |
| S2          | CI4007       | Design Thinking-7   | -                          | -   | -    | -                  | -       | -       | -       | 100     | -      | -        | 100   | 1       |
|             |              | Total               |                            | 32  | -    | -                  | -       | 30      | -       | 170     | -      |          | 200   | 16      |

**Title: Course Structure**

**FF No. 653**

**Branch:** Computer Science & Engineering (AI) **Year:** B. Tech

**A.Y.:** 2025-26

**Module:** VIII (Internship - Internship)

| Subject No. | Subject Code | Subject Name        | Teaching Scheme (Hrs/Week) |           |      | Examination Scheme |         |           |         |            |        |          |            | Credits   |
|-------------|--------------|---------------------|----------------------------|-----------|------|--------------------|---------|-----------|---------|------------|--------|----------|------------|-----------|
|             |              |                     | Th                         | Lab       | Tut. | CA                 |         | MSA       |         | ESA        |        |          | Total      |           |
|             |              |                     |                            |           |      | HA (%)             | LAB (%) | MSE (%)   | PPT (%) | ESE (%)    | GD (%) | Viva (%) |            |           |
| S1          | CI4017       | Industry Internship | -                          | 32        | -    | -                  | -       | 30        | -       | 70         | -      | -        | 100        | 16        |
| S1          | CI4019       | Project Internship  | -                          | 32        | -    | -                  | -       | 30        | -       | 70         | -      | -        | 100        | 16        |
| S1          | CI4018       | Research Internship | -                          | 32        | -    | -                  | -       | 30        | -       | 70         | -      | -        | 100        | 16        |
| S1          | CI4020       | Global Internship   | -                          | 32        | -    | -                  | -       | 30        | -       | 70         | -      | -        | 100        | 16        |
|             |              | <b>Total</b>        |                            | <b>32</b> | -    | -                  | -       | <b>30</b> | -       | <b>170</b> | -      |          | <b>200</b> | <b>16</b> |

**Title: Course Structure**

**FF No. 653**

**Branch:** Computer Science & Engineering (AI) **Year: B.Tech**      **A.Y.: 2025-26**      **Module: VII (Course Work-Internship)**

| Subject No.   | Subject Code | Subject Name      | Teaching Scheme<br>(Hrs/Week) |     |      | Examination Scheme |         |         |         |         |        |          |       | Credits |
|---------------|--------------|-------------------|-------------------------------|-----|------|--------------------|---------|---------|---------|---------|--------|----------|-------|---------|
|               |              |                   | Th                            | Lab | Tut. | CA                 |         | MSA     |         | ESA     |        |          | Total |         |
|               |              |                   |                               |     |      | HA (%)             | LAB (%) | MSE (%) | PPT (%) | ESE (%) | GD (%) | Viva (%) |       |         |
| OE1           | Linked in*   | Linked in *       |                               |     |      |                    |         |         |         | 100     |        |          | 100   | 4       |
| OE2           | CI4002       | Swayam            | 2                             | -   | -    | 10                 | -       | 30      | -       | 30      | -      | 30       | 100   | 2       |
| OE2           | CI4001       | Generative AI     | 2                             | -   | -    | 10                 | -       | 30      | -       | 30      | -      | 30       | 100   | 2       |
| Major Project | CI4008       | Major Project     | 0                             | 20  | -    | -                  | -       | 30      | -       | 70      | -      | -        | 100   | 9       |
| -             | CI4007       | Design Thinking-7 | -                             | -   | -    | -                  | -       | -       | -       | 100     | -      | -        | 100   | 1       |
|               |              | Total             | 4                             | 20  | -    | 20                 | -       | 90      | -       | 330     | -      | 60       | 500   | 16      |

**Title: Course Structure**

**FF No. 653**

**Branch: Computer Science & Engineering (AI)    Year: B.Tech    A.Y.: 2025-26    Module: VIII (Course Work- Internship)**

| Subject No. | Subject Code | Subject Name        | Teaching Scheme (Hrs/Week) |     |      | Examination Scheme |         |         |         |         |        |          |       | Credits |
|-------------|--------------|---------------------|----------------------------|-----|------|--------------------|---------|---------|---------|---------|--------|----------|-------|---------|
|             |              |                     | Th                         | Lab | Tut. | CA                 |         | MSA     |         | ESA     |        |          | Total |         |
|             |              |                     |                            |     |      | HA (%)             | LAB (%) | MSE (%) | PPT (%) | ESE (%) | GD (%) | Viva (%) |       |         |
| S1          | CI4017       | Industry Internship | -                          | 32  | -    | -                  | -       | 30      | -       | 70      | -      | -        | 100   | 16      |
| S1          | CI4019       | Project Internship  | -                          | 32  | -    | -                  | -       | 30      | -       | 70      | -      | -        | 100   | 16      |
| S1          | CI4018       | Research Internship | -                          | 32  | -    | -                  | -       | 30      | -       | 70      | -      | -        | 100   | 16      |
| S1          | CI4020       | Global Internship   | -                          | 32  | -    | -                  | -       | 30      | -       | 70      | -      | -        | 100   | 16      |
|             |              | Total               |                            | 32  | -    | -                  | -       | 30      | -       | 170     | -      |          | 200   | 16      |

**Title: Course Structure**

**FF No. 653**

**Branch: Computer Science & Engineering (AI)    Year: B.Tech    A.Y.: 2025-26    Module: VII (Internship-Course Work)**

| Subject No. | Subject Code | Subject Name        | Teaching Scheme (Hrs/Week) |     |      | Examination Scheme |         |         |         |         |        |          |       | Credits |
|-------------|--------------|---------------------|----------------------------|-----|------|--------------------|---------|---------|---------|---------|--------|----------|-------|---------|
|             |              |                     | Th                         | Lab | Tut. | CA                 |         | MSA     |         | ESA     |        |          | Total |         |
|             |              |                     |                            |     |      | HA (%)             | LAB (%) | MSE (%) | PPT (%) | ESE (%) | GD (%) | Viva (%) |       |         |
| S1          | CI4010       | Industry Internship | -                          | 32  | -    | -                  | -       | 30      | -       | 70      | -      | -        | 100   | 15      |
| S1          | CI4011       | Project Internship  | -                          | 32  | -    | -                  | -       | 30      | -       | 70      | -      | -        | 100   | 15      |
| S1          | CI4012       | Research Internship | -                          | 32  | -    | -                  | -       | 30      | -       | 70      | -      | -        | 100   | 15      |
| S1          | CI4013       | Global Internship   | -                          | 32  | -    | -                  | -       | 30      | -       | 70      | -      | -        | 100   | 15      |
| S2          | CI4007       | Design Thinking-7   | -                          | -   | -    | -                  | -       | -       | -       | 100     | -      | -        | 100   | 1       |
|             |              | Total               |                            | 32  | -    | -                  | -       | 30      | -       | 170     | -      |          | 200   | 16      |

**Title: Course Structure**

**FF No. 653**

**Branch:** Computer Science & Engineering (AI) **Year:** B.Tech      **A.Y.:** 2025-26      **Module:** VIII (Internship-Course Work)

| Subject No.   | Subject Code | Subject Name                         | Teaching Scheme<br>(Hrs/Week) |     |      | Examination Scheme |         |         |         |         |        |          |       | Credits |
|---------------|--------------|--------------------------------------|-------------------------------|-----|------|--------------------|---------|---------|---------|---------|--------|----------|-------|---------|
|               |              |                                      | Th                            | Lab | Tut. | CA                 |         | MSA     |         | ESA     |        |          | Total |         |
|               |              |                                      |                               |     |      | HA (%)             | LAB (%) | MSE (%) | PPT (%) | ESE (%) | GD (%) | Viva (%) |       |         |
| OE1           |              | LinkedIn Learning*                   | 1                             |     |      |                    |         |         |         | 100     |        |          | 100   | 2       |
| OE2           | CI4014       | Swayam Course Reinforcement Learning | 1                             | -   | -    | 10                 | -       | 30      | -       | 30      | -      | 30       | 100   | 2       |
| OE3           | CI4005       | Natural Language Processing          | 2                             | -   | -    | 10                 | -       | 30      | -       | 30      | -      | 30       | 100   | 2       |
| Major Project | CI4016       | Major Project                        | 0                             | 20  | -    | -                  | -       | 30      | -       | 70      | -      | -        | 100   | 10      |
|               |              | Total                                | 4                             | 20  | -    | 20                 | -       | 90      | -       | 230     | -      | 60       | 400   | 16      |

**LinkedIn Learning Courses\***

| Subject Code | Subject Name                                    | Subject Code | Subject Name                                  |
|--------------|-------------------------------------------------|--------------|-----------------------------------------------|
| MD4274       | Large Language Models Skill Development         | MD4282       | Natural Language Processing Skill Development |
| MD4275       | Mastering Microsoft Power BI                    | MD4283       | Prompt Engineering Skills                     |
| MD4276       | Generative AI Skills for Developers             | MD4284       | Essentials in Generative AI                   |
| MD4277       | Career in Data Analysis                         | MD4285       | Python in Finance                             |
| MD4278       | Concepts of Data Visualization and Storytelling | MD4286       | Understanding Quantum Computing               |
| MD4279       | AWS Certified Solutions Architect               | MD4287       | Foundational Maths for Machine                |
| MD4280       | IT Security Specialist                          | MD4281       | Technical Program Management                  |

**S. Y. B. Tech.  
Computer Science & Engineering  
(Artificial Intelligence)  
AY 2025-26**

**Syllabus Template**

FF No.: 654

**CI2011: Fundamentals of Data Structures**

**Credits: 3**

**Teaching Scheme:**

Theory: 2 Hours/Week

Lab: 2 Hours/Week

**Course Prerequisites:** Basic programming Skills (C/C++)

**Course Objectives:**

1. Understand the fundamentals of algorithms and data structures along with time and space complexity.
2. Implement linear data structures using sequential memory allocation techniques.
3. Apply and analyze various searching and sorting algorithm.
4. Implement and manipulate different types of linked list.
5. Understand and apply stack operations in different programming scenarios.
6. Explore and implement different types of queues and their real-world applications.

**Course Relevance:**

This is a basic Course for Computer Engineering and allied branches. This course has a high relevance in all domains of computer engineering such as in Industries; research etc. as a basic prerequisite course. Data Structures are a crucial part of computer algorithms as they allow programmers to do data management efficiently. A wise selection of data structures can improve the performance of a computer program or algorithm in a more useful way.

**Section 1: Topics/Contents**

**Unit 1- Introduction to Algorithm and Data Structures**

**(4 Hours)**

Introduction: From Problem to Program (Problem, Solution, Algorithm, Data Structure and Program). Data Structures: Data, Information, Knowledge, and Data structure, Abstract Data Types (ADT), Data Structure Classification (Linear and Non-linear, Static and Dynamic, Persistent and Ephemeral data structures). Algorithms: Problem Solving, Introduction to algorithm, Characteristics of algorithm, Algorithm design tools: Pseudo-code and flowchart. Complexity of algorithm: Space complexity, Time complexity, Asymptotic notation- Big-O, Theta and Omega, finding complexity using step count method, Analysis of programming constructs-Linear, Quadratic, Cubic, Logarithmic. Algorithmic Strategies: Introduction to algorithm design strategies- Divide and Conquer, and Greedy strategy.

**Unit II: Linear Data Structure Using Sequential Organization**

**(5 Hours)**

Concept of Sequential Organization, Overview of Array, Array as an Abstract Data Type, Operations on Array, Merging of two arrays, Storage Representation and their Address Calculation: Row major



and Column Major, Multidimensional Arrays: Two-dimensional arrays, n-dimensional arrays. Concept of Ordered List, Single Variable Polynomial: Representation using arrays, Polynomial as array of structure, Polynomial addition, Polynomial multiplication. Sparse Matrix: Sparse matrix representation using array, Sparse matrix addition, Transpose of sparse matrix- Simple and Fast Transpose, Time and Space tradeoff.

### **Unit III Searching and Sorting**

**(5 Hours)**

Searching: Search Techniques-Sequential Search/Linear Search, Variant of Sequential Search-Sentinel Search, Binary Search, Fibonacci Search, and Indexed Sequential Search. Sorting: Types of Sorting-Internal and External Sorting, General Sort Concepts-Sort Order, Stability, Efficiency, and Number of Passes, Comparison Based Sorting Methods-Bubble Sort, Insertion Sort, Selection Sort, Quick Sort, Shell Sort, Non-comparison Based Sorting Methods-Radix Sort, Counting Sort, and Bucket Sort, Comparison of All Sorting Methods and their complexities.

### **Section2: Topics/Contents**

### **Unit IV Linked List**

**(06 Hours)**

Introduction to Static and Dynamic Memory Allocation, Linked List: Introduction, of Linked Lists, Realization of linked list using dynamic memory management, operations, Linked List as ADT, Types of Linked List: singly linked, linear and Circular Linked Lists, Doubly Linked List, Doubly Circular Linked List, Primitive Operations on Linked List-Create, Traverse, Search, Insert, Delete, Sort, Concatenate. Polynomial Manipulations-Polynomial addition. Generalized Linked List (GLL) concept, Representation of Polynomial using GLL

### **Unit V Stack**

**(04 Hours)**

Basic concept, stack Abstract Data Type, Representation of Stacks Using Sequential Organization, stack operations, Multiple Stacks, Applications of Stack- Expression Evaluation and Conversion, Polish notation and expression conversion, Need for prefix and postfix expressions, Postfix expression evaluation, Linked Stack and Operations. Recursion- concept, variants of recursion- direct, indirect, tail and tree, Backtracking algorithmic strategy, use of stack in backtracking.

### **Unit VI Queue**

**(04 Hours)**

Basic concept, Queue as Abstract Data Type, Representation of Queue using Sequential organization, Queue Operations, Circular Queue and its advantages, Multi-queues, Linked Queue and Operations. Deque-Basic concept, types (Input restricted and Output restricted), Priority Queue- Basic concept, types(Ascending and Descending)

**Lab Assignment:**

1. Implement and compare Gauss's multiplication method and Karatsuba's algorithm for multiplying large integers. Track the number of recursive calls, multiplications, and additions in each method to analyze their performance. Evaluate and compare their efficiency using step counts and execution time.
2. Design and implement a greedy algorithm to solve a real-world optimization problem, such as minimizing currency notes in a digital wallet (coin change) or maximizing non-overlapping events in a scheduler (activity selection). Represent the logic using pseudo-code and a flowchart. Analyze the algorithm's time complexity (Best, Average, Worst Case) using Big-O,  $\Theta$ , and  $\Omega$  notations.
3. Implement basic array operations such as insert, delete, and merge, while simulating a 2D array using a single-dimensional array to optimize space (using both row-major and column-major formats). Use pointer arithmetic to print memory addresses of elements, helping you understand how arrays are stored in memory.
4. To handle fragmented data efficiently, implement simple transpose, fast transpose, and addition of sparse matrices using the 3-tuple representation. Analyze and compare the time and space efficiency of each method to understand how sparse matrix operations can optimize performance in real-world scenarios like handling large scale data fragmentations on distributed systems.
5. Represent polynomials using both arrays of coefficients and exponents and arrays of structures to explore different data representations. Implement polynomial addition and multiplication operations to simulate real-world computations. Analyze how the choice of representation affects the efficiency and flexibility of polynomial operations in practical applications.
6. Design a menu-driven program to implement and compare different search techniques — Linear, Sentinel, Binary, and Fibonacci Search — on sorted and unsorted datasets. Count and display the number of comparisons made by each algorithm to evaluate their efficiency.
7. In applications like stock market data analysis or real-time gaming leaderboards, sorting large datasets efficiently is crucial. Implement and compare various sorting algorithms — Bubble Sort, Insertion Sort, Quick Sort, Shell Sort, and Radix Sort — to analyze their performance on the same input array. Track and display the time taken, number of swaps, and comparisons, and generate a report with the sorted output and performance insights.
8. Use a singly linked list to represent and perform addition of polynomials, enabling dynamic storage and easy insertion of terms. Extend this to a Generalized Linked List (GLL) to handle sparse polynomial representations efficiently.
9. In real-world software like music playlists different types of linked lists are used to manage dynamic data efficiently. Implement singly, doubly, circular, and doubly circular linked lists to explore how each structure handles various operations such as create, insert, delete, traverse,

search, sort, and concatenate. Simulate practical scenarios (like adding, deleting, updating or rearranging songs to playlist) to understand the strengths and use-cases of each linked list type

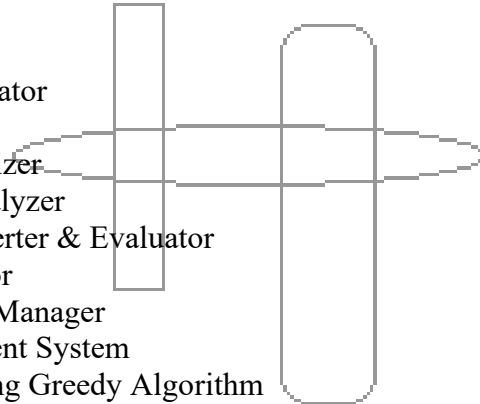
10. In calculators converting and evaluating arithmetic expressions accurately is essential. Use stack data structures to convert infix expressions to postfix and prefix formats, and then evaluate the postfix expression. Ensure your implementation handles parentheses, multi-digit operands, and input errors gracefully.

11. Solve problems like Tower of Hanoi and Factorial using both recursion and explicit stack simulation to understand how function calls are managed internally. Compare recursion depth, space usage, and manually simulate the call stack to visualize execution flow.

12. Queues are essential in real-world systems like printer spooling or traffic management. Design a menu-driven program to implement and explore different types of queues — Linear Queue, Circular Queue, Dequeue (input/output restricted), Priority Queue (ascending/descending), and Linked Queue. Understand how each queue type handles different operational constraints and efficiently manages data flow in Printers or Traffic Management.

**List of Course Project:**

- 1) Smart Polynomial Calculator
- 2) Sparse Matrix Optimizer
- 3) Search Algorithm Visualizer
- 4) Sorting Performance Analyzer
- 5) Digital Expression Converter & Evaluator
- 6) Tower of Hanoi Simulator
- 7) Dynamic Music Playlist Manager
- 8) Printer Queue Management System
- 9) Currency Minimizer Using Greedy Algorithm
- 10) Event Scheduler – Maximize Non-overlapping Events
- 11) Memory Address Mapper for Arrays
- 12) Real-Time Leaderboard using Sorting



**Text Books: (As per IEEE format)**

1. E. Horwitz , S. Sahani, Anderson-Freed, “ Fundamentals of Data Structures in C”,
2. Second Edition, Universities Press.
3. Y. Langsam, M.J. Augenstein, A.M.Tenenbaum, “Data structures using C and C++”,
4. Pearson Education, Second Edition.
5. Narasimha Karumanchi, “Data Structures and Algorithm Made Easy”, Fifth Edition, CareerMonk publication.

**Reference Books: (As per IEEE format)**

1 J. Tremblay, P. Soresan, “An Introduction to data Structures with applications”, TMHPublication, 2nd Edition.

**MOOCs Links and additional reading material:**

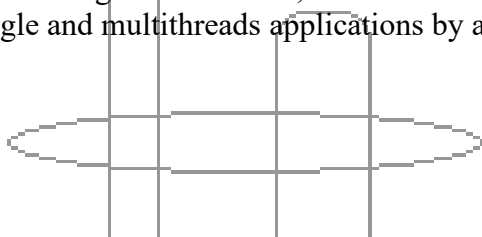
1. [www.nptelvideos.in](http://www.nptelvideos.in),
2. [www.geeksforgeeks.org](http://www.geeksforgeeks.org)
3. <https://www.youtube.com/watch?v=244YpoG1pqA&list=PLrikLQMZHuSonRoDheibeb9ffd9phWIyu&index=5>
4. <https://classroom.volp.in/>

**Course Outcomes:**

The student will be able to –

- 1) Choose and make use of single and multi-dimensional array for searching and sorting based applications.
- 2) Construct computer science applications with the help of dynamic storage representation.
- 3) Build computer science applications using stacks and queues.
- 4) Demonstrate the use of tree data structure to represent and manipulate hierarchically organized data in various applications.
- 5) Utilize graph data structure to design social media, network based and circuit applications.
- 6) Design and develop the single and multithreads applications by applying hash table and hash map techniques.

**CO-PO Map:**



| CO  | Program Outcomes (PO) |     |     |     |     |     |     |     |     |      |      | PSO   |       |       |
|-----|-----------------------|-----|-----|-----|-----|-----|-----|-----|-----|------|------|-------|-------|-------|
|     | PO1                   | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO 1 | PSO 2 | PSO 3 |
| 1   | 3                     | 2   | 2   | 2   | -   | 2   | -   | -   | 1   | -    | -    | 1     | 2     | 1     |
| 2   | 3                     | 2   | 2   |     |     |     |     | 2   | 2   | 2    | 2    | 3     |       |       |
| 3   | 3                     | 3   | 2   |     |     |     |     | 2   | 2   | 2    | 2    | 3     |       |       |
| 4   | 3                     | 3   | 2   |     |     |     |     | 2   | 2   | 2    | 2    | 3     |       |       |
| 5   | 2                     | 2   |     |     |     |     |     | 2   | 2   | 2    | 2    | 3     |       |       |
| 6   | 3                     | 3   | 2   |     |     |     |     | 2   | 2   | 2    | 2    | 3     |       |       |
| Avg | 3                     | 3   | 2   |     |     |     |     | 2   | 2   | 2    | 2    | 3     |       |       |
|     | 2.8                   | 2.6 | 2.0 |     |     |     |     | 2.0 | 2.0 | 2.0  | 2.0  | 3.0   |       |       |

**CO Attainment levels:**

Weights for attainment levels: L1 - Easy-0.75 L2 - Comfortable-0.7 L3 – Medium – 0.65

L4 – Somewhat difficult – 0.6 L5 – Difficult – 0.55

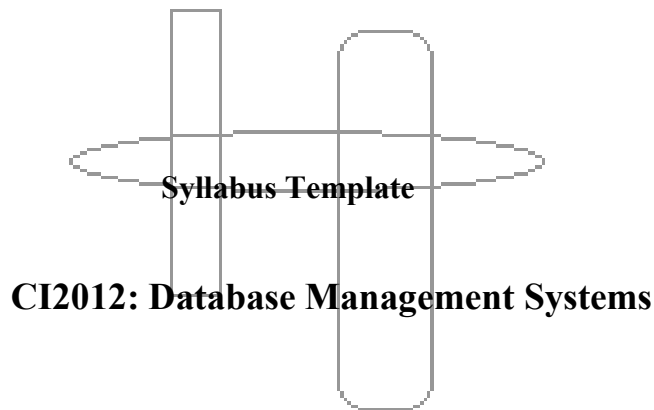
CO1 – L3, CO2– L3, CO3 – L2, CO4 – L4, CO5 – L4 and CO6 – L5

**Future Course Mapping:**

Design and Analysis of Algorithms, Operating Systems, Compiler Design, Systems Programming, Data Science and similar courses.

**Job Mapping:**

*Data Structures is must necessary part of any core programming job. Without Data structures it is not possible to be good in Competitive coding. All Industries always look for a strong knowledge in Advanced Data structures. Without learning this course, one can't imagine a job in computer/IT related industries and research.*



**FF No. : 654**

**Total Credits: 03**

**Teaching Scheme:**

**Theory:** 2 Hours/Week;

**Lab:** 2

Hours/Week

**Course Prerequisites:** Data structures, Discrete Mathematics

**Course Objectives:**

1. To understand DBMS basics, data models, and user roles.
2. To learn ER modeling and relational algebra for schema design.
3. To acquire skills in SQL and PL/SQL for database operations.
4. To understand normalization techniques to normalize the database.
5. To become familiar with transactions, concurrency, and query optimization.
6. To gain knowledge of NoSQL, MongoDB, and modern data storage trends.

### **Course Relevance:**

In today's data-driven world, efficient data management is critical for decision-making, business operations, and application development. This course equips students with a strong foundation in database concepts, models, and query languages essential for designing, developing, and managing databases. It prepares learners to handle structured and unstructured data, optimize queries, and explore modern database technologies like NoSQL and cloud databases, making them industry-ready for roles in software development, data engineering, and IT infrastructure.

### **Syllabus**

#### **Theory**

#### **Section 1: Topics/Contents**

##### **Unit 1: Introduction to DBMS and Data Models**

**(03 Hours)**

Introduction, Database Users and Administrators, Need of Database Management Systems, Characteristics of the Database, DBMS Architecture, Data Independence, View of Data, Database System Applications, Data Models: Relational Model, Hierarchical Model, Network Model, Object-based Model.

##### **Unit 2: ER Modelling and Relational Model**

**(04 Hours)**

Entity-Relationship Model: Entities, Attributes, and Relationships; Cardinality of Relationships; Strong and Weak Entity Sets; Generalization, Specialization, and Aggregation; Translating your ER model into a relational model with integrity constraints over relations. Relational Model: Concept of relations, referential integrity, and foreign keys, Relational algebra operators: selection, projection, cross product, various types of joins, division, example queries, tuple relational calculus, domain relational calculus, and converting the database specification in E/R notation to the relational schema.

##### **Unit 3: SQL and PLSQL**

**(05 Hours)**

SQL Basic, DDL, DML, DCL, DTL, Joins and Set Operations, SQL Constraints: Primary Key, Foreign Key, Unique, Not Null, Check, Default, Group by, Order by, Having Clauses, Subqueries, View, and Index. PL/SQL: Concept of Stored Procedures and Functions, Cursors, Triggers, Assertions, Roles, and Privileges.

#### **Section 2: Topics/Contents**

##### **Unit 4: Dependencies and Normal forms**

**(04 Hours)**

Importance of a good schema design, problems encountered with bad schema designs, CODD's Rules, the Concept of Functional Dependencies and Closure, Minimal Cover of Functional Dependencies, Anomalies in unnormalized and partially normalized relations, Armstrong's axioms for FD's, Decomposition: Lossless-Join Decomposition, Dependency Preservation, Multi-Valued Dependencies. Normalization: 1NF, 2NF, 3NF, BCNF, 4NF.

### **Unit 5: Transaction Management and Query Optimization**

**(04 Hours)**

Transaction Concept and ACID Properties, States of Transactions, Schedules, Serializability: Conflict and View Serializability, Concurrency Control Techniques: Lock-Based Protocols, Timestamp-Based Protocols, Deadlock Handling and Prevention, Recovery Techniques: Log-Based Recovery, Shadow Paging, Indexing: Primary, Secondary, and Multilevel Indexes, Query processing and optimization: evaluation of relational algebra expressions, query equivalence, join strategies, and query optimization algorithms.

### **Unit 6: NoSQL Databases**

**(04 Hours)**

Types of Data: Structured, Unstructured, and Semi-Structured Data CAP Theorem, NoSQL Database Introduction, Need, Features, Types of NoSQL Databases: Key-Value, Column, Document, and Graph Databases, BASE Properties, ACID vs. BASE, and a Comparative Study of RDBMS and NoSQL. MongoDB (with syntax and usage): CRUD Operations, Indexing, Aggregation, MapReduce, Cloud Databases, Distributed Databases.

### **Laboratory**

#### **List of Experiments**

1. Introduction to SQL: Create a database named StudentDB and a table named Students with fields ID, Name, Age, and Course.
2. Implementation of DDL, DML, and DQL commands: Create a table Employees, Insert records, Update a salary column, and retrieve records using SELECT.
3. SQL Joins and Set Operations: Join the Employees and Departments tables to show employee names with their department names.
4. Views and Nested Queries: Create a view called HighSalaryEmployees and write a nested query to find employees earning more than the average salary.
5. Functions and Stored Procedures: Write a function to calculate annual salary. Write a stored procedure to update salary based on designation.
6. Triggers Implementation: Create a trigger that logs insert operations into a separate Logs table.
7. ER to Relational Mapping: Convert an Entity-Relationship (ER) diagram to relational schema.
8. Normalization (up to 4NF): Normalize a given unnormalized table structure to 4NF, explaining each step.
9. Transaction Implementation and Concurrency Control: Simulate a bank transfer between two accounts using transactions.
10. Query Optimization using Indexing: Compare query execution time before and after applying an index on a large dataset.
11. NoSQL Implementation using MongoDB: Create a MongoDB collection called Products and perform insert and query operations.

12. Design a two-tier application using Java, Python, or PHP as the front-end, with MySQL or MongoDB as the back-end for data storage and retrieval.

## Syllabus

### Course Project

#### **List of Course Projects**

1. Online Course Management System: Design a system to manage online courses, student registrations, assignments, grades, and faculty dashboards. Includes user roles, course prerequisites, and submission tracking.
2. Hospital Appointment & Patient Management System: Build a DBMS to manage patient records, doctor schedules, appointments, treatments, and billing. Involves relationships among departments, doctors, patients, and reports.
3. Library Management System with Fine Calculation: Manage book check-ins/check-outs, due dates, reservations, fines, and digital catalog search. Add features like genre-based search and overdue fine tracking.
4. E-commerce Order Processing System: Handle customers, products, orders, payments, and shipping. Include stock availability, order history, and product reviews.
5. Hotel Booking and Reservation System: Design a system for guests to search, reserve, and cancel rooms, with staff and service records. Include room types, seasonal pricing, and transaction logging.
6. Student Attendance & Result Management: Track attendance and academic results of students across subjects and semesters. Enable performance analytics per student and per subject.
7. Banking and Loan Management System: Develop a database to manage user accounts, transactions, loan applications, EMI schedules, and credit scores. Requires ACID compliance, trigger-based fraud alerts, and audit trails.
8. Multi-Branch Inventory & Sales System for a Retail Chain: Manage inventory, stock transfers, branch sales, suppliers, and demand forecasting. Involves complex joins, transactions, triggers, and partitioned tables.
9. University ERP (Enterprise Resource Planning) System: Create a unified system for managing departments, students, faculty, courses, fees, payroll, and exams. Requires large-scale schema planning and role-based data access.
10. Healthcare Analytics Dashboard: Design a DBMS to collect and analyze patient data for disease prediction and hospital analytics. Incorporates views, stored procedures, triggers, and indexed performance optimization.
11. Online Job Portal with Resume Matching: Store job postings, resumes, recruiter profiles, and use query logic to match candidates with job roles. Integrates full-text search, ranking algorithms, and resume parsing.
12. Social Media Platform DBMS: Simulate a social network backend—users, posts, likes, shares, friends, and messaging. Needs recursive queries (friends of friends), normalization, and analytics.



### **Course Outcomes**

1. Describe the fundamentals of DBMS, various data models, architectures, and the responsibilities of users and administrators.
2. Make use of ER models to design and relational algebra to query database systems.
3. Implement database operations using SQL and PL/SQL constructs such as procedures, functions, and triggers.
4. Apply normalization techniques to eliminate redundancy and maintain data integrity.
5. Demonstrate transaction properties, concurrency control, and query optimization methods for efficient database performance.
6. Use knowledge of NoSQL databases, MongoDB operations, and distributed data systems to address modern storage requirements.

### **CO-PO Mapping**

| CO  | Program Outcomes (PO) |      |      |      |      |     |     |      |      |      |      | PSO   |      |      |
|-----|-----------------------|------|------|------|------|-----|-----|------|------|------|------|-------|------|------|
|     | PO1                   | PO2  | PO3  | PO4  | PO5  | PO6 | PO7 | PO8  | PO9  | PO10 | PO11 | PSO 1 | PSO2 | PSO3 |
| 1   | 2                     | 1    |      |      |      |     |     |      |      |      | 2    | 2     |      |      |
| 2   | 3                     | 2    | 2    |      | 2    |     |     | 2    |      | 2    | 2    | 3     | 2    |      |
| 3   | 3                     | 2    | 2    |      | 3    |     |     | 2    | 2    | 2    | 2    | 3     | 2    | 2    |
| 4   | 3                     | 2    | 2    | 2    |      |     |     | 2    |      | 2    | 2    | 3     | 2    |      |
| 5   | 2                     | 1    |      |      |      |     |     |      |      |      | 2    | 2     |      |      |
| 6   | 3                     | 2    | 2    |      | 3    |     |     | 2    | 2    | 2    | 2    | 3     | 2    | 2    |
| Avg | 2.67                  | 1.67 | 2.00 | 2.00 | 2.67 |     |     | 2.00 | 2.00 | 2.00 | 2.00 | 2.67  | 2.00 | 2.00 |

### **Books and E-Resources**

#### **Text Books:**

1. Abraham Silberschatz, Henry F. Korth, S. Sudarshan; “Database System Concepts”; 6th Edition, McGraw-Hill Education
2. Ramez Elmasri, Shamkant B. Navathe; “Fundamentals of Database Systems”; 7<sup>th</sup> Edition, Pearson.

#### **Reference Books:**

1. Thomas M. Connolly, Carolyn E. Begg, “Database Systems: A Practical Approach to Design, Implementation, and Management, 6th Edition ;Pearson
2. Raghu Ramakrishnan, Johannes Gehrke; “Database Management Systems”, 3rd Edition; McGraw Hill Education
3. Kristina Chodorow, MongoDB The definitive guide, O’Reilly Publications, ISBN: 978-93-5110-269-4, 2nd Edition.
4. Dr. P. S. Deshpande, SQL and PL/SQL for Oracle 10g Black Book, DreamTech.

5. Ivan Bayross, SQL, PL/SQL: The Programming Language of Oracle, BPB Publication.
6. Reese G., Yarger R., King T., Williams H, Managing and Using MySQL, Shroff Publishers and Distributors Pvt. Ltd., ISBN: 81 - 7366 - 465 – X, 2nd Edition.
7. Dalton Patrik, SQL Server – Black Book, DreamTech Press.
8. Eric Redmond, Jim Wilson, Seven databases in seven weeks, SPD, ISBN: 978-93-5023-918-6.
9. Jay Kreibich, Using SQLite, SPD, ISBN: 978-93-5110-934-1, 1st edition.

**Moocs Links and additional reading material:**

[https://onlinecourses.nptel.ac.in/noc22\\_cs91/preview](https://onlinecourses.nptel.ac.in/noc22_cs91/preview)

<https://www.datacamp.com/courses/introduction-to-sql>

[https://apexapps.oracle.com/pls/apex/f?p=44785:149:31644430784497::NO:149:P149\\_EVENT\\_ID:5571](https://apexapps.oracle.com/pls/apex/f?p=44785:149:31644430784497::NO:149:P149_EVENT_ID:5571)

**Syllabus Template**

**FF No. : 654**

**CI2013: Object Oriented Programming**

**Teaching Scheme:**

Theory: 2 Hours / Week

Laboratory: 2 Hours / Week

**Total Credits: 3**

**Course Prerequisites:**

1. C programming
2. Problem solving skills using programming language

**Course Objectives:**

1. Differentiate various programming paradigms.
2. Understand the concept of data abstraction and encapsulation, how to design C++ classes for code reuse.
3. Learn to implement class member functions, constructors and operators overloading in C++.
4. Learn Inheritance, virtual functions and dynamic binding with polymorphism.
5. Utilize file handling techniques in C++ and Java for sequential and random access, while incorporating robust, reliable and efficient data management.
6. To develop problem solving ability using Object Oriented programming constructs like multithreading

**Course Relevance:**

The Object-Oriented Programming (OOP) course using C++ is highly relevant because it introduces core programming concepts such as encapsulation, inheritance, and polymorphism, which are essential for modern software development. It helps students understand how to structure and manage complex systems effectively. The course also provides practical skills in writing maintainable and scalable code using one of the most widely used programming languages. Additionally, mastering OOP in C++ lays a strong foundation for learning other OOP languages like Java and Python.

## **Syllabus**

### **Theory**

#### **Section 1: Topics/Contents**

##### **Unit 1: Fundamentals of Object Oriented Programming (4 Hours)**

Need of object-oriented programming, Fundamentals of Object-Oriented Programming: Objects, Classes, Data Members, Methods, Namespaces, Data Encapsulation, Data Abstraction and Information Hiding, Inheritance, Polymorphism, Static and Dynamic Binding, Message Passing. Benefits of OOP, C++ as object-oriented programming Language. C++ Programming Basics, Data Types, Structure, Enumeration, Control structures, Arrays, Strings, Access specifier.

##### **Unit 2: Function and Pointer (5 Hours)**

Introducing Functions, Constructor, Destructor, Objects and Memory requirements, Static members: variables and functions, inline function, friend function. Characteristics of Constructors, Types of Constructors, Constructor Overloading, Dynamic Initialization of an Object, Constructor with Default Arguments.

Pointers: declaring and initializing pointers, indirection Operators, Memory management: new and delete, Pointers to Objects, this pointer, Arrays of pointers, Function Pointers, Pointers to pointers, Pointer to Derived classes, Passing pointers to functions.

##### **Unit 3: Inheritance and Polymorphism (5 Hours)**

Base class and derived Class, Protected members, Overriding member function, Class Hierarchies, Public and Private Inheritance, Types of Inheritance: Single, Multilevel, Hierarchical, Multiple, Hybrid, Ambiguity in multiple Inheritance, Virtual Base Class, Abstract Class, Friend Class. Introduction to polymorphism, Types of Polymorphism, Operator Overloading - concept of overloading, Overloading Unary Operators, Overloading Binary Operators, Function overloading, Runtime Polymorphism - Pointer to Base Class, Virtual function and its significance, pure virtual function, Virtual destructor

#### **Section 2: Topics / Contents**

##### **Unit 4: Exception Handling and Templates (5 Hours)**

Exception Handling - Fundamentals, other error handling techniques, simple exception handling- Divide by zero, Multiple catching, rethrowing an exception, exception specifications, user defined exceptions, processing unexpected exceptions, constructor, destructor and exception handling, exception and inheritance. Templates- The power of templates, function template, overloading Function templates and class template and Nontype parameters, template and friend generic functions, The type name and export keyword

### **Unit 5: Files and Streams**

**(5 Hours)**

Data hierarchy, Stream and files, Stream Classes, Stream Errors, Disk File I/O with Streams, File Pointers, and Error Handling in File I/O, File I/O with Member Functions, Overloading the Extraction and Insertion Operators, memory as a Stream Object, Command-Line Arguments, Printer output File Processing, Primitive Data Processing, Object Data Processing,

### **Unit 6: Object Oriented Programming with Java**

**(4 Hours)**

Introduction to Java and its OOP Features, Java vs C++, Class and Object in Java, Access modifiers: public, private, protected, default, Single and Multilevel inheritance using extends keyword, Method overriding using super keyword, Exception Handling in Java: try, catch, throw, finally.

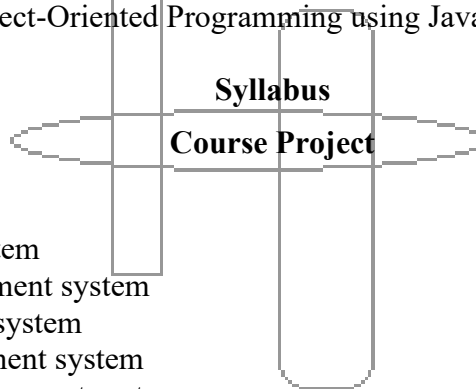
### **Laboratory**

#### **List of Experiments :**

1. Write a C++ program to implement the concept of objects, classes, constructors, destructors.
2. Write a program to implement a class for Book details as follows Private: bookid, book\_name, author, price Public: get\_details(), print\_details() Get details of some books from user and print in tabular form. Also find the total prize of all the books.
3. Write a program to extend Book class with dynamic memory allocation. (Use of “new” keyword and object pointers)
4. Design a class ‘Complex’ with data members for real and imaginary part. Provide default and Parameterized constructors. Write a program to perform arithmetic operations of two complex numbers.
5. Design and develop inheritance for a given case study, identify objects and relationships and implement inheritance wherever applicable. Employee class has Emp\_name, Emp\_id, Address, Mail\_id, and Mobile\_no as members. Inherit the classes: Programmer, Team Lead, Assistant Project Manager and Project Manager from employee class. Add Basic Pay (BP) as the member of all the inherited classes with 52% of BP as DA, 27 % of BP as HRA, 12% of BP as PF, 0.1% of BP for staff club fund. Generate pay slips for the employees with their gross and net salary.
6. Design a base class shape with two double type values and member functions to input the data and compute\_area() for calculating area of shape. Derive two classes: triangle and rectangle. Make compute\_area() as an abstract function and redefine this function in the derived class to suit their requirements. Write a program that accepts dimensions of triangle/rectangle and displays calculated area. Implement dynamic binding for given case study.
7. Design and develop a context for given case study and implement an interface for Vehicles Consider the example of vehicles like bicycle, car and bike. All Vehicles have common functionalities such as Gear Change, Speed up and apply brakes. Make an interface and put all

these common functionalities. Bicycle, Bike, Car classes should be implemented for all these functionalities in their own class in their own way.

8. Implement a program for maintaining a database of student records using Files. Students have Student\_id, name, Roll\_no, Class, marks and address. Display the data for a few students. 1. Create Database 2. Display Database 3. Delete Records 4. Update Record 5. Search Record
9. Implement a program to handle arithmetic exceptions and array index out of bounds.
10. Implement a generic program using any collection class to count the number of elements in a collection that have a specific property such as even numbers, odd number, prime number and palindromes.
11. Write a Java program to create a student class with details like Roll no, Name, Marks of five subjects, percentage, class (First, Second, etc) have following functions: parameterized constructor, destructor, Display and Calculate percentage and grade.
12. Identify commonalities and differences between Publication, Book and Magazine classes. Title, Price, Copies are common instance variables and saleCopy is a common method. The differences are, Bookclass has author and orderCopies(). Magazine Class has methods orderQty, Current issue, receiveissue(). Write a Java program to find how many copies of the given books are ordered and display total sale of publication.
13. Create a Java program to implement a Student Grading System that accepts student details such as Name, Roll No, Marks of 5 subjects, Calculates total, percentage, and grade, displays results in a tabular format. Use inheritance to separate basic student info from academic performance.
14. Demonstration of Object-Oriented Programming using Java and database through JDBC



### **List of Course Projects**

1. Electricity billing system
2. e-Healthcare management system
3. Library management system
4. Online bank management system
5. Online medical management system
6. Online quiz management system
7. Online Survey System
8. Stock management system
9. Supply chain management system
10. Hospital Management System
11. Alumni Database System
12. PayRoll System
13. Simple Inventory System
14. Employee Management System
15. Car Rental System
16. Student Grading System
17. Hotel Reservation System
18. Movie Ticket Booking System
19. Vehicle Parking Management
20. Library Fine Calculation System

### Course Outcomes

Upon completion of this course, students will be able to,

1. Understand object-oriented programming features.
2. Identify classes, objects, methods, and handle object creation, initialization, and destruction to model real-world problems.
3. Understand and implement various types of constructors and destructors.
4. Identify relationships among objects using inheritance and polymorphism principles.
5. Use files for persistent data storage to model real world applications.
6. Handle different types of exceptions, apply generic programming concepts

### CO-PO Mapping:

| CO  | Program Outcomes (PO) |     |      |     |     |     |     |     |     |      |      | PSO   |      |      |
|-----|-----------------------|-----|------|-----|-----|-----|-----|-----|-----|------|------|-------|------|------|
|     | PO1                   | PO2 | PO3  | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO 1 | PSO2 | PSO3 |
| 1   | 3                     | 2   | 2    |     |     |     |     |     |     |      |      | 3     |      |      |
| 2   | 3                     | 3   | 3    | 1   | 1   | 2   | 2   | 2   |     |      | 2    | 3     |      | 3    |
| 3   | 3                     | 2   | 3    | 1   | 1   |     | 2   |     |     |      |      | 3     |      | 3    |
| 4   | 3                     | 3   | 3    | 1   |     |     |     |     |     |      | 2    | 3     | 3    | 3    |
| 5   | 3                     | 3   | 3    | 1   | 1   |     |     |     |     |      | 2    | 3     |      | 3    |
| 6   | 3                     | 2   | 3    |     |     |     |     |     |     |      |      | 3     |      | 3    |
| Avg | 3                     | 2.5 | 2.83 | 1   | 1   | 2   | 2   | 2   |     |      | 2    | 3     | 3    | 3    |

### Future Courses Mapping:

Advanced Data structure, Operating System, Advanced C++ / Java , Object-Oriented Design (OOD), Design Patterns, Embedded Systems (with C++), CUDA Cyber Security.

### Job Mapping:

*Software Developer / Engineer, Game Developer, System Programmer, Software Architect*

### Books and E-Resources

#### **For Reference Print / E-Book -**

1. The C++ Programming Language, Bjarne Stroustrup, 4th Edition, Addison-Wesley Pearson Education .
2. Herbert Schildt , “C++: The Complete Reference”, 4th Edition.
3. E. Balaguruswamy, “Object Oriented Programming Using C++ ”, Tata McGraw Hill.

4. An Introduction to Object Oriented Programming, by Timothy A. Budd, published by Addison-Wesley.
5. Herbert Schildt, “Java: The Complete Reference”, 4th Edition

**For MOOCs and other learning Resources:**

[https://onlinecourses.nptel.ac.in/noc25\\_cs34/preview](https://onlinecourses.nptel.ac.in/noc25_cs34/preview)

[https://onlinecourses.nptel.ac.in/noc20\\_cs07/preview](https://onlinecourses.nptel.ac.in/noc20_cs07/preview)

<https://www.coursera.org/learn/packt-fundamentals-of-object-oriented-programming-c-b5fxn#modules>

[SOLID principles: implementation and examples in C++ | by Oleksandra Shershen | Medium](#)

**CI2014: Digital Systems and Microprocessor**

**Teaching Scheme:**

Theory: 02 Hours / Week; Laboratory: 02 Hours / Week;

Total Credits: 03

**Syllabus**

**Theory**

**Unit 1: Fundamentals of Digital Electronics ( 5 Hours)**

Number System, Logic Minimization – Sum of Product (SOP) form, Product of Sum (POS) form, Representation and simplification of logical functions, Minimization of SOP and POS forms using K-Maps (up to 4 variables), Don't Care Condition, and Quine-McCluskey Technique.

**Unit 2: Combinational and Sequential Circuits ( 5 Hours)**

Code converter: BCD, Excess-3, Gray code, Binary Code. Half- Adder, Full Adder, Half Subtractor, Full Subtractor, Binary Adder (IC 7483), BCD adder.

Flip-flop: SR, JK, D, T. Registers: Buffer register, shift register (SISO, SIPO, PISO & PIPO). Register Counters: Asynchronous Counter, Synchronous Counter, Ring counters, Johnson Counter, Modulus of the counter (IC 7490).

**Unit 3: ASM & PLDs ( 5 Hours)**

Algorithmic State Machines: Finite State Machines (FSM) and ASM, ASM charts, notations, construction of ASM chart and realization for sequential circuits.

PLDs: PLD, Programmable Logic Array (PLA), Programmable Array Logic (PAL), Designing combinational circuits using PLDs.

**UNIT- 4 Introduction to 80386 ( 5 Hours)**

Introduction to Intel Microprocessor – Data Bus, Address Bus, Control Bus, 8086 Architecture, 80386 DX Features and Architecture, Programmers Model, Operating modes, Addressing modes and data types.

**Unit 5: Applications Instruction Set**

**( 5 Hours)**

Data Movement Instructions, Binary Arithmetic Instructions, Decimal Arithmetic Instructions, Logical Instructions, Control Transfer Instructions, String and Character Transfer Instructions, Instructions for Block Structured Language, Flag Control Instructions, Segment Register Instructions, Miscellaneous Instructions.

**Unit 6: System Architecture, Interrupts and Exception**

**(5 Hours)**

Systems Registers - Systems flags, Memory Management registers, Control registers, Debug registers, Test registers.

Interrupts and Exceptions: Identifying Interrupts & Exceptions, Interrupt Tasks and Interrupt Procedures, Error Code, and Exception Conditions

**Syllabus**

**Laboratory**

**List of Experiments**

1. Realize Full Adder and Sub tractor using a) Basic Gates and b) Universal Gates.
2. Design and implement Code converters - Binary to Grey and BCD to Excess-3.
3. Design and Realization of BCD Adder using 4-bit Binary Adder (IC 7483).
4. Design of Ripple Counter using JK-Flip Flops.
5. Design 3 bit Synchronous Up/Down Counter using JK-Flip Flop.
6. Design and implement Mod -N counter using IC-7490.
7. Study of Shift Registers ( SISO, SIPO, PISO, PIPO)
8. Write an X86/64 ALP to accept five 64 bit Hexadecimal numbers from user and store them in an array and display the accepted numbers.
9. Write an X86/64 ALP to accept a string and to display its length.
10. Write an X86/64 ALP to find the largest of given Byte/Word/Dword/64-bit numbers.
11. Write a switch case driven X86/64 ALP to perform 64-bit hexadecimal arithmetic operations (+, -, \*, /) using suitable macros. Define procedure for each operation.
12. Write an X86/64 ALP to count number of positive and negative numbers from the array.
13. Write X86/64 ALP to convert 4-digit Hex number into its equivalent BCD number and 5-digit BCD number into its equivalent HEX number. Make your program user friendly to accept the choice from user for: (a) HEX to BCD b) BCD to HEX (c) EXIT

**Course Outcomes:**

1. Understand number systems and Simplify Boolean algebraic expressions using Minimization Technique.
2. Design and Implement Combinational and Sequential circuits
3. Design simple real-world applications using ASM and PLD.
4. Illustrate the advanced features of the 80386 microprocessor and differentiate its various operating modes and addressing modes.



5. Exhibit skill of assembly language programming for the application.
6. Differentiate various types of system registers, interrupts, and exceptions

### **Books and E-Resources**

#### **For Reference Print Book -**

1. T. L. Floyd, "Digital Fundamentals", 9<sup>th</sup> edition, Pearson International Edition.
2. R. P. Jain, "Modern digital electronics", 4<sup>th</sup> edition, TMH Publication.
3. Anand Kumar, "Fundamentals of digital circuits" 4<sup>th</sup> edition, PHI publication
4. Douglas Hall, "Microprocessors & Interfacing", McGrawHill, Revised 2<sup>nd</sup> Edition, 2006 ISBN 0-07-100462-9
5. Chris H. Pappas, William H. Murray, "80386 Microprocessor Handbooks", McGraw-Hill Osborne Media, ISBN-10: 0078812429, ISBN-13: 978-0078812422.
6. Brey, Barry B., "8086/8088, 80286, 80386 and 80486 Assembly Language Programming", Prentice Hall, ISBN: 13: 9780023142475.

#### **For Reference Electronic Book –**

1. Intel 80386 Programmer's Reference Manual 1986, Intel Corporation, Order no.: 231630-011, December 1995
2. Intel 80386 Hardware Reference Manual 1986, Intel Corporation, Order no.: 231732-001, 1986.

#### **For MOOCs and other learning Resources:**

<https://nptel.ac.in/courses/106/108/106108100/>  
<https://nptel.ac.in/courses/108/107/108107029/>  
<http://intel80386.com/386htm/toc.htm>  
<https://css.csail.mit.edu/6.858/2014/readings/i386.pdf>

## **Multidisciplinary Minor CIM001 : Discrete Mathematics**

**Credits: 3**

**Teaching Scheme Theory: 2 Hours/Week**

**Tutorial: 1 Hour/Week**

#### **Course Prerequisites:**

1. Basic knowledge of mathematics and logic

#### **Course Objectives:**

1. To provide strong basic discrete structures in mathematical reasoning and logic essential for computer science problem-solving and theoretical computation.
2. To develop recurrence relations for a wide variety of combinatorial problems and counting.

- 3.To develop a comprehensive understanding of fundamental algebraic structures including groups, rings, and fields.
- 4.To understand and apply solid foundation in the principles and techniques of number theory.
- 5.To familiarize students fundamental concepts and structures in graph theory and develop analytical skills for modeling and solving real-world problems using graphs
- 6.To understand the theory and application of tree structures in graph theory

**Course Relevance:** This is a foundational course for Computer Science and Engineering. The discrete structures play an essential role while modeling problems in computer science and engineering. The reasoning with different discrete structures is useful in understanding the underlying computer science problem more concretely. The course also builds problem solving ability.

## **Syllabus Theory**

### **Section 1: Topics/Contents**

#### **Unit 1: Set Theory And Logic**

**(4 Hours)**

Sets: Sets and Subsets, Power Set, Cartesian Product, Set Operations, Venn Diagram, Inclusion-Exclusion Principle, Computer Representation of Sets. Relations: Product Sets, Pictorial Representatives of Relations, Composition of Relations, Types of Relations, Closure Properties, Equivalence Relations, Partial Orderings, partitions, Hasse Diagram, Lattices. Logic: Propositions, Truth Tables, Tautologies, Conditional Propositions and Logical Equivalence, Arguments and Rules of Inference. Functions: Domain, range, One-to-One, Onto, and Invertible Functions, Exponential and Logarithmic Functions, Cardinality, Algorithms and Functions

#### **Unit 2: Recursion and Counting Methods**

**(5 Hours)**

Mathematical Induction, Recursion, Recurrence Relation, Solving Recurrence Relations, The Basics of Counting, rule of Sum and Product, Permutations and Combinations, Binomial Coefficients and Identities, Algorithms for generating Permutations and Combinations, The Principle of Inclusion-Exclusion, The Addition and Multiplication Rules, The Pigeon-Hole Principle.

#### **Unit 3: Algebraic Structures**

**(5 Hours)**

The structure of algebra, Algebraic Systems, Semi-Groups, Monoids, Groups, Subgroups And Their Properties -Cyclic Groups - Cosets - Permutation Groups - Lagrange's Theorem - Cayley's Theorem - Normal Subgroups - Homomorphism Of Groups - Quotient Groups –Introduction To Rings And Fields.

### **Section 2: Topics/Contents**

#### **Unit 4: Number Theory**

**(5 Hours)**

Number theory overview, Divisors - Properties of Divisibility, Division Algorithm, Greatest Common Divisor GCD and its Properties, Remainder Classes, Properties of Congruence, Solving Congruences, Euclidean Algorithm, Extended Euclidean Algorithm, Prime Factorization Theorem, Modular

Arithmetic, Euler Phi Function, Euler's Theorem, Fermat's Little Theorem, Additive and Multiplicative Inverses, Chinese Remainder Theorem.

**Unit 5: Graph Theory**

**(5 Hours)**

Graphs and Simple Graphs, Paths and Cycles, The Incidence and Adjacency Matrices, Vertex Degrees, Paths and Connection, Cycles, Hamiltonian Cycles, Subgraphs - Graph Isomorphism, Directed and Undirected graphs, Vertex types - Isolated and pendent vertices, In degree and out degree, The handshaking theorem, Types of graphs – Null graph, regular graph, complete graph, Operations of graphs – Union, Intersection, Sum of two graphs, Product of graphs, Composition, complement, Vertex And Edge Cuts - Vertex And Edge Connectivity.

**Unit 6: Trees**

**(5 Hours)**

Trees - Acyclic graph, Terminology and Characterizations of Trees, Forest, Spanning tree, Minimal Spanning, Branch of tree, Chord, Rooted tree, Co tree, Binary tree - Path length of a binary tree, Binary tree representation of general trees, Counting trees, Tree traversal - Preorder traversal, Postorder traversal, Inorder traversal, Complete binary tree - Almost complete binary tree, Representation of algebraic structure of binary trees, Infix, prefix and postfix notation of an arithmetic expression - Infix notation, Prefix notation, Postfix notation, Evaluating prefix and postfix form of an expression, Shortest path algorithms - Dijkstra's algorithm, Minimal spanning trees - Weighted graph, Minimal spanning tree, Algorithm for minimal spanning tree - Kruskal's algorithm, Prim's algorithm.

**Syllabus  
Tutorials**

**List of Tutorials**

1. Problem solving based on propositional logic
2. Problem solving based on basic set theory
3. Problem solving based on relations and functions
4. Problem solving based on basic counting principles
5. Problem solving based on properties of binomial coefficients
6. Problem solving based on permutations, combinations
7. Problem solving based on combinatorial proof technique
8. Problem solving based on double counting
9. Problem solving based on Euler's and Fermat's theorems in calculations.
10. Problem solving based on Number Theory Theorems
11. Problem solving based on Paths, Cycles, and Connectivity
12. Problem solving based on Graph Theorems and Applications
13. Problem solving based on Tree Traversals
14. Problem solving based on Spanning Trees

**Course Outcomes:**

1. Reason mathematically about elementary discrete structures (such as functions, relations, sets, graphs, and trees) used in computer algorithms and systems

2. Express mathematical properties via the formal language of propositional and predicate logic
3. Develop recurrence relations for a wide variety of combinatorial problems
4. Demonstrate use of advanced combinatorial techniques
5. Describe elementary concepts in modular arithmetic and their applications
6. Exhibit understanding of basic graph theory, tree and its applications

**CO-PO Mapping:**

| CO  | Program Outcomes (PO) |     |     |     |     |     |     |     |     |      |      | PSO   |      |      |
|-----|-----------------------|-----|-----|-----|-----|-----|-----|-----|-----|------|------|-------|------|------|
|     | PO1                   | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO 1 | PSO2 | PSO3 |
| 1   | 3                     | 1   |     | 1   |     |     |     |     |     |      | 2    | 2     |      |      |
| 2   | 3                     | 1   |     |     |     | 1   |     |     | 2   |      | 2    | 2     | 1    |      |
| 3   | 3                     | 2   | 3   |     |     |     |     |     |     |      | 2    | 2     |      |      |
| 4   | 3                     | 3   | 3   | 3   |     |     |     |     |     |      | 2    | 2     |      |      |
| 5   | 3                     | 3   | 1   |     |     | 1   | 2   |     | 1   |      | 2    | 2     | 1    |      |
| 6   | 3                     | 3   | 3   | 2   |     | 1   |     | 1   |     |      | 2    | 2     | 1    |      |
| Avg | 3                     | 2.3 | 1.6 | 1   | 0   | 0.5 | 0.3 | 0.1 | 0.5 |      | 2    | 2     | 0.5  | 0    |

**Future Courses Mapping:** Data structures, Design and analysis of algorithms, theory of computation, artificial intelligence, machine learning.

**Job Mapping:**

Wherever one wants to model a computer science problem concretely the use of discrete structures is essential. Due to abstract nature of the course, the principles learnt have wide applicability. In any job which requires algorithmic thinking, programming, use of data structures, the knowledge of discrete structures is very helpful.

**Text Books:**

1. “Discrete Mathematics and its applications” by Kenneth Rosen (William C Brown Publisher)
2. C. L. Liu and D. P. Mohapatra, “Elements of Discrete Mathematics”, 4th Edition, McGraw-Hill.
3. “Applied Combinatorics” by Alan Tucker (Wiley Publishing company)
4. “Combinatorics: Topics, techniques, algorithms” by Peter J. Cameron (Cambridge University Press)
5. Graph Theory by Reinhard Diestel (Springer Verlag Publishing Company)
6. Introduction to Graph Theory by Douglas B. West ( Prentice-Hall publishers)

**Syllabus Template**

**FF No. : 654**

**CI2007, CI2008: Engineering Design and Innovations-III, IV**

**Course Prerequisites:** Problem Based Learning

**Course Objectives:**

1. To develop critical thinking and problem-solving ability by exploring and proposing solutions to realistic/social problems.
2. To Evaluate alternative approaches, and justify the use of selected tools and methods,
3. To emphasize learning activities those are long-term, inter-disciplinary and student-centric.
4. To engage students in rich and authentic learning experiences.
5. To provide every student the opportunity to get involved either individually or as a group so as to develop team skills and learn professionalism.
6. To develop an ecosystem to promote entrepreneurship and research culture among the students

**Course Relevance:** Project Centric Learning (PCL) is a powerful tool for students to work in areas of their choice and strengths. Along with course based projects, curriculum can be enriched with semester long Engineering Design and Development courses, in which students can solve socially relevant problems using various technologies from relevant disciplines. The various socially relevant domains can be like Health care, Agriculture, Defense, Education, Smart-City, Smart Energy and Swaccha Bharat Abhiyan. To gain the necessary skills to tackle such projects, students can select relevant online courses and acquire skills from numerous sources under guidance of faculty and enrich their knowledge in the project domain, thereby achieving project centric learning. Modern world sustained and advanced through the successful completion of projects. In short, if students are prepared for success in life, we need to prepare them for a project-based world. It is a style of active learning and inquiry-based learning. Project centric learning will also redefine the role of teacher as mentor in the learning process. The PCL model focuses the student on a big open-ended question, challenge, or problem to research and respond to and/or solve. It brings students not only to know, understand and remember rather it takes them to analyze, design and apply categories of Bloom's Taxonomy.

**Preamble** - The content and process mentioned below is the guideline document for the faculties and students to start with. It is not to limit the flexibility of faculty and students; rather they are free to explore their creativity beyond the guideline mentioned herewith. For all courses of ED, laboratory course contents of "Engineering Design" are designed as a ladder to extend connectivity of software technologies to solve real word problem using interdisciplinary approach. The ladder in the form of gradual steps can be seen as below:

Industry Communication Standards, Single Board Computers and IoT, Computational Biology (Biomedical and Bioinformatics), Robotics and Drone, Industry 4.0 (Artificial Intelligence, Human Computer Interfacing, 5G and IoT, Cloud Computing, Big Data and Cyber Security etc).

**Group Structure:**

- There should be a team/group of 4-5 students.

- A supervisor/mentor teacher assigned to individual groups.
- It is useful to group students of different abilities and nationalities together.

**Selection of Project/Problem:**

- Students must focus to initiate the task/idea. The idea inception and consideration shall be from following areas as a real world problem:
- Health Care, Agriculture, Defense, Education, Smart City, Smart Energy, Swaccha Bharat Abhiyan, Environment, Women Safety.
- This is the sample list to start with. Faculty and students are free to include other areas which meet the society requirements at large.
- The model begins with the identifying of a problem, often growing out of a question or “wondering”. This formulated problem then stands as the starting point for learning. Students design and analyze the problem/project within an articulated disciplinary subject frame/domain.
- A problem can be theoretical, practical, social, technical, symbolic, cultural, and/or scientific and grows out of students’ wondering within different disciplines and professional environments. A chosen problem has to be exemplary. The problem may involve an interdisciplinary approach in both the analysis and solving phases.
- By exemplarity, a problem needs to refer back to a particular practical, scientific, social and/or technical domain. The problem should stand as one specific example or manifestation of more general learning outcomes related to knowledge and/or modes of inquiry.

**Teacher’s Role in PCL :**

- Teacher is not the source of solutions rather he will they act as the facilitator and mentor.
- To utilize the principles of problems solving, critical thinking and metacognitive skills of the students.
- To aware the group about time management.
- Commitment to devote the time to solve student’s technical problems and interested in helping students to empower them better.

**Student's Role in PCL:**

- Students must have ability to initiate the task/idea they should not be mere imitators.
- They must learn to think.
- Students working in PCL must be responsible for their own learning.
- Students must quickly learn how to manage their own learning, Instead of passively receiving instruction.
- Students in PCL are actively constructing their knowledge and understanding of the situation in groups.
- Students in PCL are expected to work in groups.
- They have to develop interpersonal and group process skills, such as effective listening or coping creatively with conflicts.

**Developing Inquiry Skills:**

- Students in PCL are expected to develop critical thinking abilities by constantly relating: What they read to do? What they want to do with that information?
- They need to analyze information presented within the context of finding answers.
- Modeling is required so that the students can observe and build a conceptual model of the required processes.
- Use the following mechanism to maintain the track of moving towards the solution.  
How effective is .....? How strong is the evidence for .....? How clear is .....?

- What are the justifications for thinking? Why is the method chosen?
- What is the evidence given to justify the solution?

**Literature Survey – To avoid reinvention of wheel:**

- It is integral part of self- directed learning
- Identify the information needed to solve a given problem or issue
- Be able to locate the needed information
- Use the information to solve the given problem effectively.
- Skills required by students in information literacy include:
  - How to prepare the search? How to carry out the research
- Sorting and assessing of information in general

**Use of Research Methodology:** - investigation, collaboration, comprehension, application, analysis, synthesize and evaluation

**Focus on following skills while working in a team to reach to solution:**

- Collaborative learning
- Interpersonal Skills
- Resources Evaluation
- Metacognitive Skills
- Reflection Skills

**EDD Sample Case Studies :-**

With the adaptation of industry communication standards, Raspberry Pi and Sensors, following projects can be taken up:

- 1) Design a deployable product for soil moisture detection
- 2) Design a deployable product for temperature detection
- 3) Design a deployable product for pressure detection
- 3) Design a deployable product smoke detection
- 4) Design a deployable product for motion detection
- 5) Design a deployable product for collision detection
- 6) Design a deployable product for sound detection

**...not limited to.....Faculty and students are free to include other areas which meet the society requirements at large.**

**Suggest an assessment Scheme:**

***Suggest an Assessment scheme that is best suited for the course. Ensure 360 degree assessment and check if it covers all aspects of Bloom's Taxonomy.***

*To focus on the higher levels of the Booms Taxonomy analyze, apply, evaluate and create.*

**Text Books: (As per IEEE format)**

- 1.A new model of problem based learning. By Terry Barrett. All Ireland Society for higher education (AISHE). ISBN:978-0-9935254-6-9; 2017
- 2.Problem Based Learning. By Mahnazmoallem, woei hung and Nada Dabbagh, Wiley Publishers. 2019.
3. Stem Project based learning and integrated science, Technology, Engineering and mathematics approach. By Robert Robart Capraro, Mary Margaret Capraro

**Reference Books: (As per IEEE format)**

- 1.De Graaff E, Kolmos A., red.: Management of change: Implementation of problem-based and project-based learning in engineering. Rotterdam: Sense Publishers. 2007.
- 2.Project management core textbook, second edition, Indian Edition , by Gopalan.
3. The Art of Agile Development. By James Shore & Shane Warden.

**MOOCs Links and additional reading material:** [www.nptelvideos.in](http://www.nptelvideos.in)



<https://worldwide.espacenet.com/>

**Course Outcomes:**

On completion of the course, learner will be able to–

- CO1: Identify the real life problem from societal need point of view
- CO2: Choose and compare alternative approaches to select most feasible one
- CO3: Analyze and synthesize the identified problem from technological perspective
- CO4: Design the reliable and scalable solution to meet challenges
- CO5: Evaluate the solution based on the criteria specified
- CO6: Inculcate long life learning attitude towards the societal problems

**CO PO Map:**

|         | Program Outcomes (PO) |     |      |      |     |     |     |     |     |      |      |      | PSO  |      |      |      |
|---------|-----------------------|-----|------|------|-----|-----|-----|-----|-----|------|------|------|------|------|------|------|
| CO/PO   | PO1                   | PO2 | PO3  | PO4  | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 | PSO1 | PSO2 | PSO3 | PSO4 |
| CO1     | 2                     | 2   | 2    | 2    |     |     |     |     | 3   |      | 2    | 2    |      |      |      | 3    |
| CO2     | 2                     | 2   | 3    | 2    | 2   |     | 2   |     | 3   |      | 2    | 2    | 3    |      | 3    | 3    |
| CO3     | 2                     | 2   | 3    | 2    | 3   |     | 2   |     | 3   |      | 2    | 2    | 3    |      | 3    | 3    |
| CO4     | 2                     | 2   | 3    | 2    | 3   | 3   |     | 2   | 3   |      | 2    | 2    | 3    | 3    | 3    | 3    |
| CO5     | 2                     | 2   | 3    | 2    | 3   | 2   |     |     | 3   |      | 2    | 2    | 3    |      | 3    | 3    |
| CO6     | 2                     | 2   | 3    | 3    | 2   |     |     |     | 3   |      | 3    | 2    | 3    |      | 3    | 3    |
| Average | 2.0                   | 2.0 | 2.83 | 2.83 | 2.6 | 2.5 | 2.0 | 2.0 | 3.0 | 1.0  | 2.16 | 2.0  | 3.0  | 3.0  | 3.0  | 3.0  |

**CO attainment levels**

CO1 -4   CO2 -2   CO3-4   CO4-5   CO5 -1   CO6-3

**Syllabus Template**

**FF No. : 654**

**SH2001, SH3001: Reasoning and Aptitude Development**

**Credits:.1**

**Lab: 2 hours/Week**

**Unit 1: English Language**

Familiarity with English Language, Ability to understand written text, spoken word and effective communication through written documents; Coverage of vocabulary to cope up with general and specific terminology, syntax and sentence structure, prevention of incorrect use leading to distortion in communication; synonyms, antonyms and contextual vocabulary, Grammar – Error identification, sentence improvement and construction, Reading Comprehension

**Unit 2: Logical Ability**

Objective interpretation of things, ability to perceive and interpret trends to make generalizations; ability to analyze assumptions behind an argument or statement; Deductive reasoning: Assessment of ability to synthesize information and derive conclusions - Coding deduction logic, Data Sufficiency, Directional Sense, Logical word sequence, Objective reasoning, Selection and decision



tables, puzzles; Inductive reasoning: Assessment of ability to learn by example, imitation or by trial – Analogy pattern recognition, Classification pattern recognition, Coding pattern recognition, Number series pattern recognition; Abductive reasoning: Critical thinking ability of seeing through logical weak links or loopholes in an argument or a group of statements; Critical reasoning: assessment of ability to think through and analyze logical arguments, assessment of ability to use logical constructs to offer reasoning in unfamiliar situations; Information Gathering and synthesis: Ability of locating information, information ordering, rule based selection and data interpretation, order and classify data, interpret graphs, charts, tables and make rule based deductions. Application of these approaches for using visual, numerical and textual data from single or multiple sources

### **Unit 3: Quantitative Ability**

Basic numbers – decimals and fractions, factorization, divisibility: HCF, LCM, Odd, even, prime and rational numbers. Application of algebra to real world, direct and inverse proportion, common applications – Speed-time -distance, Profit-loss, percentage, age relations, mixtures, other miscellaneous quantitative combination, exponentials and logarithms, permutations and combinations, probability. Spatial reasoning: Inductive – Missing portions, Sequence and series; Deductive analysis.

### **Reference Books –**

1. "English Grammar in Use" by Raymond Murphy, Cambridge University Press.
2. "Word Power Made Easy" by Norman Lewis, Goyal Publishers & Distributors.
3. "Objective General English" by S.P. Bakshi, Arihant Publications.
4. "English for Competitive Examinations" by K. Sinha, S. Chand Publishing.
5. "Essential English Grammar" by Philip Gucker, Wiley.
6. "English Idioms and Phrasal Verbs" by M.A. Yadav, Vikas Publishing House.
7. "The Oxford English Grammar" by Sidney Greenbaum, Oxford University Press.
8. "A Modern Approach to Verbal & Non-Verbal Reasoning" by R.S. Aggarwal, S. Chand Publishing, ISBN: 978-8121903409.
9. "Logical Reasoning and Data Interpretation for the CAT" by Nishit K. Sinha, Pearson India, ISBN: 978-8131709117.
10. "Logical Reasoning and Data Interpretation for the CAT" by Arun Sharma, McGraw Hill Education, ISBN: 978-0070709642.
11. "A New Approach to Reasoning Verbal and Non-Verbal" by B.S. Sijwali & Indu Sijwali, Arihant Publications, ISBN: 978-9311124692.
12. "Quantitative Aptitude for Competitive Examinations" by R.S. Aggarwal, S. Chand Publishing, ISBN: 978-8121900637.
13. "How to Prepare for Quantitative Aptitude for the CAT" by Arun Sharma, McGraw Hill Education, ISBN: 978-0070709642.
14. "The Pearson Guide to Quantitative Aptitude for Competitive Examination" by Pearson, Pearson India, ISBN: 978-8131709117.
15. "Quantitative Aptitude for Competitive Examinations" by Abhijit Guha, Tata McGraw Hill Education, ISBN: 978-0070666653.
16. "Data Interpretation & Data Sufficiency" by R.S. Aggarwal, S. Chand Publishing ISBN: 978-8121903515.
17. "Quantitative Aptitude for Competitive Examinations" by S. Chand, S. Chand Publishing, ISBN: 978-8121903423.

### **Course Outcomes –**

Upon completion of the course, the student will be able to –

1. Improve the reading, writing and verbal skills, and enhance comprehension and articulation abilities

2. Develop logical reasoning abilities, enabling them to make sound decisions in problem-solving scenarios
3. Develop mathematical aptitude as well as data interpretation abilities and use them in test cases and real world problems
4. Learn to apply approaches for optimum time-management, prioritization maximizing the accuracy
5. Learn data interpretation, apply mathematical skills to draw accurate conclusions
6. Apply their knowledge of English, reasoning and quantitative skills for planning, critical thinking and real world problems

**CO-PO Map**

|         | Program Outcomes (PO) |      |     |     |     |     |     |     |     |      |      |      |
|---------|-----------------------|------|-----|-----|-----|-----|-----|-----|-----|------|------|------|
| CO/PO   | PO1                   | PO2  | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
| CO1     | 1                     | 3    | -   | 2   | 0   | 1   | 0   | 1   | 0   | 3    | 1    | 1    |
| CO2     | 2                     | 3    | 2   | 3   | 0   | 1   | 0   | 1   | 0   | 3    | 1    | 1    |
| CO3     | 2                     | 3    | 2   | 3   | 0   | 1   | 0   | 1   | 0   | 3    | 1    | 1    |
| CO4     | 2                     | 3    | 2   | 2   | 3   | 1   | 0   | 1   | 0   | 2    | 1    | 1    |
| CO5     | 2                     | 3    | 2   | 1   | 2   | 0   | 0   | 1   | 0   | 1    | 0    | 1    |
| CO6     | 2                     | 2    | 2   | -   | 1   | 1   | 0   | 0   | 0   | 1    | 0    | 1    |
| Average | 1.83                  | 2.83 | 2.0 | 2.2 | 2.0 | 1   | 0   | 1   | 0   | 2.16 | 1.0  | 1    |

**Attainment levels: 3,3,3,3,3,4**

**Syllabus Template**

**FF No. : 654**

**CI2006, CI3010: Design Thinking 3,5**

**Credits: 01**

**Teaching Scheme:** Tutorial 01 Hr/week

**Course Prerequisites:** Problem Based Learning, Project Centric Learning

**Course Objective:**

To provide ecosystem for students and faculty for paper publication and patent filing

**Section 1: Topics/Contents**

What is research?

Importance of Paper Publication and Patents

Structure of Paper

Journal Publication

Publication in conference  
Literature Review  
Research Paper Writing  
Journal Ratings and Evaluation  
How to rate a Journal?  
Intellectual property (IP)  
Research Ethics  
Entrepreneurship

### **Section 2: Topics/Contents**

Structure of The paper  
Journal List (Top 50 Journals)  
Selection of the journal  
Use of various online journal selection tools  
Plagiarism checking  
Improving contents of the paper  
Patent drafting  
Patent search  
Filing of patent  
Writing answers to reviewer questions  
Modification in manuscript  
Checking of publication draft

**Course Outcome:** [Publication of paper or patent]

The student will be able to

1. Understand the importance of doing Research
2. Interpret and distinguish different fundamental terms related to Research
3. Apply the methodology of doing research and mode of its publication
4. Write a Research Paper based on project work
5. Understand Intellectual property rights
6. Use the concepts of Ethics in Research
7. Understand the Entrepreneurship and Business Planning

**Syllabus Template**

**FF No.: 654**

**CI2017: Advanced Data Structures and Algorithms**

**Credits: 3**

**Teaching Scheme Theory: 2 Hours/Week**

**Lab: 2 Hours/Week**

**Course Prerequisites:** Fundamentals of Data Structures

**Course Objectives:**

1. To interpret concept of Tree Data Structure
2. To build Graphs and represent them in various ways
3. To apply hashing technique on data
4. To analyse Algorithms and working of divide and conquer method
5. To correlate Greedy and Dynamic Programming methods
6. To implement Backtracking and Branch and bound techniques

**Course Relevance:**

The course Advanced Data Structures is highly relevant to the Computer Engineering / Information Technology curriculum as it provides in-depth knowledge of efficient data organization, algorithm design, and complexity analysis, which are fundamental to solving large-scale and real-time computational problems. This course equips students with the ability to design, analyze, and implement advanced data structures and algorithms such as balanced trees, graphs, heaps, hashing techniques, and optimization algorithms. These concepts form the backbone of modern applications including databases, operating systems, compiler design, artificial intelligence, machine learning, networking, and distributed systems.

By emphasizing algorithmic efficiency, correctness proofs, and problem-solving strategies, the course strengthens students' analytical and logical thinking skills. It also prepares students for technical interviews, competitive programming, higher studies, and research, while supporting lifelong learning and adaptation to emerging technologies, in alignment with NBA and Outcome-Based Education (OBE) requirements.

**Section 1: Topics/Contents**

**Unit 1: Trees**

**(5 Hours)**

Linear and non-Linear data structures, **Tree representation**, Tree & Binary tree with their properties, converting general tree to Binary tree, **Expression Tree**, **Binary tree traversal**: Inorder, Preorder and

Postorder, Binary Search Tree, Binary search tree operations, **Binary Search Tree** – Analysis, Height, Optimality, AVL trees. Red-Black Trees, B-Trees & B+ Trees

**Unit II: Graphs (5 Hours)**

Basic concepts, **storage representation for graphs**, Adjacency matrix, adjacency list, **Traversal Techniques**: Depth First Search, Breadth First Search, BFS & DFS (Iterative + Recursive)

**Minimum Spanning Tree**- Prim's algorithm, Kruskal's algorithm, Dijkstra's shortest path algorithm, Bellman-Ford Algorithm, Floyd-Warshall Algorithm, A\* Search Heuristic

**Unit III: Heap and Hashing Techniques (5 Hours)**

**Symbol table**, Heap data structure, min and max heap, heap sort, applications of heap (Priority Queue), **Advanced Heaps Hash Table**, Scattered tables, hash functions, characteristics of good hash function, different methods to transform keys to hash values, Collision resolution techniques, Fibonacci Heap, Universal Hashing, Double Hashing, Load Factor & Rehashing

**Section2: Topics/Contents**

**Unit IV: Introduction to Algorithms (05 Hours)**

**Algorithms**: Introduction, Algorithm specifications-pseudo code conventions, Algorithm analysis-Asymptotic Notations, Master Theorem and Substitution method to solve recurrence relation, Recursion Tree Method

**Divide and Conquer**: General method, Quick sort, Merge Sort and their time complexities.

**Unit V: Greedy Algorithm and Dynamic Programming (05 Hours)**

**Greedy Algorithm**: General method, Knapsack Problem, Job sequencing with deadlines, coin change problem. Huffman Coding

**Dynamic Programming**: General method, 0/1 Knapsack Problem, Optimal Binary Search Tree, Travelling Salesperson Problem, Matrix Chain Multiplication (IIT/MIT standard), Longest Common Subsequence (LCS)

**Unit VI: Backtracking and Branch and Bound (05 Hours)**

**Backtracking**: General method, Knapsack Problem, 8 Queens Problem, Sudoku / Constraint-Based Search Graph colouring.

**Branch and Bound**: General Method, Bounding Function Design, Knapsack Problem, Travelling Salesperson Problem, , Knapsack using B&B, TSP using B&B

**List of Practical's:**

1. A. Create Binary tree and perform following operations:
  - a. Insert
  - b. Display
  - c. Find Depth of a tree
  - d. Display leaf-nodes
  - e. Create a copy of a tree.  
B. Implement binary search tree and perform following operations:
  - a. Insert b. Delete c. Display d. Search e. BFS (Level wise print)

**BST Traversals & Height Check**

Construct a Binary Search Tree (BST) from a given set of keys.

Perform recursive and iterative traversals, compute the tree height, and verify whether the BST is height-balanced.

2. Construct an expression tree from postfix expression and perform recursive and non- recursive traversals
3. AVL Tree with Rotations  
Implement insertion and deletion operations in an AVL Tree.  
After each update, identify and display the rotation performed (LL, LR, RL, or RR).
4. Consider a friend's network on face book social web site. Model it as a graph to represent each node as a user and a link to represent the friend relationship between them. Store data such as date of birth, number of comments for each user.
  1. Find who is having maximum friends
  2. Find who has post maximum and minimum comments
  3. Find users having birthday in this monthHint: (Use adjacency list representation and perform DFS and BFS traversals).
5. B-Tree / B+ Tree Insertion  
Insert a sequence of keys into a B-Tree or B+ Tree of specified order.  
Show all node splits, promotions, and the final hierarchical structure.
6. You are given an undirected weighted graph with nodes and edges. The nodes are numbered from 1 to n. Find the total weight of the minimum spanning tree, as well as one specific minimum spanning tree using Prims/Kruskal's algorithm. Note that there may be multiple different minimum spanning trees. You need to construct any one of them.

7. A student wants to travel from Source (A) to destination (B). Student books a cab from A to reach B. Calculate a shortest path by avoiding real time traffic to reach destination.  
Shortest Path Algorithms: Dijkstra, Bellman–Ford & Floyd–Warshall. Compare the resulting shortest path costs and detect any negative cycle conditions.
8. Read the marks obtained by students of second year in an online examination of particular subject. Find out maximum and minimum marks obtained in that subject.  
Use heap data structure. Analyse the algorithm.
9. Consider a telephone book database of N clients. Make use of a hash table implementation to quickly look up client's telephone number. Make use of two collision handling techniques and compare them using number of comparisons required to find a set of telephone numbers
10. A. Read the Credits obtained to students in different club activities and sort the credits using merge sort.  
B. Implement quick sort for sorting students credits they received in last 2 years and display the names of top 3 students.

8. Hash Table with Chaining & Open Addressing

Implement a hash table using chaining and open addressing (linear/quadratic/double hashing). Record and compare collisions, load factor, and probe counts for different techniques.

11. Given a set of items with weights and values, and a knapsack capacity, find the maximum total value that can be obtained by selecting fractions of items using the Greedy strategy.
12. There are 5 jobs, each with a deadline and associated profit. The Select a subset of jobs that maximize the total profit while ensuring that each selected job is completed within its deadline. Display the selected jobs with their deadlines.

| Job | Deadline | Profit |
|-----|----------|--------|
| A   | 2        | 100    |
| B   | 1        | 19     |
| C   | 2        | 27     |
| D   | 1        | 25     |
| E   | 3        | 15     |

13. Given a 2D matrix  $tsp[][]$ , where each row has the array of distances from that indexed city to all the other cities and -1 denotes that there doesn't exist a path between those two indexed cities. The task is to print minimum cost in TSP cycle.  
Input:  
 $tsp[][] = \{-1, 10, 15, 20\}, -$   
 $\{10, -1, 35, 25\},$

{15, 35, -1, 30},  
{20, 25, 30, -1}};

14. Place each queen one by one in different rows, starting from the topmost row. While placing a queen in a row, check for clashes with already placed queens. For any column, if there is no clash then mark this row and column as part of the solution by placing the queen. In case, if no safe cell found due to clashes, then backtrack (i.e, undo the placement of recent queen) and return false.

**Text Books:** *(As per IEEE format)*

1. Computer Algorithms by Ellis Horowitz, Sartaj Sahni, Sanguthevar Rajasekaran 2<sup>nd</sup> edition
2. Aho, J Hopcroft, J.Ulman, “Data Structures and Algorithms”, Pearson Education, 1998, ISBN-0-201-43578-0
3. Data Structures and Algorithms in Java by Robert Lafore 2<sup>nd</sup> edition

**Reference Books:** *(As per IEEE format)*

1. Y. Langsam, M. Augenstein, A. Tannenbaum, "Data Structures using C and C++", 2nd Edition, Prentice Hall of India, 2002, ISBN-81-203-1177-9.
2. M. Weiss, —Data Structures and Algorithm Analysis in C++, 4th edition, Pearson Education, 2002, ISBN-81-7808-670-0.
3. G. A.V, PAI , “Data Structures and Algorithms “, McGraw Hill, ISBN -13: 978-0-07-066726-6
4. Introduction to Algorithms 3rd Edition by Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, & Clifford Stein, MIT Press, Cambridge MA USA, ISBN 978-81-203-4007-

**MOOCs Links and additional reading material:**

1. NPTEL Course “Data Structures And Algorithms”, IIT Delhi, Prof. Naveen Garg
2. <https://nptel.ac.in/courses/106102064>
3. NPTEL Course “Programming and Data Structure”, IIT Kharagpur, Dr.P.P.Chakraborty  
<https://nptel.ac.in/courses/106105085>

**Course Outcomes:**

Upon completion of the course, students will be able to

CO1: Apply advanced tree and balanced search structures (AVL, RB-Tree, B/B+) to perform efficient searching, insertion, and deletion with complexity justification.

CO2: Analyze and implement graph algorithms (MST, shortest path, connectivity) using correct representations and efficiency trade-offs.

CO3: Apply heaps, priority queues, and hashing techniques (universal hashing, collision resolution) to design efficient dictionary and scheduling structures.

CO4: Solve recurrences, analyze algorithms, and implement divide-and-conquer strategies with correctness and complexity proofs.



CO5: Apply greedy and dynamic programming methods to solve optimization problems with appropriate problem decomposition and DP table formulation.

CO6: Use backtracking and branch-and-bound techniques to solve combinatorial search problems with pruning and optimization strategies.

**CO-PO Map:**

|         | Program Outcomes (PO) |     |     |     |     |     |     |     |     |      |      |      | PSO   |      |       |
|---------|-----------------------|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|-------|------|-------|
| CO/PO   | PO1                   | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 | PSO 1 | PSO2 | PSO 3 |
| CO1     | 2                     | 2   | 2   | 2   | 2   | 1   | –   | –   | –   | 1    | –    | 2    | -     | -    | -     |
| CO2     | 3                     | 3   | 3   | 3   | 2   | 1   | –   | –   | –   | 1    | –    | 3    | 2     | -    | 1     |
| CO3     | 3                     | 2   | 2   | 2   | 2   | 1   | –   | –   | –   | 1    | –    | 2    | 2     | -    | -     |
| CO4     | 3                     | 3   | 3   | 3   | 2   | 1   | –   | –   | –   | 1    | –    | 3    | 2     | 2    | 2     |
| CO5     | 3                     | 3   | 3   | 2   | 1   | 2   | –   | –   | 1   | 2    | –    | 3    | -     | 2    | 2     |
| CO6     | 3                     | 2   | 3   | 2   | 2   | 2   | –   | –   | 1   | 2    | –    | 3    | -     | 2    | 2     |
| Average | 3                     | 3   | 3   | 2   | 2   | 1   | –   | –   | 1   | 1    | –    | 3    | 2     | 2    | 2     |

**CO Attainment levels:**

Weights for attainment levels: L1 - Easy-0.75 L2 - Comfortable-0.7 L3 – Medium – 0.65

L4 – Somewhat difficult – 0.6 L5 – Difficult – 0.55

CO1 –, CO2 –, CO3 –, CO4 –, CO5 – and CO6 –

**Future Courses Mapping:**

The knowledge and skills acquired in this course serve as a prerequisite and strong support for advanced courses such as Design and Analysis of Algorithms, Database Management Systems, Operating Systems, Computer Networks, Artificial Intelligence, Machine Learning, and Distributed Systems. Mastery of these concepts enables students to better understand algorithmic efficiency, system performance, and intelligent decision-making in subsequent courses.

**Job Mapping:**

The competencies developed through this course prepare students for roles that require strong algorithmic thinking and efficient data handling. Graduates are well-equipped for positions such as Software Engineer, Backend Developer, Data Engineer, Algorithm Engineer, System Software Developer, and Research Engineer, where optimized data structures and algorithms are critical for performance-driven applications.

**Syllabus Template**

**FF No.: 654**

**CI2018: Machine Learning**

**Credits: 3**

Teaching Scheme Theory: 2 Hours/Week

Lab :2 Hours/Week

**Course Prerequisites:** Probability and Statistics

**Course Objectives:**

- 1.To introduce the fundamental concepts of Machine Learning.
- 2.To develop the ability to apply data transformation for effective model preparation.
- 3.To equip students with the skills to evaluate machine learning models using appropriate binary and multi-class classification metrics.
- 4.To provide understanding and hands-on exposure to various regression techniques, cost functions, optimization strategies, and methods to address generalization issues such as overfitting and underfitting.
- 5.To enable students to implement and compare supervised classification algorithms, including SVM, KNN, Decision Trees, Random Forest, and Naïve Bayes.
- 6.To introduce unsupervised learning techniques and clustering algorithms, enabling students to analyze data patterns and structure using different clustering methods.

**Course Relevance:**

This course has high relevance across various domains such as industry, research, and academia, as it provides the foundational concepts required for intelligent systems. Machine Learning plays a crucial role in enabling systems to learn from data, identify patterns, and make predictions or decisions without explicit programming. A sound understanding of machine learning algorithms, data preprocessing, and model evaluation allows engineers to develop efficient, scalable, and data-driven solutions. The appropriate selection and application of machine learning techniques significantly enhance the performance, accuracy, and adaptability of modern computer applications across domains such as healthcare, finance, automation, and smart systems.

**Section 1: Topics/Contents**

**Unit I: Introduction to Machine Learning**

**(5 Hours)**

Basics of Machine Learning: Definitions, ML vs AI vs Data Science, Real life applications. Types of Learning: Supervised Learning Unsupervised Learning, Semi-Supervised Learning, Reinforcement Learning. Types of Data: Qualitative and Quantitative data, Data Sampling Techniques. Concept of Feature: Feature construction, Feature Selection and Transformation, Dataset Preparation: Training Vs. Testing Dataset. Dataset Preparation: Training Vs. Testing Dataset, Identifying Missing values and approaches, Noisy Data Extraction, Data Cleaning as a process, Data reduction

**Unit II: Data Pre-processing and Transformation**

**(5 Hours)**

Data Transformation and Discretization: Data Transformation by Normalization, Discretization by Binning Discretization by Histogram Analysis Discretization by Cluster.  
Dataset Validation Techniques: Hold-out, k-fold Cross validation, Leave-One-Out Cross-Validation (LOOCV). Dimensionality Reduction: Introduction, Subset Selection, Principal Components Analysis, Factor Analysis, Multidimensional Scaling, Linear Discriminant Analysis  
Models of Machine learning: Geometric model, Probabilistic Models, Logical Models, Grouping models, Parametric and non-parametric models

### **Unit III: Model Evaluation and Classification Metrics**

**(4 Hours)**

Binary Classification: Metrics for Evaluating Classifier Performance - Confusion Matrix, Accuracy, Precision, Recall, ROC Curves, AUC, F-Measure. Multi-class Classification: Metrics for Evaluating Classifier Performance - Per-class Precision and Per-Class Recall, weighted average precision and recall -with example, Handling more than two classes

### **Section2: Topics/Contents**

### **Unit IV: Supervised Learning - Regression Models**

**(05 Hours)**

Introduction, Univariate Regression – Model Representation, Least-Square Method for finding values of the regression coefficients, Multiple Regression, Logistic Regression. Multivariate Regression – Model Representation, Polynomial Regression. Performance enhancement with Regularization techniques. Regularized Regression – Overview of Ridge and Lasso Regression  
Cost Functions: MSE, MAE, R-Square, Performance Evaluation, Optimizing Simple Linear Regression with Gradient Descent Algorithm. Generalization Issues - Overfitting Vs. Underfitting, Bias Vs. Variance.

### **Unit V: Supervised Models – Classification Models**

**(05 Hours)**

SVM: Introduction to SVM, Computing Support Vector Classifier for classification, Soft Margin SVM, Introduction to various SVM Kernel to handle non-linear data – RBF, Gaussian, Polynomial, Sigmoid: Introduction to K-Nearest neighbour. Decision trees: Overview, decision tree algorithm, evaluating a decision tree using Gini Index and Entropy, Random Forest: Bagging, Boosting. Naive Bayes: Bayes Theorem, Naïve Bayes Classifier, smoothing, diagnostics. Diagnostics of classifiers

### **Unit VI: Unsupervised Learning**

**(04 Hours)**

Cluster Analysis: Basic Concepts and Methods, Proximity Matrices, Clustering Algorithms: K-mean, Gaussian Mixtures as Soft K-means Clustering Partitioning Methods: k-Means, A Centroid Based Methods: k-Medoids, Hierarchical Methods: Agglomerative versus Divisive Hierarchical Clustering, Dendrogram for hierarchical clustering

### **Lab Assignment**

1. Perform the following operations using Python on suitable data sets:
  - a. Read data from different formats (like csv, xls)
  - b. Find Shape of Data
  - c. Find Missing Values

- d. Find data type of each column
  - e. Finding out Zero's
  - f. Indexing and selecting data, sort data,
  - g. Describe attributes of data, checking data types of each column,
  - h. counting unique values of data, format of each column, converting variable data type (e.g. from long to short, vice versa)
2. Perform the following operations using Python on the data sets:
    - a. Compute and display summary statistics for each feature available in the dataset. (e.g. minimum value, maximum value, mean, range, standard deviation, variance and percentiles
    - b. Illustrate the feature distributions using histogram.
    - c. Data cleaning, Data integration, Data transformation, Data model building (e.g. Classification)
  3. Visualize the data using Python by plotting the graphs for assignment no. 1 and 2. Consider a suitable data set. Use Scatter plot, Bar plot, Box plot, Pie chart, Line Chart.
  4. Use a multifeature dataset Wine dataset. Apply normalization and Principal Component Analysis. Plot the first two principal components and interpret dimensionality reduction. UCI Machine Learning Repository : <https://archive.ics.uci.edu/datasets/>
  5. Predict the price of the Uber ride from a given pickup point to the agreed drop-off location. Perform following tasks: 1. Pre-process the dataset, 2. Identify outliers. 3. Check the correlation. 4. Implement linear regression and ridge, Lasso regression models. 5. Evaluate the models and compare their respective scores like R<sup>2</sup>, RMSE, etc. Dataset link: <https://www.kaggle.com/datasets/yasserh/uber-fares-dataset>
  6. Use the diabetes data set from UCI and Pima Indians Diabetes data set for performing the following: a. Univariate analysis: Frequency, Mean, Median, Mode, Variance, Standard Deviation, Skewness and Kurtosis b. Bivariate analysis: Linear and logistic regression modeling c. Multiple Regression analysis d. Also compare the results of the above analysis for the two data sets. Dataset link: <https://www.kaggle.com/datasets/uciml/pima-indians-diabetes-database>
  7. A bank wants to predict whether a customer will default on credit card payment based on age, income, education, and payment history.
    - a. Preprocess data (cleaning, encoding, scaling, train-test split).
    - b. Build SVM models using: Linear, Polynomial, RBF kernels.
    - c. Build KNN models with k = 3, 5, 7.
    - d.
    - e. Compare model accuracy and confusion matrices.

Dataset: Default of Credit Card Clients :

<https://www.kaggle.com/datasets/uciml/default-of-credit-card-clients-dataset>

8. Apply appropriate ML algorithm on a dataset collected in a cosmetics shop showing details of customers to predict customer response for special offer. Create confusion matrix based on above data and find
  - a. Accuracy

- b. Precision
- c. Recall
- d. F-1 score

9. Write a program to do following:

This dataset gives the data of Income and money spent by the customers visiting a shopping mall. The data set contains Customer ID, Gender, Age, Annual Income, Spending Score. Therefore, as a mall owner you need to find the group of people who are the profitable customers for the mall owner. Apply at least two clustering algorithms (based on Spending Score) to find the group of customers.

- a. Apply Data pre-processing
- b. Perform data-preparation (Train-Test Split)
- c. Apply Machine Learning Algorithm
- d. Evaluate Model.

Data Set: <https://www.kaggle.com/shwetabh123/mall-customers>

10. Assignment on Regression technique.

Download temperature data from the link below.

<https://www.kaggle.com/venky73/temperaturesof-india?select=temperatures.csv>

This data consists of temperatures of INDIA averaging the temperatures of all places month wise. Temperatures values are recorded in CELSIUS

- a. Apply Linear Regression using a suitable library function and predict the Month-wise temperature.
- b. Assess the performance of regression models using MSE, MAE and R-Square
- c. metrics
- d. Visualize a simple regression model.

11. Assignment on Classification technique. Every year many students give the GRE exam to get admission in foreign Universities. The data set contains GRE Scores (out of 340), TOEFL Scores (out of 120), University Rating (out of 5), Statement of Purpose strength (out of 5), Letter of Recommendation strength (out of 5), Undergraduate GPA (out of 10), Research Experience (0=no, 1=yes), Admitted (0=no, 1=yes). Admitted is the target variable.

Data Set: <https://www.kaggle.com/mohansacharya/graduate-admissions>

The counsellor of the firm is supposed check whether the student will get an admission or not based on his/her GRE score and Academic Score. So to help the counsellor to make appropriate decisions, build a machine learning model classifier using a Decision tree to predict whether a student will get admission or not. Apply Data pre-processing (Label Encoding, Data Transformation....) techniques if necessary.

- a. Perform data-preparation (Train-Test Split)
- b. Apply Machine Learning Algorithm
- c. Evaluate Model.

12. A telecom company wants to predict whether a customer will churn based on demographics, account info, and usage patterns.

- a. Preprocess data (missing values, encoding, scaling, train-test split)
- b. Train a Random Forest classifier and tune hyperparameters
- c. Evaluate model: Confusion Matrix, Accuracy, Precision, Recall, F1-score
- d. Analyse feature importance

Telco Customer Churn (Kaggle): <https://www.kaggle.com/datasets/blastchar/telco-customer-churn>

13. Implement K-Means clustering on Iris.csv dataset. Determine the number of clusters using the elbow method. Dataset Link: <https://www.kaggle.com/datasets/uciml/iris>

14. A retail company wants to group customers based on Annual Income and Spending Score for targeted marketing. Dataset: Mall Customer Segmentation

(Kaggle): <https://www.kaggle.com/datasets/shwetabh123/mall-customers>

- a. Preprocess data (missing values, encoding, scaling)
- b. Apply Hierarchical Clustering and plot dendrography
- c. Visualize clusters (Income vs Spending Score)
- d. Interpret and describe each customer segment

14) Mini project is to be performed in a group of 3 to 4 students.

Develop a mini project in a group using different predictive models' techniques to solve any real-life problem. Use different frameworks of ML Frameworks such as Tensorflow, PyTorch, Scikit-learn – Keras.

Data Set References:

1. <https://aikosh.indiaai.gov.in/home/datasets/all>
2. Amazon Dataset:- <https://registry.opendata.aws/>
3. <http://archieve.ics.uci.edu/ml/index.php>.
4. Google's Data Search Engines :-<https://datasetsearch.research.google.com/>
5. Microsoft Dataset: - <https://msropendata.com/>
6. Awesome Public Dataset: - <https://github.com/awesomedata/awesome-public-datasets>
7. Scikit-learn Dataset: -<https://scikit-learn.org/stable/>
8. Computer Vision Datasets: - <https://www.visual.io/discovery>
9. Indian Government Datasets: -<https://data.gov.in/>
10. US Government Datasets: - <https://www.data.gov/>

11. Northern Ireland Public Sector Dataset: - <https://www.opendatani.gov.uk/>
12. European Union Open Data Portal: - <https://data.europa.eu/euodp/data/dataset>

### **List of Course Project**

1. Explainable AI-Based Credit Risk Assessment Using SHAP and LIME
2. Hybrid Ensemble Model for Traffic Accident Severity Prediction
3. Early Detection of Alzheimer's Disease Using ML on MRI-Derived Features
4. Imbalanced Data Handling for Fraud Detection Using Advanced Sampling Techniques
5. Time-Series Forecasting of Energy Consumption Using LSTM and ML Models
6. Multi-Class Skin Disease Classification Using Machine Learning and Feature Engineering
7. Anomaly Detection in Network Traffic Using Unsupervised Learning Techniques
8. Comparative Analysis of Feature Selection Techniques for High-Dimensional Datasets
9. Demand Forecasting System Using Machine Learning with Seasonal Decomposition
10. Fake News Detection Using Hybrid NLP and Machine Learning Models
11. Predictive Maintenance of Industrial Equipment Using Machine Learning
12. Emotion Recognition from Speech Using Machine Learning Techniques
13. Stock Market Volatility Prediction Using Machine Learning and Technical Indicators
14. Medical Diagnosis System Using Ensemble Learning Techniques
15. Personalized Recommendation System Using Hybrid Collaborative and Content-Based Filtering
16. Federated Learning Framework for Privacy-Preserving Medical Data Classification
17. Explainable Intrusion Detection System Using Machine Learning
18. Multi-Step Time Series Forecasting Using Hybrid ARIMA-LSTM Models
19. Bias and Fairness Analysis in Machine Learning Classification Models
20. Graph-Based Recommendation System Using Machine Learning
21. Automated Resume Skill Extraction and Matching Using Machine Learning
22. Predictive Analysis of Air Quality Index Using Machine Learning Models
23. Semi-Supervised Learning for Fake Review Detection
24. Machine Learning-Based Early Warning System for Financial Risk Prediction
25. Feature Optimization and Model Compression for Efficient ML Deployment

### **Text Books: (As per IEEE format)**

Data Science and Machine Learning: Mathematical and Statistical Methods By D.P. Kroese, Z.I. Botev, T. Taimre, R. Vaisman, Chapman and Hall/CRC, Boca Raton, 2019.

### **Reference Books: (As per IEEE format)**

Introducing Data Science, Davy Cielen, Aron D.B Meysman. MANNING publishing Data Science and Machine Learning, Publisher: Sigma Data Systems, United States, ISBN: 978-1655848049

### **List of MOOC / NPTEL Courses:**



1. Introduction to Machine Learning: <https://nptel.ac.in/courses/106/106/106106139/>
2. Machine Learning: <https://nptel.ac.in/courses/106/106/106106202/>
3. Machine Learning for Science and Engineering applications: <https://nptel.ac.in/courses/106/106/106106198/>
4. Introduction to Machine Learning:
5. <https://nptel.ac.in/courses/106/105/106105152/>

### Course Outcomes:

The student will be able to –

CO1: Understand the fundamental concepts of Machine Learning, types of learning, data types, data preprocessing, and dataset preparation.

CO2: Apply data transformation, normalization, discretization, validation techniques, and dimensionality reduction for effective model building.

CO3: Evaluate machine learning models using appropriate binary and multi-class classification metrics.

CO4: Apply and analyze various regression techniques, cost functions, gradient descent, and generalization issues.

CO5: Implement and compare different supervised classification algorithms such as SVM, KNN, Decision Trees, Random Forest, and Naïve Bayes.

CO6: Apply clustering algorithms and interpret outputs from unsupervised learning methods.

### CO-PO Map:

|         | Program Outcomes (PO) |     |     |     |     |     |     |     |     |      |      |      | PSO  |      |      |
|---------|-----------------------|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|------|------|------|
| CO/PO   | PO1                   | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 | PSO1 | PSO2 | PSO3 |
| CO1     | 3                     | 2   | 2   |     |     |     |     | 2   | 2   | 2    | 2    | 3    | 3    |      |      |
| CO2     | 3                     | 3   | 2   |     |     |     |     | 2   | 2   | 2    | 2    | 3    | 3    |      |      |
| CO3     | 3                     | 3   | 2   |     |     |     |     | 2   | 2   | 2    | 2    | 3    | 3    |      |      |
| CO4     | 2                     | 2   |     |     |     |     |     | 2   | 2   | 2    | 2    | 3    | 2    |      |      |
| CO5     | 3                     | 3   | 2   |     |     |     |     | 2   | 2   | 2    | 2    | 3    | 3    |      |      |
| CO6     | 3                     | 3   | 2   |     |     |     |     | 2   | 2   | 2    | 2    | 3    | 3    |      |      |
| Average | 2.8                   | 2.6 | 2.0 |     |     |     |     | 2.0 | 2.0 | 2.0  | 2.0  | 3.0  | 2.8  |      |      |

### CO Attainment levels:

Weights for attainment levels: L1 - Easy-0.75 L2 - Comfortable-0.7 L3 – Medium – 0.65

L4 – Somewhat difficult – 0.6 L5 – Difficult – 0.55

CO1 – L3, CO2– L3, CO3 – L2, CO4 – L4, CO5 – L4 and CO6 – L5

### Future Course Mapping:



Data Science, Deep Learning, Artificial Intelligence, Big Data Analytics, Natural Language Processing, Computer Vision, Pattern Recognition, Business Intelligence, Intelligent Systems, and similar advanced courses.

**Job Mapping:**

Machine Learning Engineer, Data Scientist, AI Engineer, Data Analyst, Business Intelligence Analyst, NLP Engineer, Computer Vision Engineer, Research Assistant, Software Engineer (AI/ML), Big Data Analyst, and related roles in industry and research organizations.

**Syllabus Template**

FF No.: 654

**CI2019: Data Communication and Networking**

**Credits: 3**

Teaching Scheme Theory: 2 Hours/Week

Lab: 2 Hours/Week

Course Prerequisites: Digital Electronics

Course objectives:

1. To understand the fundamental concepts of Data Communication.
2. Identify and compare transmission media, connecting devices, and network topologies
3. To illustrate the functions of OSI and TCP/IP Protocol.
4. Apply error detection, error correction, flow control, and multiple access techniques
5. Understand network, transport, and application layer protocols

**Section1: Topics/Contents**

**Unit I – Introduction to Data Communications**

**(5 Hours )**

Process of Data Communication and its components: Transmitter, Receiver, Medium, Message, Protocol. Analog and Digital: Analog and Digital Data, Analog and Digital Signals, Periodic and Non-periodic Signal.

Transmission Impairment: Attenuation, Distortion, Noise

Data Rate Limits: Noiseless Channel: Nyquist Bit Rate, Noisy Channel: Shannon Capacity, Using Both Limits

Performance: Bandwidth, Throughput, Latency (delay), Bandwidth-delay Product, Jitter.

Modes of Communication (Simplex, Half Duplex, Full Duplex).

**Unit II – Transmission Media and Addressing ( 5 Hours)**

Guided Media: Twisted-Pair, Coaxial and Fiber-Optic Cable Unguided Media: Radio Waves, Microwaves, Infrared

Connecting devices: Hubs, Repeaters, Active hubs, Bridges, Switches, Routers, Gateways LAN, MAN, WAN, Topologies- star, mesh, bus, hybrid etc.

Addressing: Physical & logical Addresses, Port Addresses, Specific Addresses

**Unit III – Layered Model and Topologies (3 Hours)**

The OSI Model: Layered Architecture, peer-to-peer Processes, Encapsulation of Layers in the OSI Model, TCP/IP Protocol Suite

**Unit IV– Error Control and Data Link Control (6 Hours)**

Design Issues: Services to Network Layer, Framing. Types of errors: Single bit error and burst error, Redundancy, detection versus correction, forward error correction versus retransmission,

ARQ strategies: Error Detection and correction, Parity Bits, Hamming Codes and CRC. Flow

Control Protocols: Unrestricted Simplex, Stop and Wait, Go back to N, Selective repeat Point-to-point Protocol (PPP) ,Sliding Window Protocol.

MAC Sub layer: Multiple Access Protocols: Pure and Slotted ALOHA, CSMA, WDMA, CSMA/CD, CSMA/CA

**Unit V– Network Layer (5 Hours)**

Introduction: Functions of Network layer. ~~Switching Techniques~~: Circuit switching, Message

Switching, Packet Switching. IP Protocol: Classes of IP (Network addressing), IPv4 , IPv6, Network Address Translation, Sub-netting, CIDR. Network layer Protocols: ARP, RARP, ICMP,

IGMP. Network Routing and Algorithms: Shortest Path First, Distance Vector Routing

**Unit VI– Transport and Application Layer (6 Hours)**

Process to Process Delivery, Services, Socket Programming. Elements of Transport Layer Protocols:

Addressing, Connection establishment, Connection release, Flow control and buffering. Transport

Layer Protocols: TCP and UDP. Application Layer- HTTP, DNS, Email: SMTP, MIME, POP3,

Webmail, FTP, TELNET, DHCP, Application layer protocols interaction with end-user applications.

**Reference books:**

1. Andrew S. Tenenbaum,” Computer Networks”, Pearson
2. Behrouz A. Forouzan,” Data Communications and Networking”
3. Douglas E. Comer, “Computer Networks and Internets”, Pearson Education
4. Godbole Achyut, “Data Communication and Networks”, Tata McGraw Hill Publication

**List of Assignments:**

1. Network Representation on Network Simulator tool(packet Tracer)
2. Study of DCE, DTE, straight and cross cables
3. Identify MAC and IP addresses (physically as well as using packet tracer tool).
4. Demonstrate the different types of topologies and types of transmission media by using a packet tracer tool.
5. Test with basic network tools. Read the man pages of ifconfig, ping, traceroute, arp, dig and nslookup. Write the solution of the following problems into the report and also paste the screenshots of Terminal into the report.
  - (a) Find the IP address of your machine using 'ifconfig' command.
  - (b) Use 'ping' command and find out ( i.) the average RTT(round trip time). (ii.) the %packet loss. (iii.) size of packet that is sent to www.iitk.ac.in server. (iv.) size of packet that is received by your machine.
  - (c) Use 'dig www.vit.edu' command and find out i. the ip address of vit.edu ii. the ip addresses of local DNS servers of your institute
  - (d) Use 'traceroute www.vit.edu' and find out i. number of hops in between your machine and iitk.ac.in server. ii. the ip address of your network gateway of your subnet.
  - (e) Use 'arp -an' command to find out the MAC address of the device that is performing as your network gateway.
6. Design and configure static and dynamic routing in a simulated network environment using packet Tracer
7. Implement the data link layer framing methods such as character, character-stuffing and bit stuffing.
8. Write a program to compute CRC code for the polynomials.
9. Develop a simple data link layer that performs the flow control using the sliding window protocol, and loss recovery using the Go-Back-N mechanism.
10. Implement Dijkstra's algorithm to compute the shortest path through a network.
11. Implement distance vector routing algorithm for obtaining routing tables at each node.
12. Wireshark
  - i. Packet Capture Using Wire shark
  - ii. Starting Wire shark

- iii. Viewing Captured Traffic
- iv. Analysis and Statistics & Filters.

13. Write the client server programs using TCP Berkeley socket primitives for wired /wireless network for following

- a. to say Hello to Each other
- b. File transfer

14. Write the client server programs using UDP Berkeley socket primitives for wired /wireless network for following

- a. to say Hello to Each other
- b. Calculator (Trigonometry)

**NPTEL / MOOC course:**

“Data Communication”, IIT Kharagpur Prof. Ajit Pal

<https://nptel.ac.in/courses/106105082>

**Course Outcomes: -**

CO1: Apply fundamental concepts of data communication to identify different types of signals, transmission impairments, and performance metrics

CO2: Differentiate between guided and unguided transmission media. Select appropriate communication channels, networking devices, and addressing schemes for computer networks.

CO3: Describe and analyze layered network architectures (OSI & TCP/IP).

CO4: *Implement* framing techniques and *evaluate* error detection and correction mechanisms for reliable data communication.

CO5: Analyze network-layer functions, addressing mechanisms, and routing techniques.

CO6: Implement transport layer services and analyze application layer protocols for reliable end-user communication

**CO-PO Map:**

|       | Program Outcomes (PO) |     |     |     |     |     |     |     |     |      |      |      | PSO  |      |      |
|-------|-----------------------|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|------|------|------|
| CO/PO | PO1                   | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 | PSO1 | PSO2 | PSO3 |
| CO1   | 3                     | 2   | -   | -   | -   | 1   | -   | -   | -   | -    | -    | 1    | 3    | 1    | 1    |
| CO2   | 2                     | 3   | 2   | -   | -   | 3   | -   | -   | -   | -    | -    | 1    | 3    | 2    | 1    |
| CO3   | 3                     | 2   | -   | -   | -   | 2   | -   | -   | -   | -    | -    | 1    | 3    | 2    | 1    |

|         |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|---------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO4     | 2 | 3 | 2 | 2 | 2 | 3 | – | – | – | – | – | 1 | 2 | 3 | 2 |
| CO5     | 2 | 3 | 2 | 2 | 2 | 3 | – | – | – | – | – | 1 | 3 | 2 | 2 |
| CO6     | 2 | 3 | 3 | 3 | 2 | 3 | – | – | – | 1 | – | 1 | 2 | 3 | 3 |
| Average |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |

## **Syllabus Template**

FF No.: 654

### **CI2020: Full Stack Development**

#### **Credits: 3**

Teaching Scheme Theory: 2 Hours/Week

Lab: 2 Hours/Week

**Course Prerequisites:** Object-Oriented Programming Concepts, Fundamentals of Web Technologies, Database Management Systems

#### **Course Objectives:**

1. To introduce basic web concepts, HTML5 structure, CSS styling, and responsive layouts.
2. To develop the ability to write dynamic and interactive web applications using JavaScript, ES6 features, and DOM manipulation techniques.
3. To build interactive and responsive web applications using jQuery and Bootstrap.
4. To equip students with backend development skills using Node.js/Express.js, including API creation and database integration.
5. To introduce modern component-based frameworks such as Angular/React and TypeScript to build scalable and modular front-end applications.
6. To enable students to develop dynamic, component-based web applications using React.js

#### **Section1:**

#### **Topics/Contents**

##### **Unit 1: Web Essentials**

**(05 Hours)**

Introduction to the Web: Evolution of the Web: Web 1.0, Web 2.0, Web 3.0, Internet basics: Clients, servers, IP, DNS, hosting, browser architecture & rendering process, HTTP & HTTPS: Request/Response, Methods, Status Codes, URL Structure, and Query Parameters.

Developer Tools: Inspecting elements, understanding the console & network tab, working with sources & debugging. Introduction to Web Project Structure: Folder organization: assets, scripts, styles, linking external files. HTML5 Basics: HTML5 elements: headings, colors, fonts, links, frames, lists, tables, images, Forms & default validation, Block vs inline elements, Semantic elements (header, nav, section,

footer), Difference between HTML & HTML5, HTML5 media: audio, video. CSS Basics: Why CSS? Types of CSS, Applying CSS: inline, internal, external, properties, classes, nested (child) selectors, Colors, text, background, border, margin, padding, box model.

Layout + Animation: Positioning: flex, grid, inline, block, CSS animation & transition.

### **Unit 2: JavaScript Development**

**(05 Hours)**

JavaScript Fundamentals: Syntax, variables, datatypes, operators, conditionals, loops, functions, objects, arrays, and built-in objects (Math, Date, and String) ES6 Modern JavaScript Features: let, const, arrow functions, spread/rest operator, for/of, map & set, classes, promises, symbols, default parameters, String methods: includes(), startsWith(), endsWith() DOM Manipulation: Document Object Model, selecting and modifying elements, Event handling, modifying CSS dynamically, DOM tree traversing.

### **Unit 3: JQuery and Bootstrap**

**(05 Hours)**

jQuery: Introduction to jQuery: Need for jQuery, jQuery Syntax: Selectors, filters, DOM traversal, DOM manipulation: adding/removing elements, modifying CSS & attributes, Event Handling: Click, hover, keyboard, and form events. Effects & Animations: hide/show, fade, slide. Bootstrap (Industry Requirement): Why Bootstrap? CSS vs. Bootstrap, Bootstrap grid system, responsive design, Bootstrap components: button, table, lists, cards, cross-platform UI using Bootstrap

### **Section2: Topics/Contents**

### **Unit 4: Backend Development with Node.js or Express.js**

**(06 Hours)**

Node.js Core: Introduction to Node.js, environment setup, Node.js events, Node.js functions, Built-in modules (fs, http, path, os, events), Express.js: Introduction to Express, Routing, Middleware, Creating APIs, Database Integration: Express with MongoDB or Express with MySQL/SQLite Applications: Front-end + back-end integration, Building a chat application, Introduction to task managers (npm scripts, nodemon, and PM2) and unit testing basics.

### **Unit 5: Frontend Framework: Angular**

**(05 Hours)**

Angular: Angular Architecture, Project Structure, Lifecycle, Modules & Components, Data Binding, Directives & Pipes, Services & Dependency Injection, Routers, Angular Forms

TypeScript: Type annotations & inference, basic types, interfaces, classes, inheritance, enums, generics, Type guards, unions, intersections, decorators, modules & namespaces, advanced types, declaration files, and tooling.

### **Unit 6: Frontend Framework: React.js**

**(04 Hours)**

React.js: Introduction to React, React Components, JSX, Render Function, Component API, Component Lifecycle, State, Props, Passing Data from Parent to Child, Passing Data from Child to Parent, Component Styling, Refs, Keys, React Router.

**Text Books:** *(As per IEEE format)*

1. Achyut Godbole, Atul Kahate; Web Technologies: TCP/IP to Internet Application Architectures; 1st Edition, McGraw Hill Education.
2. Chris Northwood; The Full Stack Developer; 2nd Edition, Apress, 2023

**Reference Books:** *(As per IEEE format)*

1. Ethan Brown; Web Development with Node and Express; 2nd Edition, O'Reilly Media, 2022
2. Alex Banks, Eve Porcello; Learning React; 3rd Edition, O'Reilly Media.
3. Brad Dayley, Brendan Dayley, Caleb Dayley, "Node.js, MongoDB and Angular Web Development", 2nd Edition, Addison-Wesley
4. Ethan Brown; Learning JavaScript: JavaScript Essentials for Modern Application Development; 4th Edition, O'Reilly Media.
5. Bear Bibeault, Yehuda Katz, Aurelio De Rosa; jQuery in Action; 3rd Edition, Manning Publications
6. Jeremy McPeak & Paul Wilton, "Beginning JavaScript", Wrox Publication, ISBN-13: 978-0470525937
7. Maximilian Schwarzmüller; React Key Concepts; 2nd Edition, Packt Publishing

**List of Practical's:**

1. Design a multipage personal portfolio website using HTML5 semantic elements and CSS. The website should include headings, images, tables/lists, navigation links, forms with default validation, and proper use of classes, colors, margins, padding, and box model.
2. Build a multimedia webpage that includes HTML5 audio and video elements, styled using external CSS. Add captions, controls, and create a clean layout using padding, borders, and background properties.
3. Write JavaScript programs demonstrating variables, datatypes, operators, conditionals, loops, functions, arrays, and objects.
4. Create a webpage and use DOM manipulation to change text, style elements, and handle button click events.
5. Build a Random Quote Generator using JavaScript that displays a new quote each time a button is clicked.
6. Create an interactive webpage using jQuery that performs DOM manipulation, handles click/hover events, and applies effects such as hide/show, fade, and slide on selected elements.
7. Build a fully responsive webpage using Bootstrap with a grid layout and components (button, table, list, card).
8. Create a Node.js server using built-in modules (http, fs, path) that serves static HTML pages and demonstrates event handling and file operations.
9. Build a RESTful API using Express.js integrated with MongoDB/MYSQL/SQLite, implement routing and middleware, and connect it to a simple front-end to perform CRUD operations.
10. Create a simple Angular project with two components and display data using interpolation and property/event binding.



11. Create an Angular service to share data between components and set up routing with at least two navigable pages.
12. Create a React application with multiple components, demonstrating state and props for dynamic content rendering
13. Implement event handling in a React app, such as button clicks, form submissions, and input change events.
14. Build a multi-page React application using React Router and apply CSS or inline styling to components for a polished UI.

**List of Project areas:**

1. AI-Driven Adaptive Learning Platform: Design and develop a full-stack platform that dynamically personalizes learning content using user behavior analytics and performance metrics.
2. Scalable Microservices-Based E-Commerce System: Build an e-commerce application using microservices architecture with service discovery, API gateway, distributed database, and fault tolerance.
3. Real-Time Collaborative Code Editor: Develop a browser-based collaborative coding platform with real-time synchronization, conflict resolution, version history, and user authentication.
4. Blockchain-Based Secure Document Management System: Design a decentralized web application for document storage and verification using blockchain to ensure immutability and traceability.
5. Smart Traffic Monitoring and Prediction System: Build a full-stack system that collects real-time traffic data, applies predictive analytics, and visualizes congestion patterns on interactive maps.
6. Multi-Tenant SaaS Project Management Tool: Develop a secure, multi-tenant project management application with tenant isolation, role-based access control, billing, and usage analytics.
7. IoT-Enabled Smart Energy Management Platform: Create a full-stack solution that integrates IoT sensor data with cloud services to monitor, analyze, and optimize energy consumption in real time.
8. High-Availability Online Examination System: Design a fault-tolerant exam platform supporting thousands of concurrent users with load balancing, auto-scaling, and anti-cheating mechanisms.
9. Real-Time Financial Trading Dashboard: Build a full-stack web application that streams live market data, performs analytics, and visualizes trends with low-latency updates.
10. Secure Healthcare Data Exchange System: Develop a full-stack platform for interoperable healthcare data exchange using encryption, audit trails, and compliance-aware access controls.
11. Intelligent Supply Chain Optimization System: Design a web-based system that integrates demand forecasting, inventory optimization, and logistics tracking using data analytics.
12. Federated Authentication and Authorization System: Build a full-stack identity management system supporting SSO, OAuth 2.0, JWT, multi-factor authentication, and fine-grained access control.

Course Outcomes: After completion of the course, student will be able to

- CO1. To implement structured and responsive web pages using HTML5, CSS, and layout techniques.
- CO2. To use JavaScript, ES6 features, and DOM manipulation techniques for constructing interactive and dynamic web content.



CO3. To employ jQuery and Bootstrap to design responsive and interactive user interfaces.

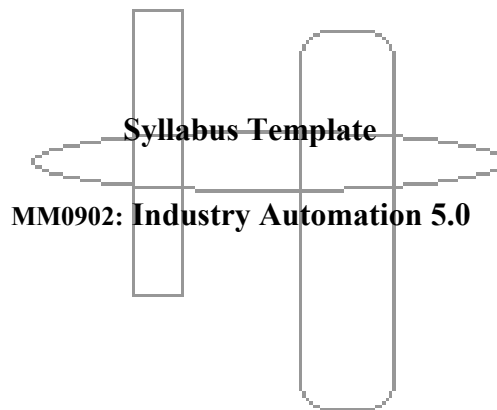
CO4. To analyze backend processes by building APIs and integrating databases using Node.js and Express.js.

CO5. To construct modular, component-driven applications using Angular or React, along with TypeScript constructs.

CO6. To implement dynamic, component-based web applications using React.js.

### CO-PO Mapping

|         | Program Outcomes (PO) |      |      |      |      |      |      |      |      |      |      |      | PSO  |      |      |
|---------|-----------------------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|
| CO/PO   | PO1                   | PO2  | PO3  | PO4  | PO5  | PO6  | PO7  | PO8  | PO9  | PO10 | PO11 | PO12 | PSO1 | PSO2 | PSO3 |
| CO1     | 3                     | 2    | 2    |      | 2    |      |      |      | 2    | 2    | 2    | 2    | 3    | 2    | 2    |
| CO2     | 3                     | 2    | 2    | 1    | 2    |      |      |      | 1    | 1    | 1    | 1    | 3    | 2    | 2    |
| CO3     | 3                     | 2    | 2    | 1    | 2    |      |      |      | 2    | 2    | 2    | 2    | 3    | 2    | 2    |
| CO4     | 3                     | 3    | 3    | 2    | 3    |      |      | 1    | 2    | 2    | 2    | 2    | 3    | 2    | 2    |
| CO5     | 3                     | 2    | 2    | 2    | 3    |      |      |      | 2    | 2    | 2    | 2    | 3    | 2    | 2    |
| CO6     | 3                     | 2    | 2    | 2    | 2    | 1    | 1    |      | 2    | 2    | 2    | 2    | 3    | 2    | 2    |
| Average | 3.00                  | 2.17 | 2.17 | 1.60 | 2.33 | 1.00 | 1.00 | 1.00 | 1.83 | 1.83 | 1.83 | 1.83 | 3.00 | 2.00 | 2.00 |



FF No.: 654

### Teaching Scheme

**Credits: 3**

Lectures: 2 Hrs/week

Tutorial: 1 Hrs/week

### Examination Scheme

MSE (O): 35 Marks

ESE (O): 35 Marks

TUTORIAL: 30 Marks

**Prerequisites:** Basic knowledge of electronics and sensors, Fundamentals of computer networks, Basic programming understanding (C/Python)

### Course Objectives:

1. To introduce concepts of Industrial Automation and Industry 5.0.
2. To explain IoT architecture, sensors, actuators, and communication technologies.
3. To understand industrial protocols, cloud monitoring, and data handling.
4. To explore industrial automation systems such as SCADA, predictive maintenance, and digital twins.
5. To understand security, privacy, and safety in IIoT environments.
6. To analyze real-world case studies of Industry 5.0 applications.

## **Topics/Contents**

### **Section1:**

#### **UNIT 1: Introduction to Industrial Automation & Industry 5.0 (5 Hr)**

Industrial Automation: Meaning, Types, Need, Evolution: Industry 1.0 → 5.0, Features of Industry 4.0: Smart Sensors, Connectivity, Automation, Big Data, Role of IoT, AI, Cyber-Physical Systems, Real industrial examples (manufacturing, robotics, energy)

#### **UNIT 2: IoT Fundamentals for Industrial Automation (5 Hr)**

IoT Definition, Characteristics, Architecture, IoT layers: Device → Network → Cloud, M2M (Machine-to-Machine) communication, IoT Frameworks & Platforms, Identifiers in IoT (UID, EPC, IP Addressing) Overview of simple hardware: Arduino, ESP32, Basics of sensor interfacing, Introduction to Raspberry Pi board: architecture, features, GPIO pins, operating system support, networking capability, role as edge device / IoT gateway in industrial automation.

#### **UNIT 3: Industrial Sensors, Actuators & Communication (4 Hr)**

Sensors: Temperature, Proximity, Vibration, Pressure, Load, Angle, Speed, Non-contact: IR, Ultrasonic, Laser sensors, Actuators: Motors, Relays, Solenoids, Industrial Mechanisms, Communication Technologies: Wi-Fi, Bluetooth, ZigBee (Basic), MQTT Publish/Subscribe model, Edge Controllers & Industrial Gateways

### **Section2: Topics/Contents**

#### **UNIT 4: IoT Protocols & Cloud in Industry 5.0 (5 Hr)**

IoT Protocols: IPv6 basics, lightweight protocols – 6LoWPAN, RPL, REST, AMQP, MQTT, CoAP, simple overview of text/image transfer protocols. Cloud & IIoT Platforms: IIoT basics, cloud monitoring using ThingSpeak, Blynk, Firebase – dashboards, data logs, alerts, remote device control. Arduino Cloud and Node-RED: Device connectivity, cloud dashboards, flow-based programming for industrial IoT applications. Cloud Service Providers: Study of AWS, Azure, GCP IoT platforms.

#### **UNIT 5: Automation Systems in Industry 5.0 (5 Hr)**

Smart Manufacturing Concepts, Predictive Maintenance (Basic Idea), Smart Energy Monitoring, Digital Twin (Concept Only): SCADA: Architecture, Components, Use Cases Selection of industrial computational hardware: Raspberry Pi, Jetson Nano, ESP32, Microcontrollers

**UNIT 6: Security, Safety & Industrial Case Studies**

**(4 Hr)**

Security in IIoT: Threats & Challenges, Authentication mechanisms, Basic encryption & password protection, Privacy regulations and policies, Safety in Industry 5.0: Safety sensors, Emergency switches, alarms. Case Studies: Industry 5.0 use cases in manufacturing, healthcare, logistics, Breakdown of real IIoT systems, Students analyze and present case studies (tutorial-based)

**Text Books:**

1. Alasdair Gilchrist, "Industry 4.0: The Industrial Internet of Things", Apress.
2. Rajesh Singh, Anita Gehlot, "Internet of Things", McGraw Hill.
3. William Buchanan, "Industrial Control Systems & SCADA Security", CRC Press.
4. Arshdeep Bahga and Vijay Madisetti, "Internet of Things: A Hands-on Approach", (Universities Press)
5. Pethuru Raj and Anupama C. Raman, "The Internet of Things: Enabling Technologies, Platforms, and Use Cases", (CRC Press)

**Reference Books:**

1. Jay Lee, Behrad Bagheri, Hung-An Kao, "A Cyber-Physical Systems Architecture for Industry 5.0".
2. Adrian McEwen, Hakim Cassimally, "Designing the Internet of Things", Wiley.
3. Salesforce IoT & IBM Industrial IoT documentation (online resources).
4. Adrian McEwen, Hakim Cassimally "Designing the Internet of Things", Wiley
5. Ovidiu Vermesan & Peter Friess "Internet of Things Applications From Research and Innovation to Market Deployment", ISBN:987-87-93102-94, River Publishers.
6. Joe Biron and Jonathan Follett, "Foundational Elements of an IoT Solution," by Joe Biron

**List of Tutorials:**

1. Industry 1.0 to 5.0 Evolution
2. IoT Architecture & Components
3. Smart Sensors & Connectivity Features in Industry 5.0
4. Classification of Industrial Sensors
5. Actuators & Microcontroller Interfacing Basics
6. IoT Protocol Comparison
7. Short-Range & Long-Range Communication Technologies
8. MQTT Publish/Subscribe Model
9. Industrial Sensors & Actuators Study
10. Cloud Monitoring Platforms
11. IIoT Threats & Vulnerabilities
12. Privacy & Security Policies in IIoT
13. Predictive Maintenance Techniques
14. Digital Twin & Industry 4.0 Case Studies
15. Sensor & protocol demonstration

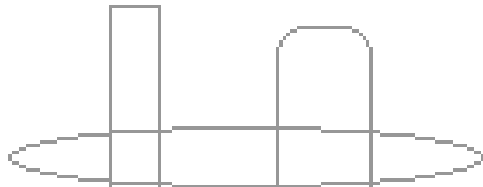
16. Cloud dashboard activity
17. Case study analysis
18. Simple device communication tasks
19. Mini conceptual assignments

**Course Outcomes:** After completion of the course, student will be able to

- 1.Explain the concepts of Industrial Automation and Industry 5.0.
- 2.Describe IoT fundamentals and related architectures used in industrial environments.
- 3.Identify and compare industrial sensors, actuators, and communication modules.
- 4.Apply industrial IoT protocols and cloud platforms for data monitoring.
- 5.Explain automation systems such as SCADA, predictive maintenance, and hardware selection.
- 6.Analyze security, safety, and case studies in Industrial IoT applications.

**CO-PO Mapping:**

|         | Program Outcomes (PO) |      |      |      |      |      |      |      |      |      |      |      | PSO  |      |      |
|---------|-----------------------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|
| CO/PO   | PO1                   | PO2  | PO3  | PO4  | PO5  | PO6  | PO7  | PO8  | PO9  | PO10 | PO11 | PO12 | PSO1 | PSO2 | PSO3 |
| CO1     | 3                     | 2    | –    | –    | –    | 2    | –    | –    | –    | –    | –    | 1    | 2    | 1    | 1    |
| CO2     | 3                     | 2    | –    | –    | 2    | –    | –    | –    | –    | –    | –    | 1    | 2    | 2    | 1    |
| CO3     | 3                     | 2    | 2    | –    | 3    | –    | –    | –    | –    | –    | –    | –    | 3    | 2    | 1    |
| CO4     | 2                     | 3    | 3    | –    | 3    | –    | –    | –    | –    | –    | –    | –    | 3    | 3    | 1    |
| CO5     | 2                     | 2    | 3    | –    | 2    | 1    | –    | –    | –    | –    | –    | –    | 2    | 3    | 2    |
| CO6     | 1                     | 3    | –    | 2    | –    | 3    | 2    | 2    | 2    | 2    | –    | 2    | 1    | 2    | 3    |
| Average | 2.33                  | 2.33 | 2.67 | 2.00 | 2.60 | 2.00 | 2.00 | 2.00 | 2.00 | 2.00 | –    | 1.40 | 2.17 | 2.17 | 1.50 |



**T.Y. B. Tech.**  
**Computer Science & Engineering**  
**(Artificial Intelligence)**  
**AY 2025-26**

**Syllabus Template**

**FF No.: 654**

**CI3001: Computer Network Technology**

**Credits: 4**

**Teaching Scheme Theory: 2 Hours/Week**

**Tutorial: 1 Hours/Week**

**Lab: 2 Hours/Week**

**Course Prerequisites:** Operating System, Database Management Systems

**Course Objectives:**

1. To learn transmission mediums, networking devices and topologies used in the Internet
2. To learn networking standards, IP packet switching and routing used in the Internet
3. To learn transport layer and application layer protocols used in the Internet
4. To learn front end technologies for website development
5. To learn single page applications development using REACT
6. To learn REST API based enterprise website development using REACT, Node JS, Spring Boot with different database technologies

**Course Relevance:**

The key technology of the information age is communications. Computer network is a truly global area of study, both because the technology enables global communication over telephone lines and the Internet. Computer Networks and web technologies are the backbone of all IT infrastructures and their applications in the world. These technologies and applications often emerge in communication within countries of countries and spread rapidly around the world. Most of the jobs available in the IT industries are web technology related.

**Section 1: Topics/Contents**

**Unit-I Networking Fundamentals and Physical Layer**

**04 Hours**

Network Organizations and Architectures: What is computer Networks, Network Topologies: Mesh, Star and Hierarchical, Types of Computer Networks: LAN, MAN, WAN, PAN, Internet, internet and Intranet. Client-Server; Peer To Peer. Network Architecture Modes: Infrastructure and Ad-hoc mode. Reference Models: OS and TCP/IP. Design Issues for Layers.

Physical Layer: Transmission Mediums: Air, Vacuum, Cat5, Cat5e, Cat6, Cat6a, Cat7, Cat8, OFC - Single and Multicore.

Networking Devices Wired and Wireless: NIC, Repeater, Bridge, Switch, Modem, Router, Gateways and Access Point.

### **Unit-II Medium Access Control and Network Layer**

**04 Hours**

Medium Access Control: Legacy Standard : 10 Mbps IEEE 802.3 Standard(Ethernet), High Speed Ethernet Standards: Fast, Gigabit and 10Gigabit.

Wireless Standards: IEEE 802.11a/b/g/n/ac, IEEE 802.15, IEEE 802.15.4 and IEEE 802.16 Standards, CSMA/CA

Switching Techniques and IP Addressing: Circuit, Message and Packet Switching. Logical Addressing: IPv4 and IPv6

Network Layer Protocols: Internet Protocol(IP), Internet Control Message Protocol(ICMP)

### **Unit-III Transport Layer and Application Layer**

**05 Hours**

Transport Layer Protocols: Transmission Control Protocol (TCP), User Datagram Protocol (UDP)

Services: Berkeley Sockets, Connection Establishment, Connection Release

Application Layer: Domain Name System (DNS) and File Transfer Protocol (FTP)

WWW: Hyper Text Transfer Protocol (HTTP1.1/1.2/2.0) and HTTPS with SSL.

Email: SMTP, MIME, POP3 and Webmail.

### **Section2: Topics/Contents**

### **Unit-IV Client-Side Technologies**

**09 Hours HTML5:**

structure of html document, HTML elements: headings, paragraphs, line break, links, frames, lists, tables, images and forms CSS3.0: Styles, colors, fonts and Text Alignments

Java Script: Basics of Document Object Model (DOM), Variable Declarations : Using var, let, and const, Reserved Keywords, Objects and Classes, Understanding Functions: Declarations, Expressions, Arrow Functions, Event Handling- Browser Events and Event Listeners, Form Validation, AJAX

React

Introduction to React, React component, JSX, Render function, Component API, Component lifecycle, State, Props, Mixins, Component composition, Pass data from parent to child, Pass data from child to parent, Component styling, Forms, Events, Refs, Keys, Router, Flux, Redux

### **Unit-V Spring Boot**

**05 Hours**

Spring Framework, Spring Boot Framework, Installing Spring Boot, Build Tool Maven/Gradle/Ant, Core Features, Spring Security, Web Applications, JPA for database connectivity, working with SQL and NoSQL, Messaging, Testing, Deploying Spring Boot Applications, Monitoring and Testing. POSTMAN Tool for API testing.

### **Unit-VI Node JS**

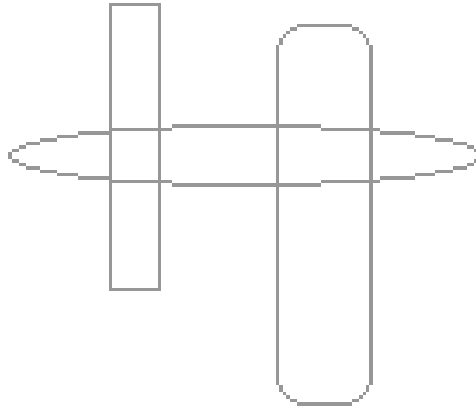
**03 Hours**

Introduction to Node JS, Installation of Node JS, Node JS Modules, Node Package Manager (NPM), Creating Web server, File System, Express JS, Serving Static Resources, Database connectivity.

### **List of Tutorials (13):**

### **List of Tutorials (13):**

1. Use of XML in web development
2. Use of JSON in web development
3. Learning JQuery
4. Learning JQuery
5. Learning Bootstrap
6. Learning Bootstrap
7. Learning PHP
8. Learning PHP
9. Learning MySQL
10. Learning MySQL
11. Learning Mongo DB
12. Learning Mongo DB
13. Learning REST API



### **List of Practical's (Minimum Six):**

### **Unit-I and Unit II: Use two turn of lab**

1a) Setting up small wired computer network :

Set up a small wired network of 2 to 4 computers using Hub/Switch/. It includes Preparation of Cables and setting up wired network.

1b) Setting up small wireless computer network and hands-on networking command:

Set up a small wired network of 2 to 4 computers using access point and ask students to access it on their wireless gadgets.

Hands on for network commands - ping, pathping, ipconfig/ifconfig, arp, netstat, nbtstat, nslookup, route, traceroute/tracert, nmap.



### **Unit-II MAC and Network Layer**

2) Write a program to find the shortest path using Dijkstra Equation for Link State Routing Protocol which is used by Open Shortest Path First Protocol (OSPF) in the Internet for the network flow provided by instructor.

### **Unit-III Transport Layer and Application Layers**

3a) Write the client server programs using TCP Berkeley socket primitives for wired /wireless network for following

- a. to say Hello to Each other
- b. File transfer

3b) Write the client server programs using UDP Berkeley socket primitives for wired /wireless network for following

- a. to say Hello to Each other
- b. Calculator (Trigonometry)

3c) Understanding protocol stack of Intranet

Analyze packet formats of Ethernet, IP, TCP and UDP captured through Wireshark for wired networks.

### **Unit-IV Client-Side Technologies**

4) Design and develop a website using toggleable or dynamic tabs or pills with bootstrap and JQuery to show the relevance of SDP, EDI, DF and Course projects in VIT.

### **Unit-V Springboot**

5) Design and develop a responsive website to prepare one semester result of VIT students using REACT, Springboot and MySQL/ MongoDB/Oracle. Take any four subjects with MSE Marks (30%) ESE Marks (70%).

### **Unit-VI NODE JS**

6) Design and develop a responsive website for an online book store using REACT, Node JS/ PHP and MySQL/ MongoDB/Oracle having 1) Home Page 2) Login Page 3) Catalogue Page: 4) Registration Page: (database)

### **List of Course Project areas: Networking**

- 1. Design and deploy website for TCP based Multithreaded HTTP client server for accessing student activity data in the institute.
- 2. Design and deploy website for TCP based Multithreaded FTP client server to share institute level notices.
- 3. Design and deploy website for UDP based Multithreaded TFTP client server for your class

4. Design and deploy website for TCP based Multithreaded SMTP and POP3 mail client server for your campus.
5. Design and deploy website for TCP based Multithreaded Chat client server for your class.
6. Design and deploy website for UDP based Multithreaded Chat client server for your class.
7. Design and deploy website for UDP based Multithreaded Audio Conferencing client server for computer engineering department.
8. Design and deploy website for UDP based Multithreaded Video Conferencing client server for computer department
9. Design and deploy website to demonstrate implementation of RIP/OSPF/BGP using Packet Tracer
10. Design and deploy website to simulation of AODV routing protocol using Packet Tracer/ NS3/OMNet

**List of Course Project areas: Web Technology**

1. Develop a responsive web application for Student Grievance System
2. Develop a responsive web application for Workflow Management System for MNC
3. Develop a responsive Gaming Website
4. Develop a responsive web application to help farmers to solve their farming problems
5. Develop a responsive web application for GST Billing Software for Small Business
6. Develop a responsive web application for online Crime Reporting System using PHP
7. Develop a responsive web application for online College Voting System
8. Develop a responsive web application for online Loan Processing System for Farmers.
9. Develop a responsive web application for restaurant food order management
10. Develop a responsive web application for e-book shop
11. Develop a responsive web application for on-line music store
12. Develop a responsive web application for guest visiting management to your society
13. Develop a responsive web application for web search engine

**Assessment Scheme:** Suggest an Assessment scheme that is best suited for the course. Ensure 360 degree assessment and check if it covers all aspects of Bloom's Taxonomy.

Assessment scheme covers following aspects of Modified Blooms Taxonomy:

L2 Understanding, L3 Apply, L3 Design, L3 Apply, L4 Analyze and L5 Evaluate

Laboratory Continuous Assessment : 100 Marks converted to 10 Marks

Course Project: End Semester Examination: 100 Marks converted to 20 Marks

Laboratory: End Semester Examination: 100 Marks converted to 50 Marks

Comprehensive Viva Voce: End Semester Examination: 100 Marks converted to 20 Marks

**Text Books (Networking) : (As per IEEE format)**

1. Andrew S. Tanenbaum, "Computer Networks", 5th Edition, PHI, ISBN 81-203-2175-8.
2. Kurose, Ross "Computer Networking a Top Down Approach Featuring the Internet", Pearson; 6th edition (March 5, 2012), ISBN-10: 0132856204

3. Frouzan B., "Data Communications and Networking", 5th edition, Tata McGraw- Hill, Publications, 2006

**Reference Books (Networking) : (As per IEEE format)**

1. Matthew S. Gast "802.11 Wireless Networks", O'Reilly publications; 2nd Edition.
2. C. Siva Ram Murthy and B. S. Manoj, "Ad Hoc Wireless Networks: Architectures and Protocols" Prentice Hall, 2004
3. Holger Karl and Andreas Willing, "Protocols and Architectures for Wireless Sensor Networks", Wiley, ISBN: 0-470-09510-5

**Text Books (Web Technology): (As per IEEE format)**

1. Kumar, A., Web technologies, CRC press, 2019
2. Gupta, R., Internet & Web Technologies, Engineering Handbook, 2019
3. Martin, M.G., Programming for Beginners: 6 Books in 1 – Swift+PHP+Java+Javascript+Html+CSS: Basic Fundamental Guide for Beginners, independently published, 2018
4. Learning PHP, MySQL & JavaScript: With jQuery, CSS & HTML5, O'Reilly Media; 5th edition, 2018
5. Kohli, S., Web Technologies, PPB Publications, 2015
6. Adam Bretz & Colin J Ihrig, "Full Stack Javascript Development with MEAN", SPD, First Edition 2015, Indian Reprint September 2015
7. Giulio Zambon, "Beginning JSP, JSF and Tomcat", Apress Publication, Second Edition, 2013
8. Jeremy McPeak & Paul Wilton, "Beginning JavaScript", Wrox Publication, Fifth Edition, 2015
9. Jeffrey C. Jackson, "Web Technologies: A Computer Science Perspective", Second Edition, Pearson Education, 2007, ISBN 978-0131856035.
10. Robert W. Sebesta: Programming the World Wide Web, 4th Edition, Pearson education, 2008

**Reference Books (Web Technology) : (As per IEEE format)**

1. Marty Hall, Larry Brown, "Core Web Programming", Second Edition, Pearson Education, 2001, ISBN 978-0130897930.
2. H.M. Deitel, P.J. Deitel and A.B. Goldberg, "Internet & World Wide Web How To Program", Third Edition, Pearson Education, 2006, ISBN 978-0131752429.
3. Chris Bates: Web Programming Building Internet Applications, 3rd Edition, Wiley India, 2006.
4. Xue Bai et al: The web Warrior Guide to Web Programming, Thomson, 2003

**MOOCs Links and additional reading material:**

1. [www.w3.org](http://www.w3.org)
2. HTML, The Complete Reference
3. [www.htmlref.com](http://www.htmlref.com)
4. [w3schools.org](http://w3schools.org)
5. [php.net/](http://php.net/) <https://jquery.com/>
6. [developer.mozilla.org/en-US/docs/AJAX](http://developer.mozilla.org/en-US/docs/AJAX)
7. [www.tutorialspoint.com/css/](http://www.tutorialspoint.com/css/)
8. PHP: Data Structures - Manual -----
9. [docs.spring.io/spring-boot/docs/current/reference/html/](http://docs.spring.io/spring-boot/docs/current/reference/html/)
10. [nodejs.org/en](http://nodejs.org/en)
11. [react.dev](http://react.dev)

**MOOCs Links and additional reading material:**

[www.nptelvideos.in](http://www.nptelvideos.in), [www.coursera.com](http://www.coursera.com), [www.udemy.com](http://www.udemy.com)

### Course Outcomes:

The student will be able to –

1. Select topology, essential components of physical layer and networking devices to design computer networks.
2. Build wired and wireless intranet with correct communication and service frameworks.
3. Develop Client-Servers by the means of correct standards, protocols and technologies
4. Build single page applications using REACT as a reusable UI component technology
5. Write Web API/RESTful API application programming interface to communicate with Springboot as a server side technology.
6. [ **Group Assignment**] Design and develop three tier enterprise application using client side, server side and back end technologies

### CO-PO Map:

|         | Program Outcomes (PO) |      |     |      |     |      |     |     |      |      |      |      | PSO  |      |      |
|---------|-----------------------|------|-----|------|-----|------|-----|-----|------|------|------|------|------|------|------|
| CO/PO   | PO1                   | PO2  | PO3 | PO4  | PO5 | PO6  | PO7 | PO8 | PO9  | PO10 | PO11 | PO12 | PSO1 | PSO2 | PSO3 |
| CO1     | 3                     | 3    | 2   | 1    | -   | 2    | -   | -   | 1    | -    | -    | 1    | -    | -    | -    |
| CO2     | 3                     | 2    | 3   | 1    | 3   | 2    | 2   | 2   | 1    | -    | -    | 2    | 3    | 3    | -    |
| CO3     | 3                     | 3    | 3   | 1    | 3   | 2    | -   | 2   | 1    | -    | -    | 2    | 3    | 3    | -    |
| CO4     | 3                     | 3    | 3   | 1    | 3   | 2    | -   | 2   | 1    | -    | -    | 1    | 3    | 3    | -    |
| CO5     | 3                     | 2    | 3   | 1    | 3   | 2    | -   | 2   | 1    | -    | -    | 1    | 3    | 3    | -    |
| CO6     | 3                     | 3    | 3   | 2    | 3   | 3    | -   | 3   | 3    | 2    | 2.0  | 3    | 3    | 3    | 2    |
| Average | 3                     | 2.83 | 3   | 1.16 | 3   | 2.16 | 2   | 2.2 | 1.33 | 2.0  | 2.0  | 1.66 | 3    | 3    | 2    |

Attainment Levels: 2,3,4,3,4, 4

### CO Attainment levels:

Weights for attainment levels: L1 - Easy-0.75 L2 - Comfortable-0.7 L3 – Medium – 0.65  
L4 – Somewhat difficult – 0.6 L5 – Difficult – 0.55

CO1 – L1, CO2 – L3, CO3 – L4, CO4 – L3, CO5 – L4 and CO6 – L4

### Future Course Mapping:

High Speed Networks, Wireless Networks, Mobile Networks, Network Security, Cyber Security, Cloud Computing, Distributed System, Mobile Application Development

### Job Mapping:

Network Engineer, Network Stack Developers, Application Developer, Software Engineer, Web Developer, IT Engineer, UI Developer

**Syllabus Template**

**FF No. : 654**

**CI3002: Design and Analysis of Algorithms**

**Credits: 4.....**

**Teaching Scheme Theory: 2 Hours/Week**

**Tutorial: 1 Hours/Week**

**Lab: 2 Hours/Week**

**Course Prerequisites:** Basic courses on programming, data structures, discrete structures, theory of computing.

**Course Objectives:**

1. Students will gain understanding of asymptotic notations and will be able to apply suitable mathematical techniques to find asymptotic time and space complexities of algorithms.
2. Students will develop the ability to formulate computational problems in the abstract and mathematically precise manner.
3. Student will gain understanding of different algorithm design paradigms such as divide and conquer, dynamic programming, greedy, backtracking and will apply suitable paradigm for designing algorithms for computational problems.
4. Students will develop understanding of notions of NP-hardness and NP-completeness and their relationship with the intractability of decision problems.
5. Students will design randomized, approximation algorithms for some computational problems.
6. Students will be able to incorporate algorithm design principles, data structures and provide efficient solutions for complex computational problems.

**Course Relevance:**

This is a foundational course for Computer science and Engineering. This course develops algorithmic thinking capability of students. Designing algorithms using suitable paradigm and analysing the algorithms for computational problems has a high relevance in all domains where computer science plays a crucial role (equally in Industry as well as research). This course is also an essential prerequisite for advanced domain specific algorithmic courses such as Algorithmic Graph Theory, Algorithmic Number Theory, Computational Geometry, Motion planning and Robotics, etc, to give a few examples. Once the student gains expertise in Algorithm design and in general gains ability of Algorithmic thinking, it facilitates in systematic study of any other domain (in computer science or otherwise) which demands logical thinking. This course is also relevant for students who want to pursue

research career in theory of computing, computational complexity theory, advanced algorithmic research.

### **Section 1: Topics/Contents**

#### **Unit-I Basic introduction and time and space complexity analysis [4 Hours]**

Asymptotic notations (Big Oh, small oh, Big Omega, Theta notations). Best case, average case, and worst-case time and space complexity of algorithms. Overview of searching, sorting algorithms. [Cache optimization](#), Adversary lower bounds (for the comparison-based sorting algorithms). Using Recurrence relations and Mathematical Induction to get asymptotic bounds on time complexity. Master's theorem and applications.

#### **Unit-II Divide and Conquer [4 Hours]**

General strategy, [Application of divide and conquer for solution of some computational problems like](#): Quick sort, Merge sort, Finding a majority element, Order statistics (randomized and deterministic algorithms), Efficient algorithms for Integer arithmetic (Euclid's algorithm, Karatsuba's algorithm for integer multiplication, fast exponentiation).

#### **Unit-III Dynamic Programming [6 Hours]**

General strategy, [Application of dynamic programming for solution of some computational problems like](#): computing Fibonacci numbers, binomial coefficients, Matrix Chain multiplication, Coin change problem, 0-1 Knapsack, Traveling Salesperson Problem, Optimal Binary Search Tree construction, Shortest paths in directed acyclic graphs, All pair shortest path algorithm, Longest increasing subsequence problem, Largest independent set for trees.

### **Section 2: Topics/Contents**

#### **Unit-IV Greedy and Backtracking strategy [4 Hours]**

**Greedy:** General strategy, Analysis and correctness proof of minimum spanning tree and shortest path algorithms, fractional knapsack problem, conflict free scheduling, Huffman coding.

**Backtracking:** General strategy, [backtracking strategy for some problems like](#): n-queen problem, graph coloring, subset sum problem, vertex cover, independent set, solution of SUDOKU puzzle.

(Note: for all the strategies discussed in Units II, III, and IV the student is expected to understand the essence of the strategies and not just the specific applications and should be able to apply suitable strategies for fresh problems)

#### **Unit-V Introduction to complexity classes and NP-completeness [6 Hours]**

Complexity classes P, NP, coNP, and their interrelation, Notion of polynomial time many one reductions reduction, Notion of NP-hardness and NP-completeness, Cook-Levin theorem and implication to P versus NP question, NP-hardness of halting problem. NP-Complete problems (some

selected examples from - Satisfiability problem, Circuit-SAT, 3-CNF SAT, vertex cover problem, independent set problem, clique problem, Hamiltonian-circuit problem, subset sum problem, Integer Linear Programming (ILP)). [Brief introduction to Linear Programming and modeling NP-complete problems using ILP.](#)

## **Unit-VI Introduction to Randomized and Approximation algorithms**

**[4 Hours]**

Introduction to randomness in computation, Las-Vegas and Monte-Carlo algorithms, Abundance of witnesses/solutions and application of randomization, solving SAT for formulas with “many” satisfying assignments, randomized quick sort, Karger’s Min-cut algorithm, coupon collector problem,

Introduction to Approximation algorithms for NP-optimization problems, Approximation algorithm for Vertex Cover, metric Traveling-Sales-Person Problem (metric-TSP), Hardness of approximation for TSP. [Introduction to quantum computation.](#)

### **List of Tutorials (12):**

1. Problem solving based on asymptotic notations, solution of recurrences.
2. Proving correctness of algorithms: some techniques
3. Problem solving based on Divide and Conquer strategy (Binary search interesting applications, counting inversions)
4. Advanced problem solving based on Divide and Conquer strategy (Discrete Ham-Sandwich theorem, efficient algorithm for Josephus problem)
5. Problem solving based on Dynamic Programming strategy (Largest sum contiguous block and generalizations, Optimal binary search tree (OBST) construction)
6. Advanced problem solving based on Dynamic Programming strategy (Winning strategy for two player games, Variants of shortest path algorithms)
7. Problem solving based on Greedy strategy with emphasis on proof of correctness.
8. Problem solving based on Backtracking strategy.
9. reducing NP problems to Integer Linear Programming.
10. Problem solving based on complexity classes, NP-completeness.
11. Problem solving based on Randomized Algorithms
12. Problem solving based on Approximation Algorithms

### **List of Practical’s (Minimum Six):**

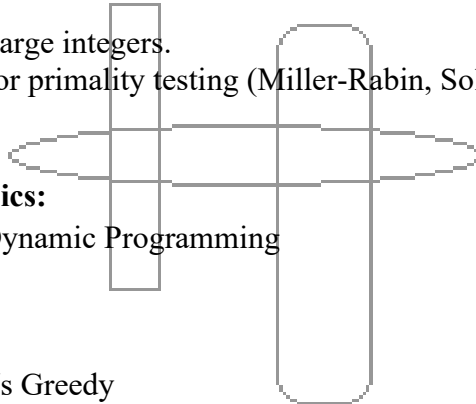
1. Assignment based on some simple coding problems on numbers, graphs, matrices.
2. Assignment based on Divide and Conquer strategy (e.g., majority element search, finding kth rank element in an array)
3. Assignment based on Divide and Conquer strategy (e.g., efficient algorithm for Josephus problem using recurrence relations, fast modular exponentiation)
4. Assignment based on Dynamic Programming strategy (e.g., Matrix chain multiplication, longest increasing subsequence)
5. Assignment based on Dynamic Programming strategy (e.g., All pair shortest path, Traveling Salesperson problem)



6. Assignment based on Greedy strategy (e.g., Huffman encoding)
7. Assignment based on Backtracking (e.g., graph coloring, n-queen problem)
8. Assignment based on analysis of quick sort (deterministic and randomized variant)
9. Assignment based on Las-Vegas and Monte-Carlo algorithm for majority element search.
10. Assignment based on factor-2 approximation algorithm for metric-TSP.

**List of Course Project topics:**

1. Applications of A\* algorithm in gaming
2. Pac-Man game
3. File compression techniques
4. Solution of Maze (comparing the backtracking based solution and Dijkstra's algorithm)
5. Different exact and approximation algorithms for Travelling-Sales-Person Problem
6. Creation of Maze using backtracking
7. Knight tour algorithms
8. Network flow optimization and maximum matching
9. AI for different games such as minesweeper, shooting games, Hex, connect-4, sokoban, etc
10. SUDOKU solver
11. Graph theoretic algorithms
12. Computational Geometry Algorithms
13. AKS primality testing
14. Algorithms for factoring large integers.
15. Randomized algorithms for primality testing (Miller-Rabin, Solovay-Strassen)
16. Slider puzzle game



**List of Course Seminar Topics:**

1. Divide and Conquer Vs Dynamic Programming
2. Greedy strategy
3. NP-hardness
4. Backtracking strategy
5. Dynamic Programming Vs Greedy
6. Computational Complexity
7. Philosophical relevance of P Vs NP question
8. Complexity classes
9. Space complexity
10. Compression Techniques
11. Real world applications of Graph theoretic algorithms
12. Approximation algorithms
13. Hardness of approximation
14. Pseudorandom number generators

**List of Home Assignments Topics:**

*List of Design Based Home Assignments:*

1. Problem solving based on Divide and Conquer strategy
2. Problem solving based on Dynamic Programming strategy



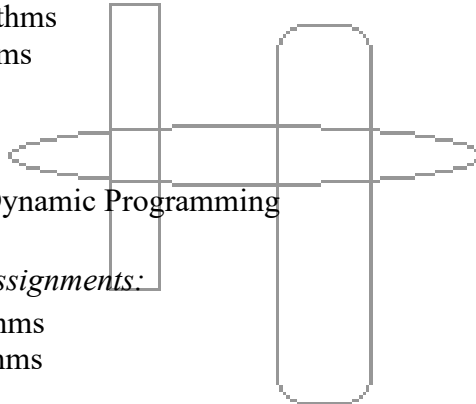
3. Problem solving based on Greedy strategy
4. Problem solving based on Backtracking strategy
5. Problems on Randomized Algorithms
6. Problems on Approximation Algorithms
7. Problems on NP completeness

*List of Case Study Based Home Assignments:*

1. AKS primality test
2. Quadratic sieve factoring algorithm
3. Huffman Encoding, LZW encoding
4. Network flow optimization algorithms
5. Approximation algorithms for TSP
6. Cook-Levin theorem and its relationship with intractability of computational problems
7. Sorting techniques

*List of Blog Based Home Assignment:*

1. Approximation Algorithms
2. Randomized Algorithms
3. Computational Geometry Algorithms
4. Number Theoretic Algorithms
5. Graph Theoretic Algorithms
6. P Vs NP Problem
7. Complexity classes
8. Greedy Algorithms
9. Divide and Conquer Vs Dynamic Programming



*List of Survey Based Home Assignments:*

1. Primality Testing Algorithms
2. Integer Factoring Algorithms
3. NP-complete problems
4. Compression Techniques
5. Shortest Path Algorithms
6. Algorithms for finding Minimum Weight Spanning Tree

**Assessment Scheme:** Suggest an Assessment scheme that is best suited for the course. Ensure 360 degree assessment and check if it covers all aspects of Bloom's Taxonomy.

**Laboratory Continuous Assessment:** 100 Marks converted to 10 Marks

**Course Project:** End Semester Examination: 100 Marks converted to 20 Marks

**Presentation:** End Semester Examination: 100 Marks converted to 20 Marks

**Theory:** End Semester Examination (Written): 60 Marks converted to 30 Marks

**Comprehensive Viva Voce:** End Semester Examination: 100 Marks converted to 20 Marks

**Text Books:** (As per IEEE format)

1. Cormen, Leiserson, Rivest and Stein "Introduction to Algorithms", PHI 3rd edition, 2009. ISBN 81-203-2141-
2. Jon Kleinberg, Eva Tardos "Algorithm Design", Pearson, 1st edition, 2005. ISBN 978-81-317-0310-6  
Dasgupta, Papadimitriou, Vazirani "Algorithms" McGraw-Hill Education; 1 edition (September 13, 2006), ISBN-10: 9780073523408, ISBN-13: 978-0073523408

**Reference Books:** (As per IEEE format)

1. Motwani, Raghavan "Randomized Algorithms", Cambridge University Press; 1 edition (August 25, 1995), ISBN-10: 0521474655, ISBN-13: 978-0521474658
2. Vazirani, "Approximation Algorithms", Springer (December 8, 2010), ISBN-10: 3642084699, ISBN-13: 978-3642084690 Gerd Keiser, MC Graw Hill International edition, optical fiber communication, third edition

**MOOCs Links and additional reading material:**

1. [www.nptelvideos.in](http://www.nptelvideos.in)

**Course Outcomes:**

The student will be able to –

1. To formulate computational problems in abstract and mathematically precise manner
2. To design efficient algorithms for computational problems using appropriate algorithmic paradigm
3. To analyze asymptotic complexity of the algorithm for a complex computational problem using suitable mathematical techniques.
4. To establish NP-completeness of some decision problems, grasp the significance of the notion of NP-completeness and its relationship with intractability of the decision problems.
5. To understand significance of randomness, approximability in computation and design randomized algorithms for simple computational problems and design efficient approximation algorithms for standard NP-optimization problems.
6. To incorporate appropriate data structures, algorithmic paradigms to craft innovative scientific solutions for complex computing problems.

**CO-PO Map:**

|         | Program Outcomes (PO) |     |      |     |     |     |     |     |     |      |      |      | PSO  |      |      |
|---------|-----------------------|-----|------|-----|-----|-----|-----|-----|-----|------|------|------|------|------|------|
| CO/PO   | PO1                   | PO2 | PO3  | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 | PSO1 | PSO2 | PSO3 |
| CO1     | 2                     | 3   | 2    | 1   |     |     | 2   |     |     |      |      | 2    | 3    |      | 2    |
| CO2     | 2                     | 3   | 3    | 1   |     |     | 2   |     |     |      |      | 2    | 3    |      | 2    |
| CO3     | 2                     | 3   | 2    | 1   |     |     | 2   |     |     |      |      | 2    | 3    |      | 2    |
| CO4     | 2                     | 3   | 3    | 1   |     |     | 2   |     |     |      |      | 2    | 3    |      | 2    |
| CO5     | 2                     | 3   | 3    | 1   |     |     | 2   |     |     |      |      | 2    | 3    |      | 2    |
| CO6     | 2                     | 3   | 3    | 1   |     |     | 2   |     |     |      |      | 2    | 3    |      | 2    |
| Average | 2                     | 3   | 2.66 | 1.0 |     |     | 2.0 |     |     |      |      | 2.0  | 3    |      | 2    |

**CO Attainment levels:**

Weights for attainment levels: L1 - Easy-0.75 L2 - Comfortable-0.7 L3 – Medium – 0.65  
L4 – Somewhat difficult – 0.6 L5 – Difficult – 0.55

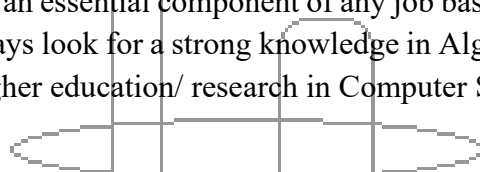
CO1 – L1, CO2 – L3, CO3 – L2, CO4 – L3, CO5 – L4 and CO6 – L5

### **Future Course Mapping:**

Advanced Algorithms, Computational Complexity, Computational Geometry, Algorithmic Number Theory, Algorithmic Graph Theory

### **Job Mapping:**

Algorithm design lie at heart of any Computer Science/Engineering application. Once the student gains expertise in Algorithm design and in general gains ability of Algorithmic thinking, it facilitates in systematic studying any other domain (in computer science or otherwise) which demands logical thinking. Algorithm design is an essential component of any job based on programming. All Industries in computer Engineering always look for a strong knowledge in Algorithm design and Data structures. If student wants to pursue higher education/ research in Computer Science, this course is must.



**Syllabus Template**

**FF No.: 654**

**CI3003: Artificial Neural Networks**

**Credits: 4**

**Teaching Scheme Theory: 2 Hours/Week**

**Tutorial: 1 Hours/Week**

**Lab: 2 Hours/Week**

**Course Prerequisites:** Mathematics for AI and Artificial Intelligence

### **Learning Objectives:**

1. To learn how to design AI based learning system and their evaluation methods
2. To learn ANN learning types and ANN training process
3. To learn Single Layer Perceptron Classifiers
4. To learn Multilayer Feedforward Networks
5. To learn Competitive learning Neural Network
6. To learn Optimization of Neural Networks

**Course Relevance:**

Artificial Neural Networks are at the forefront of the AI revolution, with applications spanning across multiple domains, driving technological advancements, and creating significant career opportunities. Understanding ANNs is crucial for anyone looking to engage with modern AI technologies, contribute to innovative research, or develop cutting-edge solutions in various fields.

**Section 1: Topics/Contents**

**Unit-I: Learning System and evaluation parameters [4 Hours]**

Well posed learning problem, Designing a learning system, Issues in AI learning. Hypothesis Space, Hypothesis functions, Hypothesis Evaluation, Bias, Variance, Underfitting, Overfitting, Inductive bias, Evaluation, Training, Testing, Cross-validation. Error Analysis, Error Metrics, Precision and recall.

**Unit II: Introduction to Artificial Neural Networks (ANNs) [5 Hours]**

Neural Computation, History of Artificial Neural Systems Development, Biological Neurons and Their Artificial Models, Models of Artificial Neural Networks, Neural Network Learning Rules, Perceptron Learning Algorithms. Supervised learning, unsupervised learning, and reinforcement learning, Training Process: Overview of the neural network training process and the role of training data., Initialization of Weights: Importance of initializing weights and common strategies (e.g., random, Xavier, He initialization). Hebbian Learning, Competitive, Boltzmann Learning,

**Unit III: Single Layer Perceptron Classifiers [5 Hours]**

Classification Model, Features, and Decision Regions, Discriminant Functions, Linear Machine and Minimum Distance Classification, Nonparametric Training Concept, Training and Classification Using the Discrete Perceptron: Algorithm and Example, Single-Layer Continuous Perceptron Networks for Linearly Separable Classifications, Multicategory Single-Layer Perceptron Network

**Section2: Topics/Contents**

**Unit IV: Multilayer Feedforward Networks [5 Hours]**

Linearly Nonseparable Pattern Classification, Delta Learning Rule for Multi-Perceptron Layer, Generalized Delta Learning Rule, Feedforward Recall and Error Back-Propagation Training, Learning Factors, Classifying and Expert Layered Network, Multilayer Networks, Backpropagation algorithm, case study to implement MLP.

**UNIT V Competitive learning Neural Network [5 Hours]**

Components of CL network, Pattern clustering and feature mapping network, ART networks, Features of ART models, character recognition using ART network. Self-Organization Maps, (SOM): Two Basic Feature Mapping Models, Self-Organization Map, SOM Algorithm, Properties of Feature Map,

Computer Simulations, Learning Vector Quantization, Adaptive Pattern Classification. Application and analysis of ART1 & ART2, Case study

**Unit VI: Optimization of Neural Networks**

**[4 Hours]**

Data Preparation and Preprocessing, Weight Initialization Techniques, Loss Functions (Mean Squared Error (MSE) and Cross-Entropy Loss.) and Backpropagation, Learning Rate strategies and its Importance, Gradient Descent Variants (SGD, Adam, RMSprop), Overfitting and Regularization Techniques (L1, L2, Dropout), Hyper-parameter Tuning, Handling Vanishing and Exploding Gradients

**List of Tutorials (12):**

- 1.Examples on Bias and Variance
- 2.Examples on Overfitting and Inductive bias
- 3.Examples on Precision and recall
- 4.Examples on ANN learning
- 5.Examples on ANN learning
- 6.Examples on Single Layer Perceptron Classifiers
- 7.Examples on Single Layer Perceptron Classifiers
- 8.Examples on Multilayer Layer Perceptron Classifiers
- 9.Examples on Multilayer Layer Perceptron Classifiers
- 10.Examples on Competitive learning Neural Network
- 11.Examples on Competitive learning Neural Network
- 12.Examples on ANN optimization

**List of Practical's (Minimum Six):**

1. Write a program to visualize popular activation functions used in neural networks, aiding in understanding their behaviour and suitability for different tasks.
2. Write a program to use a Perceptron neural network to recognize even and odd numbers represented in ASCII form, illustrating the basic principles of a single-layer neural network.
3. Write a program to demonstrate the perceptron learning law with decision regions, visually showcasing how a perceptron learns and separates different classes.
4. Write a program to implement the training process of an Artificial Neural Network, covering forward propagation and backpropagation, fundamental steps in neural network training.
5. Write a program to create a neural network architecture from scratch, focusing on multi-class classification with customizable parameters such as hidden layers, neurons, non-linearity, and optimization algorithm.
6. Write a program to illustrate an ART (Adaptive Resonance Theory) neural network, showcasing its self-organizing and adaptive capabilities.
7. Write a program to implement of AND/NAND gate using feed forward neural network.
8. Write a program to implement of OR/ NOR gate using feed forward neural network.
9. Write a program to implement of Ex-OR gate using feed forward neural network.

**List of Course Project areas:**

1. Security: Use of ANN in Cryptographic Applications.
2. Banking: Use of ANN for Credit Scoring System
3. Music: Use of ANN for Automatic Music Generation
4. Autonomous Vehicle: Use of ANN for Vision and Control in Autonomous Flying Vehicle.
5. GPS: USE of ANN in Global Positioning System.
6. Recognition: Use of ANN for Handwritten Recognition.
7. Stock Market: Use of ANN for stock Market prediction.
8. Training System: Use of ANN for Web based Training Systems
9. Vehicle Security: Use of ANN to build Vehicle Security System

**List of Topics for PPT/Seminar:**

1. ANN for Cryptographic Applications.
2. ANN for Credit Scoring System
3. ANN for Automatic Music Generation
4. ANN for Vision and Control in Autonomous Flying Vehicle.
5. ANN in Global Positioning System.
6. ANN for Handwritten Recognition.
7. ANN for stock Market prediction.
8. ANN for Web based Training Systems
9. ANN to build Vehicle Security System

**Assessment Scheme:** Suggest an Assessment scheme that is best suited for the course. Ensure 360 degree assessment and check if it covers all aspects of Bloom's Taxonomy.

Assessment scheme covers following aspects of Modified Blooms Taxonomy:

L2 Understanding, L3 Apply, L3 Design, L3 Apply, L4 Analyze and L5 Evaluate

**Laboratory Continuous Assessment:** 100 Marks converted to 10 Marks

**Course Project:** End Semester Examination: 100 Marks converted to 20 Marks

**Seminar/Presentation:** End Semester Examination: 100 Marks converted to 20 Marks

**Comprehensive Viva Voce:** End Semester Examination: 100 Marks converted to 20 Marks

**Theory:** End Semester Examination (Written): 60 Marks converted to 30 Marks

**Text Books (Networking) :** (As per IEEE format)

1. *Introduction to Artificial Neural Systems*, JACEK M. ZURADA, Jaico Publishing House; First Edition, 1994
2. *Neural Networks, Fuzzy Logic and Genetic Algorithms: Synthesis & Applications*, S.Rajasekaran, G. A. Vijayalakshami, PHI, 2007
3. *Neural Networks and Learning Machines* by Simon S. Haykin, Pearson; 3rd edition 12 March 2009

**Reference Books (Networking) :** (As per IEEE format)

1. *Hands-On Machine Learning with Scikit-Learn and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems* by GeronAurelien
2. *Foundations of Neural Networks, Fuzzy Systems, and Knowledge Engineering*, Nikola K. Kasabov, MIT Press, 1998
3. <https://towardsdatascience.com/convolutional-neural-networks-explained-9cc5188c4939>

### **MOOCs Links and additional reading material:**

1. NPTEL Course “Neural Networks and Application”, Prof. SomnathSengupta, IIT Kharagpur Link of the Course:<https://nptel.ac.in/courses/117105084>

### **Course Outcomes:**

The student will be able to –

1. Decide on learning system and their evaluation parameters
2. Select appropriate ANN learning type and training process
3. Propose suitable Single Layer Perceptron Classifier
4. Propose suitable Multilayer Feedforward Networks
5. Apply Competitive learning Neural Network
6. Design, Implement, Evaluate and Optimize ANN based applications

### **CO-PO Map:**

|         | Program Outcomes (PO) |      |     |     |     |      |     |     |      |      |      |      | PSO  |      |      |
|---------|-----------------------|------|-----|-----|-----|------|-----|-----|------|------|------|------|------|------|------|
| CO/PO   | PO1                   | PO2  | PO3 | PO4 | PO5 | PO6  | PO7 | PO8 | PO9  | PO10 | PO11 | PO12 | PSO1 | PSO2 | PSO3 |
| CO1     | 3                     | 2    | 2   | 2   | -   | 2    | -   | -   | -    | -    | -    | 1    | 2    | 1    |      |
| CO2     | 3                     | 3    | 3   | 3   | 3   | 2    | 2   | 2   | 1    | -    | -    | 2    | 3    | 2    |      |
| CO3     | 3                     | 3    | 2   | 2   | 3   | 2    | -   | 2   | -    | -    | -    | 2    | 3    | 2    |      |
| CO4     | 3                     | 3    | 2   | 2   | 3   | 2    | -   | 2   | 1    | -    | -    | 2    | 3    | 2    |      |
| CO5     | 3                     | 3    | 3   | 3   | 3   | 2    | -   | 2   | 1    | -    | -    | 3    | 3    | 3    |      |
| CO6     | 3                     | 3    | 3   | 3   | 3   | 3    | -   | 3   | 3    | 2    | 2.0  | 3    | 3    | 3    |      |
| Average | 3                     | 2.83 | 2.5 | 2.5 | 3   | 2.16 | 2   | 2.2 | 1.33 | 2.0  | 2.0  | 2.16 | 2.83 | 2.16 |      |

Attainment Levels: 2,3,4,4,4,5

### **CO Attainment levels:**

Weights for attainment levels: L1 - Easy-0.75 L2 - Comfortatble-0.7 L3 – Medium – 0.65  
L4 – Somewhat difficult – 0.6 L5 – Difficult – 0.55

CO1 – L2, CO2 – L3, CO3 – L4, CO4 – L4, CO5 – L4 and CO6 – L5

### **Future Course Mapping:**

Advanced course on Deep learning including Auto encoders and Boltzmann machines, Reinforcement Learning etc

**Job Mapping:**

AI Scientist, AI System Designer, AI Developer, AI Data Analyst

**Syllabus Template**

**FF No. : 654**

**CI3004: Cloud Computing**

**Credits: 4.....**

**Teaching Scheme Theory: 2 Hours/Week**

**Tutorial: 1 Hours/Week**

**Lab: 2 Hours/Week**

**Course Prerequisites:** Operating Systems, Computer Networks, Database Management System

**Course Objectives:**

1. To become familiar with cloud computing and its ecosystem
2. To acquire basics of virtualization and its importance
3. To evaluate in-depth analysis of Cloud Computing capabilities and its services.
4. To configure and implement storage services.
5. To analyze different cloud-based services to meet a set of given requirements.
6. To design security aspects for cloud computing

**Course Relevance:** Cloud computing to enable transformation, business development and agility in an organization.

**SECTION-I Topics and Contents:**

**Unit-I Introduction to Cloud Computing**

**[4 Hrs]**

Recent trends in computing, Cluster computing, Distributed computing, Evolution of cloud computing, Cloud versus traditional architecture, Cloud Computing Architecture, Google Cloud architecture, Infrastructure as a Service (IaaS), Platform as a Service (PaaS), Software as a Service (SaaS), Public cloud, Private cloud, Hybrid cloud, Community cloud

**Unit-II Virtualization**

**[6 Hrs]**



Introduction to virtualization, Different approaches to virtualization, Hypervisors, Machine Image, Virtual Machine (VM), Compute options in the cloud, Exploring IaaS with Compute Engine, Configuring elastic apps with auto scaling, Basics of virtualization and implementation challenges. System virtualization technologies-architectures and internals. KVM, Xen, VMware. [Amazon Elastic Compute Cloud EC2 as computing service.](#)

Memory virtualization-virtualization techniques, ballooning, deduplication and sharing. Network and storage virtualization, Virtual machine migration and replication techniques pre-copy and post-copy techniques, applicability to system availability.

### **Unit-III Cloud Services**

**[4 Hrs]**

Service Oriented Architecture (SOA), Web services, Web 2.0, Web OS. Introduction to IaaS, PaaS, SaaS. Cloud Platform and Management, Exploring PaaS with App Engine, Event driven programs with Cloud Functions, Containerizing and orchestrating apps with Google Kubernetes Engine Software as a Service (SaaS) Docker flow, orchestration with Docker, dynamic linking and legacy linking of containers. The GCP Console, understanding projects, Billing in GCP, Install and configure Cloud SDK, Use Cloud Shell, GCP APIs.

### **SECTION-II Topics and Contents:**

### **Unit-IV Cloud Storage**

**[4 Hrs]**

Storage options in the cloud, ~~Structured and unstructured storage~~ in the cloud, unstructured storage using Cloud Storage, SQL managed services, Exploring Cloud SQL, Cloud Spanner as a managed service, NoSQL managed service options, Cloud Datastore, a NoSQL document store, Cloud Bigtable as a NoSQL option. OpenStack: NOVA, Neutron, Keystone Cinder, Swift and Glances, VMware Suit, Apache Cloud Stack, [Data Lakes](#), [Snowflake](#).

### **Unit-V Service Management**

**[4 Hrs]**

Service Level Agreements (SLAs), Billing and accounting, Billing in GCP Cloud Security: Introduction to security in the cloud, the shared security model, Encryption options, Authentication and authorization with Cloud IAM, Identify Best Practices for Authorization using Cloud IAM., Introduction to configuration and management tools Ansible, Architecture of DevOps.

### **Unit-VI Cloud Network and Security**

**[6 Hrs]**

Introduction to networking in the cloud, defining a Virtual Private Cloud, Public and private IP address basics, Google's network architecture, Routes and firewall rules in the cloud, Multiple VPC networks, building hybrid clouds using VPNs, interconnecting, and direct peering, Different options for load balancing. Introduction to security in the cloud, the shared security model, Encryption options

### **List of Tutorials (Any Thirteen)**

**List of Tutorials:**

**Unit-I Introduction to Cloud Computing**

- 1) Install VirtualBox/VMware Workstation with different Linux or Windows Operating Systems.
- 2) Study Google Cloud Architecture.

**Unit-II Virtualization**

- 3) Find a procedure to launch virtual machine
- 4) Find a procedure to transfer the files from one virtual machine to another virtual machine.

**Unit-III Cloud Services**

- 5) Simulate a cloud scenario using CloudSim and run a scheduling algorithm that is not present in CloudSim.
- 6) Install Google App Engine. Create hello world app and other simple web applications using python/java.

**Unit-IV Cloud Storage**

- 7) Launch the Web Applications using GAE launcher.
- 8) Install Hadoop single node cluster and run simple applications like wordcount.

**Unit-V Service Management**

- 9) Use AWS Pricing Calculator: Create estimate for EC2 Compute cost for VM instance. Use region closest to you. Find On demand cost and compare the pricing for other regions.

**Unit-VI Cloud Network and Security**

- 9) Launch EC2 instance and explore Public/Private/Elastic IP

**Practical's:**

**List of Practical's (Any Six)**

**Unit-I Introduction to Cloud Computing**

- 1) To setup AWS accounts and launch instances.

**Unit-II Virtualization**

- 2) To install an OS using VirtualBox/ VMWare Workstation. Add Storage to create new virtual disk.
- 3) To Deploy Virtual Machine on hypervisor such as KVM, ESXi. Take Backup and Migrate them.

**Unit-III Cloud Services**

- 4) To use Infrastructure as a Service to facilitates for creating and deleting compute resources. Create network and attach volumes to run instances.
- 5) To install docker on window/linux and build docker image from docker hub.
- 6) Deploy a stateless/stateful application on Kubernetes cluster.

**Unit-IV Cloud Storage**

- 7) To work on different Cloud Storage Services.

**Unit-V Service Management**

- 8) To create login into AWS and use S3 Bucket Service for storage.

## **Unit-VI Cloud Network and Security**

9) Develop elastic services for dynamic load scenario using AWS APIs. Build load balancer and explore on scalability, fault detection and performance.

### **Course Projects:**

#### **List of Course Project Topics**

1. Creating Google Account to store files and programs.
2. Creating Account to Store Images.
3. Creating a Warehouse Application in SalesForce.com
4. Creating an Application in SalesForce.com using Apex programming Language.
5. To study and implement Web services in SOAP for JAVA Applications.
6. Implementation of Para-Virtualization using VMWare 's Workstation/ Oracle's Virtual Box and Guest Operator System.
7. Installation and Configuration of Hadoop.
8. AWS Case Study: Amazon.com.
9. Case Study of Google App Engine.
10. Case Study of Face book.

### **Seminars:**

#### **List of Course Seminar Topics**

1. Storage Cost Optimization on Cloud.
2. Cloud Security and Cryptography
3. Infrastructure As A Code (IAC)
4. Cloud Computing in Healthcare
5. Serverless
6. Deployment of Microservices in Kubernetes Engine
7. RPA Using AWS Cloud
8. Cloud Trends In Supporting Ubiquitous Computing
9. Mobile Cloud Computing
10. Modern Data Center Architecture

### **Group Discussion:**

#### **List of Group Discussion Topics**

1. Data Storage Security in Cloud
2. Cloud Services for SMB's.
3. Monitoring Services Provided by GCP and AWS.
4. Docker and Kubernetes.
5. SaaS vs FaaS (Function as a service).
6. Hybrid Cloud.

7. GCP Vs AWS Web Service Architecture.
8. Cloud based security issues and threats.
9. Authentication and identity.
10. Future of Cloud-Based Smart Devices.

**List of Home Assignments:**

**List of Design Based Home Assignments**

1. Serverless Web App to order taxi rides using AWS lambda.
2. Deploying App on Kubernetes.
3. Serverless web Application (GCP Cloud Functions).
4. Demonstration of EBS, Snapshot, Volumes.
5. Single Node Cluster Implementation (Hadoop).

**List of Case Study Based Home Assignments**

1. PayU Migration to AWS.
2. Cloud object storage.
3. Deployment and Configuration options in AWS.
4. Deployment and Configuration options in Microsoft Azure.
5. Deployment and Configuration options in GCP.

**List of Blog Based Home Assignment**

1. Comparing design of various cloud computing platforms.
2. AWS EKS and Google Cloud Functions.
3. App Engine.
4. Cloud Endpoints.
5. Cloud Pub/Sub.

**List of Survey Based Home Assignments**

1. Disaster Recovery in Cloud Computing.
2. Cloud Economics.
3. Data archiving solutions.
4. Salesforce.
5. Dropbox.

**Assessment Scheme:** Suggest an Assessment scheme that is best suited for the course. Ensure 360 degree assessment and check if it covers all aspects of Bloom's Taxonomy.

**Laboratory Continuous Assessment:** 100 Marks converted to 10 Marks

**Course Project:** End Semester Examination: 100 Marks converted to 20 Marks

**Presentation:** End Semester Examination: 100 Marks converted to 20 Marks

**Theory:** End Semester Examination (MCQ): 60 Marks converted to 30 Marks

**Comprehensive Viva Voce: End Semester Examination: 100 Marks converted to 20 Marks**

**Text Books: (As per IEEE format)**

1. Judith Hurwitz, R.Bloor, M.Kanfman, F.Halper, "Cloud Computing for Dummies", Wiley, India.
2. Ronald Krutz and Russell Dean Vines, "Cloud Security", Wiley-India
3. Gautam Shroff. "Enterprise Cloud Computing", Cambridge

**Reference Books: (As per IEEE format)**

1. Barrie Sosinsky, "Cloud Computing Bible", Wiley India
2. Anthoy T Velte, et.al, "Cloud Computing : A Practical Approach", McGraw Hill.
3. Michael Miller, "Cloud Computing", Que Publishing.
4. Tim Malhar, S.Kumaraswamy, S.Latif, "Cloud Security & Privacy", SPD, O'REILLY
5. Scott Granneman, "Google Apps", Pearson

**MOOCs Links and additional reading material:**

<https://nptel.ac.in/courses/106/105/106105167/>  
[https://swayam.gov.in/nd1\\_noc20\\_cs55/preview](https://swayam.gov.in/nd1_noc20_cs55/preview)  
<https://www.coursera.org/specializations/cloud-computing>  
<https://azure.microsoft.com/en-in/overview/what-is-cloud-computing/>  
<https://aws.amazon.com/what-is-cloud-computing/>  
<https://www.ibm.com/in-en/cloud/learn/cloud-computing>

**Course Outcomes:**

**On the completion of course, student will able to**

1. Describe the main concepts, key technologies, strengths, and limitations of cloud computing and the possible applications for state-of-the-art cloud computing
2. Explain the architecture and infrastructure of cloud computing, including SaaS, PaaS, IaaS, public cloud, private cloud, hybrid cloud, etc.
3. Identify problems, and explain, analyze, and evaluate various cloud computing solutions.
4. Choose the appropriate technologies, algorithms, and approaches for the related issues.
5. Display new ideas and innovations in cloud computing.
6. Collaboratively research and write a paper on the state of the art (and open problems) in cloud computing.

**CO-PO Map:**

| CO/PO | Program Outcomes (PO) |     |     |     |     |     |     |     |     |      |      |      | PSO  |      |      |
|-------|-----------------------|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|------|------|------|
|       | PO1                   | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 | PSO1 | PSO2 | PSO3 |

|         |      |      |      |      |      |      |      |      |      |      |      |      |     |     |      |
|---------|------|------|------|------|------|------|------|------|------|------|------|------|-----|-----|------|
| CO1     | 2    | 1    |      |      | 1    |      |      |      |      |      |      |      | 2   | 2   | 3    |
| CO2     | 2    | 2    | 1    | 1    | 1    |      |      |      |      |      |      |      | 2   | 2   | 3    |
| CO3     | 3    | 2    | 2    | 2    | 2    |      | 3    | 3    |      |      |      |      | 2   | 2   | 3    |
| CO4     | 3    | 2    | 2    | 2    | 3    | 3    |      |      | 3    |      |      |      | 2   | 2   | 3    |
| CO5     | 3    | 3    | 1    | 3    | 3    |      |      |      | 1    |      | 2    |      | 2   | 2   | 3    |
| CO6     | 2    | 2    | 1    | 3    | 1    |      |      |      |      | 3    |      | 3    | 2   | 2   | 2    |
| Average | 2.50 | 2.00 | 1.40 | 2.20 | 1.83 | 3.00 | 3.00 | 3.00 | 2.00 | 3.00 | 2.00 | 3.00 | 2.0 | 2.0 | 2.83 |

**CO attainment levels:**

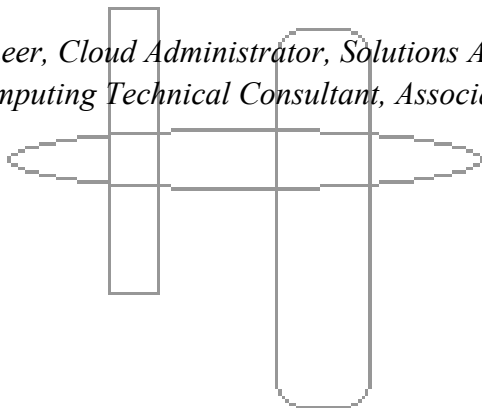
**Attainment Levels:1,2,3,5,4,3**

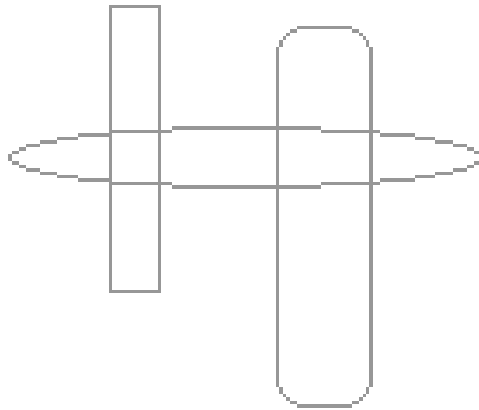
**Future Course Mapping:**

*After completing this course different certifications courses in cloud be taken such as AWS, Azure, Google cloud certifications. One can go for higher studies in specialization of cloud computing and allied subjects*

**Job Mapping:**

*Cloud Architect, Cloud Engineer, Cloud Administrator, Solutions Architect - Cloud Computing - AWS / Kubernetes, Cloud Computing Technical Consultant, Associate Cloud Computing Engineer, Cloud Computing Trainer*





**T. Y. B. Tech. Computer Science &  
Engineering (Artificial Intelligence)**  
**AY 2025-26**

**Module VI Course Content**

**Syllabus Template**

**FF No. : 654**

**CI3007: Software Engineering**

**Credits: 4**

**Teaching Scheme Theory: 2 Hours/Week**

**Tutorial: 1 Hours/Week**

**Lab: 2 Hours/Week**

**Course Prerequisites:**

**Course Objectives:**

1. To summarize capabilities and impact of software development process models and justify process maturity through application of Software Engineering principles and practices,
2. To differentiate feasible and competing system requirements, indicating correct real world problem scope and preparing stepwise system conceptual model,
3. To formulate system specifications by analyzing user-level tasks and compose software artifacts using agile principles, practices and scrum framework,
4. To compose system analysis and design specifications using UML diagrams,
5. To design a system architecture and map it with a suitable architectural style,
6. To comprehend the nature of design patterns and apply these patterns in system design.

**Course Relevance:**

Given that Software Engineering is built upon the foundations of Computer Science as well as Computer Engineering, a Software Engineering curriculum can be focused on two perspectives - a Computer Science-first or Software Engineering-first perspective. Software engineering spans the entire software lifecycle. It involves creating high-quality and reliable programs in a systematic, controlled, and efficient manner using formal methods for specification, analysis, design and evaluation of proposed systems. It requires suitable software development techniques and processes that successfully scale to large applications, which should satisfy timing, size, and security requirements all within acceptable application/project budgets and deadlines. For these reasons,



Software Engineering requires both the analytical and descriptive tools and techniques developed in Computer Science and the rigor that the Computer Engineering discipline brings to the reliability and trustworthiness of the systems that software developers design and implement, while working as a cohesive team.

### **Section 1: Topics / Contents**

#### **Unit-I Software Engineering Paradigms:**

**05 Hours**

Process Models: Code-and-Fix Model, Waterfall Model, Rapid Application Development Model, Incremental Model, Evolutionary Model and Others.

#### **Unit-II Requirements Engineering:**

**05 Hours**

Requirements Engineering Tasks, Requirement Elicitation Techniques, Functional, Non- Functional and Domain Requirements, Requirements Characteristics, Eliminating Requirement Ambiguities, Conflict Identification and Resolution, Requirement Qualities, Requirement Specification, System Scope Determination and Feasibility Study.

#### **Unit-III Agile Methodology:**

**04 Hours**

Landscape of Agile and Planned Methods, Definition - Scrum, Scrum Origins, Scrum Framework, Agile Principles, Sprints, Requirements, User Stories, Product Backlog, Roles: Product Owner, Scrum Master, Development Team, Managers, Scrum Team Structures, Scrum Planning.

### **Section 2: Topics/Contents**

#### **Unit-IV Static and Dynamic Interaction Modeling:**

**05 Hours**

Static Behavior: Use Case, Use Case Diagram, Class Diagram, Component Diagram, Deployment Diagram, Dynamic Behavior: Sequence Diagram, Collaboration Diagram, Activity Diagram, Communication Diagram, Interaction Diagrams.

#### **Unit-V Software Architecture Design:**

**05 Hours**

Design Model, Design Qualities, Characteristics of Design Activities, Design Principles, Cohesion and Coupling, Software Architecture Vs Software Design, Software Reuse, Design Heuristics, Layered Architecture, Client-Server Architecture, Pipe-Filter Architecture, Model-View Controller Architecture.

#### **Unit-VI Design Patterns:**

**04 Hours**

Definition, Describing Design Pattern,  
Creational Patterns: Abstract Factory, Builder, Factory Method, Prototype  
Structural Patterns: Adapter, Bridge, Composite, Decorator, Façade,  
Behavioral Patterns: Chain of Responsibility, Command, Interpreter.

### **List of Tutorials:**

1. Requirement Engineering,
2. System Requirement Specification,
3. Scrum Artifacts,
4. User Stories and Use Cases,
5. Product Backlog Development,
6. Burn-up and Burn-down Chart Development and Management,
7. Software System Analysis and Design: UML Static Diagram,
8. Software System Analysis and Design: UML Dynamic Diagram,
9. Software Architecture Design,
10. Use of Design Patterns,
11. Software Testing,
12. Automated Testing,
13. Project Management Techniques.

**List of Practicals (Minimum SIX):**

1. To prepare a Statement Of Work (SOW) document, which addresses the vision, goals and objectives of the real-world problem.
2. To prepare a Software Requirement Specification (SRS) document, based on several types of system requirements, such as functional and non-functional requirements.
3. To document a product backlog for the project aimed at maintaining a prioritized queue of project requirements.
4. To develop a Sprint-plan and Sprint-design indicating detailed activity planner accommodating user story points.
5. To prepare Class Collaboration-Responsibility (CRC) cards for the Conceptual classes traced from the system analysis phase.
6. To develop a static structure of the target system with a Class Diagram using all components of it.
7. To decompose and organize the problem domain area into broad subject areas and identify the use cases to show them in a Use Case Diagram.
8. To depict the dynamic behavior of the target system using Sequence Diagram. The Sequence diagram should be based on the scenarios generated by the inter-object Communication.
9. To depict the dynamic behavior using a detailed Activity Diagram.
10. To prepare an Architecture Diagram with appropriate design patterns. Suitable Architectural Styles shall be selected and the structural elements shall be well-documented.

**List of Course Projects:**

1. Automated Parking Lot Identifier,
2. Healthcare Software,
3. Financial Application,
4. Appraisal System,
5. Smart Project Administrative System,
6. Translator for Agriculture System,

7. Development of Applications using Agile Methodology,
8. Development of SMART Mobile Applications,
9. Graphics-based Password Identification System
10. System Security Application

**List of Course Seminar Topics:**

1. Mobile Apps and App Store Analysis,
2. Automated Reasoning Techniques,
3. Autonomic and Self-Adaptive System,
4. Component-based Software Engineering,
5. Computer-Supported Cooperative Work (CSCW),
6. Configuration Management and Deployment,
7. Crowd-Sourced Software Engineering,
8. Cyber-Physical System,
9. Data-driven Software Engineering,
10. Dependability, Safety and Reliability.

**List of Home Assignments:**

**Design:**

1. Software Visualization
2. Specification and Modeling Languages
3. Tools and Environments
4. Traceability
5. Ubiquitous and Pervasive Software Systems

**Case Study:**

1. Software Economics and Metrics
2. Machine Learning in Software Engineering
3. Software Evolution and Maintenance
4. Software Modeling and Design
5. Software Product Lines

**Blog**

1. Mining Software Engineering Repositories
2. Model-driven Engineering
3. Parallel, Distributed and Concurrent systems
4. Recommendation Systems
5. Refactoring

**Surveys**

1. Reverse Engineering
2. Safety-Critical Systems
3. Security, Privacy and Trust
4. Software Architecture
5. Software Reuse
6. Software Testing

**Assessment Scheme:** Ensures 360 degree assessment and covers all aspects of Bloom's Taxonomy.

**Laboratory Continuous Assessment:** 100 Marks converted to 10 Marks

**Course Project:** End Semester Examination: 100 Marks converted to 20 Marks

**Group Discussion:** End Semester Examination: 100 Marks converted to 20 Marks

**Theory:** End Semester Examination (MCQ): 60 Marks converted to 30 Marks

**Comprehensive Viva Voce:** End Semester Examination: 100 Marks converted to 20 Marks

**Text Books:** (As per IEEE format)

1. Ian Sommerville, 'Software Engineering', Pearson, 10th Edition, 2017, ISBN-13: 978-9332582699.
2. Kenneth Rubin, 'Essential SCRUM: A Practical Guide To The Most Popular Agile Process', Addison-Wesley, 2012, ISBN-13: 978-0-13-704329-3.
3. Tom Pender, 'UML Bible', John Wiley & Sons, 2003, ISBN - 0764526049

**Reference Books:** (As per IEEE format)

1. Soren Lauesen, 'Software Requirements: Styles and Techniques', Addison Wesley, 2002, ISBN 0201745704.
2. Dean Leffingwell, 'Agile Software Requirements', Addison-Wesley, 2011, ISBN-13: 978-0-321-63584-6.
3. Grady Booch, James Rumbaugh, Ivar Jacobson, 'Unified Modeling Language User's Guide', 2nd Edition, Addison-Wesley 2005, ISBN – 0321267974.
4. Erich Gamma, Richard Helm, Ralph Johnson, 'Design Patterns: Elements of Reusable Object-Oriented Software', Addison-Wesley Professional, 1994, ISBN-13: 978-0201633610.
5. Paul Clements, Felix Bachmann, Len Bass, David Garlan, 'Documenting Software Architectures: Views and Beyond', Addison-Wesley Professional, 2003, ISBN-13: 9780201703726.

**MOOCs Links and additional reading material:**

[www.nptelvideos.in](http://www.nptelvideos.in), [www.coursera.com](http://www.coursera.com), [www.udemy.com](http://www.udemy.com)

**Course Outcomes:**

The student will be able to –

1. Compare Software Development Process Models and justify process maturity through application of Software Engineering principles and practices,
2. Differentiate competing and feasible system requirements identifying problem scope in the real-world,
3. Apply agile principles and practices through scrum framework,
4. Design UML diagrams through efficient system analysis, using identified design specifications
5. Formulate system architecture as per a suitable architectural style,
6. Apply relevant design patterns for effective system design.

**CO-PO Map:**

|         | Program Outcomes (PO) |      |      |     |     |     |     |     |     |      |      |      | PSO  |      |      |
|---------|-----------------------|------|------|-----|-----|-----|-----|-----|-----|------|------|------|------|------|------|
| CO/PO   | PO1                   | PO2  | PO3  | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 | PSO1 | PSO2 | PSO3 |
| CO1     | 2                     | 3    | 2    |     |     |     |     |     |     |      |      | 3    | 2    | 3    |      |
| CO2     | 2                     | 3    | 2    |     |     |     |     |     |     |      |      | 3    | 2    | 3    |      |
| CO3     | 3                     | 2    | 3    |     | 3   | 2   | 2   | 2   |     |      | 2    | 3    | 2    | 3    |      |
| CO4     | 3                     | 2    | 3    |     | 3   | 2   | 2   | 2   | 2   |      | 2    | 3    | 2    | 3    |      |
| CO5     | 3                     | 2    | 3    |     | 3   | 2   | 2   | 2   |     |      | 2    | 3    | 2    | 3    |      |
| CO6     | 3                     | 2    | 3    |     | 3   | 2   | 2   | 2   |     | 3    | 2    | 3    | 2    | 3    |      |
| Average | 2.66                  | 2.33 | 2.66 |     | 3   | 2   | 2   | 2   | 2.0 | 3.0  | 2.0  | 3.0  | 2.0  | 3.0  |      |

**CO Attainment levels:**

Weights for attainment levels: L1 - Easy- 0.75    L2 - Comfortable - 0.7    L3 – Medium – 0.65  
L4 – Somewhat difficult – 0.6    L5 – Difficult – 0.55

CO1 – L1, CO2 – L2, CO3 – L3, CO4 – L4, CO5 – L4 and CO6 – L5

**Future Course Mapping:**

*Software testing and Quality Assurance, Service-Oriented Software*

**Job Mapping:**

*Application Architect, Project Designer, SCRUM Role Player, Project Manager*

**Syllabus Template**  
**CI3008: Cyber Security and Blockchain**

**FF No. : 654**

**Credits: 4.....**

**Teaching Scheme Theory: 2 Hours/Week**

**Tutorial: 1 Hours/Week**

**Lab: 2 Hours/Week**

**Course Prerequisites: Computer Networks**

**Course Objectives:**

1. Apply cryptographic techniques and security protocols to secure systems and networks.
2. Identify, resolve, and mitigate programming bugs and cyber threats.
3. Design secure systems using blockchain technology and ensure application security.
4. Understand and apply cloud security and physical security principles.
5. Integrate AI in cyber security and develop business continuity and disaster recovery plans.
6. Implement ethical hacking practices and perform effective penetration testing.

**Course Relevance:**

Cyber Security teaches how to protect operating systems, networks, and data from cyber attacks, monitor systems, and mitigate threats, aiming to develop skills to prevent attacks and protect data privacy.

**Section 1: Topics/Contents**

**Information security**

- **Key Security Properties:** Confidentiality, Integrity, Availability.
- **Risk Management:** Understanding governance policies, frameworks, laws, regulations, guidelines, and compliance.

- **Symmetric Key Cryptography:** Role of random numbers and nonce in security, importance of prime numbers, GCD, Euclid's Algorithm, Extended Euclid's algorithm.
- **Data Encryption Standard (DES):** Block cipher, stream cipher, Feistel structure, block cipher modes, S-DES, attacks on DES, S-AES, AES.
- **Public Key Cryptography:** RSA algorithm, key generation, attacks on RSA.
- **Elliptic Curve Cryptography (ECC):** Elliptic curves over real numbers and  $Z_p$ , elliptic curve arithmetic.

## Network Security

- **Certificates and Hashing:** Properties of hash functions, HASH + SALT, hashing algorithms (SHA1, SHA2).
- **Authentication and Authorization:** Network access control (SHA-512, Kerberos, and multifactor authentication).
- **Transport-Level Security:** Web security considerations, Secure Sockets Layer (SSL), Transport Layer Security (TLS), HTTPS standard, Secure Shell (SSH) application, IPSec.
- **Application Security:** Security by design, writing secure code, static and dynamic application security testing (SAST and DAST), interactive application security testing (IAST), Integrated Security in DevOps, OWASP, Application Security Services,

## Section 2: Topics/Contents

### Cyber Attacks and **Penetration Testing (06 Hours)**

- **Cyber Ethics:** Threats, threat modeling, injections, sniffing, and types of attacks.
- **Security Vulnerabilities:** risk, attack types, countermeasures.
- **Protocol Vulnerabilities:** DoS and DDoS, session hijacking, ARP spoofing.
- **Software Vulnerabilities:** Phishing, buffer overflow, cross-site scripting attack, ransomware, SYN-flooding, SQL-injection, DNS poisoning.
- **Penetration Testing:** Difference from automated vulnerability scans, objectives and limitations of a pen test, scoping and planning pen tests, executing pen tests and managing findings. Introduction to SDL (Secure Development Lifecycle) – Merging Security into SDLC,

### **Physical Security and Forensics (04 Hours)**

- **Physical Security:** Physical access types, crime prevention through environmental design (CPTED).
- **IoT Security:** Definitions of OT, IoT, IIoT, and ICS, most widely used protocols in IoT environments (MQTT and CoAP).
- **Business Continuity (BC):** RTP/RPO, RTO, MTPD, ISO 22301 standard for business continuity management, importance, differences between BCMS and DRMS, risk management, testing, maintenance., Operation Resilience,
- **Digital Forensics:** Introduction to digital forensics, data recovery, OS forensics, email crimes and violations, cyber forensics.

## Cloud Security

**Principles / Key Concepts of Cloud Security:** Overview of cloud security principles and key concepts.

- **Threats and Risks in Cloud Security:** Diverse types of threats and risks associated with cloud security.
- **Importance of Security Measures in Cloud Security:** Importance of implementing security measures in cloud environments.
- **Solutions for Cloud Security:** Effective solutions to address cloud security challenges.

**Role of AI in Cyber Security:** Examination of how AI is integrated into cyber security.

- **Challenges and Opportunities of AI in Cyber Security:** Analysis of the challenges and opportunities presented by AI in the field of cyber security.

## Blockchain

- **Decentralized Systems & Distributed Ledger Technology:** Blockchain computing power, hash, and Merkle tree with hands-on examples.
- **Use-Cases of Blockchain:** Different types of blockchain including public and private blockchain, consensus and types of consensus with examples.
- **Smart Contracts:** Need for smart contracts, developing smart contracts, programming basics of Solidity (data types) and advanced Solidity, EVM in relation to smart contracts and gas price, running and debugging smart contracts in Remix, deploying and debugging smart contracts with Truffle.

## List of Tutorials (13)

1. Mathematical background for cryptography: modulo arithmetic, GCD (Euclid's algorithm), algebraic structures (Groups, Rings, Fields, Polynomial Field).
2. Chinese remainder theorem.
3. Diffie-Hellman key exchange: Algorithm, Key exchange protocol, Attack.
4. ECC over Diffie-Hellman key exchange.
5. Study of certificates and hashing algorithms.
6. Network access control and transport-level security.
7. Security by design and writing secure code.
8. Static and dynamic application security testing.
9. Study of Snort.
10. Nessus: a Security Vulnerability scanning tool.



11. Metasploit/Ollydbg.
12. Testing for Brute Force Password.
13. Testing for SQL Injection.
14. Computer forensics, Facebook forensic, mobile forensic, cyber forensic, digital forensic.
15. Source Code Analysis Tools.
16. OWASP Zed Attack Proxy (ZAP).
17. Study of various types of Blockchain, Connecting the Metamask wallet with the local Ganache network.
18. Simulation of Blockchain.
19. Creating Smart Contract using Solidity and Remix IDE.
20. Study of DOA and DAPP.

### **List of Practicals (Minimum Six)**

#### **Section-I:**

- Simplified DES implementation.
- Simplified AES implementation.
- Encryption and Decryption by RSA algorithm.
- Implementation of ECC over Diffie-Hellman Key Exchange Protocol.
- Implementation of authentication algorithms.
- Implementation of SHA.

#### **Section-II:**

- Acquisition of System Information/ RAM/Volume Shadow Copy/Detecting Encryption in information.
- Vulnerabilities finding in Mobile/ computer/ digital devices.
- Forensic of Disc Image/ Registry/ Meta data/ RAM.
- Digital forensic of images.
- Forensics of Video alteration.
- Implement and demonstrate the use of the following in Solidity: Variable, Operators, Loops, Decision Making, Strings, Arrays, Enums, Structs.
- Implement and demonstrate the use of the following in Solidity: Functions, Function Modifiers, View functions, Pure Functions, Mathematical functions, Cryptographic functions.
- Use Geth to configure a private Blockchain node in our machine.
- Cryptography in Blockchain, Merkle root tree hash.
- Creating Transactions using Solidity and Remix IDE.

- Case Study on Hyperledger Fabric.

**List of Course Project areas:**

Course Project 01 Statement: Design a System to develop a analyzer which will differentiate between different vulnerability and packets entered using it. This system will detect the intrusions coming through the vulnerabilities.

Course Project 02 Statement: Securing Video Conferencing App for online meetings

Course Project 03 Statement: Steganography for Image/Video/Files

Course Project 04 Statement: Secure Image display on online social media.

Course Project 05 Statement: Secure transfer of government subsidies to farmers/BPL people/ students etc

Course Project 06 Statement: Authentication of users for various applications for integrity, availability, confidentiality.

Course Project 07 Statement: Implementing a system for detecting the modification of videos/images on social media

Course Project 08 Statement: Secure App for online exams detecting Keystroke and camera movements.

Course Project 09 Statement: A system to detect the difference between the voice edited in the audio/video

Course Project 10 Statement: A System to check the vulnerabilities in the websites.

Course Project 11 Statement: Decentralized (Uber)Peer to Peer Carpooling

Course Project 12 Statement: Decentralized Skill Verification System

Course Project 13 Statement: Decentralized talent acquisition (like Nokari.com)

Course Project 14 Statement: Decentralized gaming DAPP(earn coin through game)

**List of Course Seminar Topics**

Seminar 01 Statement: Blockchain architecture and its implementation

Seminar 02 Statement: Cloud Security

Seminar 03 Statement: Mobile Security

Seminar 04 Statement: IoT and Security Issues/ Security Models for IoT

Seminar 05 Statement: Dark web

Seminar 06 Statement: Docker Security

Seminar 07 Statement: Access control methods for online social media and various organizations

Seminar 08 Statement: Security of Android Vs IOS

Seminar 09 Statement: Machine learning and SCADA Security

Seminar 10 Statement: Security Applications for Smart

### **List of Design Based Home Assignments**

**HA\_D 01** Statement: Design a secure system using cryptography techniques for security of multimedia files.

**HA\_D 02** Statement: Design a secure system using steganography for hiding data files in image/video

**HA\_D 03** Statement: Design a system for educational institutes using authentication and authorization techniques, also give details about the access control policies that must be implemented for the design of system by various places.

**HA\_D 04** Statement: Design a secure system using SSL/TLS/IPSec for the various organizations

**HA\_D 05** Statement: Design a system for the analysis of cyber crime using various cyber forensic techniques and compare each technique with respect to integrity, confidentiality, availability

### **List of Case Study Based Home Assignments**

**HA\_CS 01** Statement: How to improve the security of social media? Write a detail case study

**HA\_CS 02** Statement: Find out the vulnerability issues in educational institutes websites/online systems and give solutions to these problem. Perform a detailed case study of the various issues.

**HA\_CS 03** Statement: Write a detail case study about the banking security flows and solutions to these flows.

**HA\_CS 04** Statement: Give a detail case study of the antivirus system giving the flows and solutions to it.

**HA\_CS 05** Statement: Perform the detail case study of various operating systems used for mobile devices and give a secure solution to one for widely used OS.

### **List of Blog Based Home Assignment**

**HA\_Blog 01** Statement: Dark Web

**HA\_Blog 02** Statement: Crypto currency and Economy

**HA\_Blog 03** Statement: Cybercrime and solutions

**HA\_Blog 04** Statement: Authentication and Access control for social media

**HA\_Blog 05** Statement: Cyber forensic and Cyber laws

### **List of Survey Based Home Assignments**

**HA\_Survey 01** Statement: Survey on various blockchain related issues/ cryptocurrency/ application systems developed using blockchain

**HA\_Survey 02** Statement: Survey on various authentication and access control methods for different applications

**HA\_Survey 03** Statement: Steganography and Biometric Systems for authentication

**HA\_Survey 04** Statement: Survey of various attacks and its effect on Indian economy and its analysis

**HA\_Survey 05** Statement: Problems over Integer Lattices: A Study

**Assessment Scheme:** Suggest an Assessment scheme that is best suited for the course. Ensure 360 degree assessment and check if it covers all aspects of Bloom's Taxonomy.

**Laboratory Continuous Assessment:** 100 Marks converted to 10 Marks

**Course Project:** End Semester Examination: 100 Marks converted to 20 Marks

**Presentation:** End Semester Examination: 100 Marks converted to 20 Marks

**Theory:** End Semester Examination (Written): 60 Marks converted to 30 Marks

**Comprehensive Viva Voce:** End Semester Examination: 100 Marks converted to 20 Marks

**Text Books:** (As per IEEE format)

1. William Stallings, "Cryptography and Network Security-Principles and Practices" 6th Edition, Pearson Education, 2014, ISBN13:9780133354690.
  2. Bernard Menezes, "Network Security and Cryptography", 1st Edition, Cengage Learning, 2010, ISBN 81-315-1349-1.
  3. Raef Meeuwisse, "Cybersecurity for Beginners", 2nd Edition, Cyber Simplicity, 2017, ISBN-9781911452157
  4. Ambadas Tulajadas Choudhari, Arshad Sarfarz Ariff, Sham M R, "Blockchain for Enterprise Application Developers" Willey publications, ISBN: 9788126599967, 2020
- Hyperledger Fabric - <https://www.hyperledger.org/projects/fabric>

**Reference Books:** (As per IEEE format)

1. M. Speciner, R. Perlman, C. Kaufman, "Network Security: Private Communications in a Public World", Prentice Hall, 2002
2. Michael Gregg, "The Network Security Test Lab: A Step-By-Step Guide", Dreamtech Press, 2015, ISBN-10:8126558148, ISBN-13: 978-8126558148.
3. Matt Bishop, "Computer Security: Art and Science", 1st Edition, Pearson Education, 2002, ISBN 0201440997.
4. Charlie Kaufman, Radia Perlman and Mike Spencer, "Network security, private communication in a public world", 2nd Edition, Prentice Hall, 2002, ISBN 9780130460196.
5. V.K. Pachghare, "Cryptography and Information Security", 2nd Edition, PHI, 2015, ISBN-978-81-203-5082-3.
6. Mastering Blockchain: Deeper insights into decentralization, cryptography, Bitcoin, and popular Blockchain frameworks by Bashir, Imran, 2017.

Arvind Narayanan, Joseph Bonneau, Edward Felten, Andrew Miller, and Steven Goldfeder. Bitcoin and cryptocurrency technologies: a comprehensive introduction. Princeton University Press, 2016

**MOOCs Links and additional reading material:**

1. Cryptography And Network Security By Prof. Sourav Mukhopadhyay, IIT Kharagpur  
Cryptography and Network Security - Course (nptel.ac.in)
2. Information Security and Cyber Forensics By Prof. Pratosh Bansal Devi Ahilya Vishwavidyalaya, Indore, Information Security and Cyber Forensics - Course (swayam2.ac.in)
3. Blockchain and its Applications By Prof. Sandip Chakraborty, Prof. Shamik Sural IIT Kharagpur  
Blockchain and its Applications - Course (nptel.ac.in)

**Course Outcomes:**

The student will be able to –

1. Demonstrate cryptographic techniques using a mathematical approach by examining nature of attack.
2. Design a secure system for protection from the various attacks for 7 layer model by determining the need of security from various departments of an organization
3. Justify various methods of authentication and access control for application of technologies to various sections of industry and society.
4. Identify and establish different attacks on the system.
5. Estimate future needs of security for a system by researching current environment on a continuous basis for the benefit of society.
6. Analyze the need of Decentralized system and implement using blockchain technology.

**CO-PO Map:**

| CO/PO   | Program Outcomes (PO) |     |     |     |     |     |     |     |     |      |      |      | PSO  |      |      |
|---------|-----------------------|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|------|------|------|
|         | PO1                   | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 | PSO1 | PSO2 | PSO3 |
| CO1     | 3                     | 3   |     | 3   | 2   | 2   |     | 3   | 1   |      |      |      | 3    |      | 1    |
| CO2     | 3                     | 2   | 3   | 2   |     |     | 2   | 3   | 3   | 3    | 2    | 1    |      |      |      |
| CO3     | 2                     | 3   | 3   |     | 1   | 2   |     | 3   | 1   |      |      |      |      |      | 1    |
| CO4     | 3                     | 3   | 1   | 3   | 3   | 3   | 3   | 3   | 1   |      |      | 3    |      | 3    | 1    |
| CO5     | 2                     | 2   | 3   | 2   | 1   |     | 2   |     | 3   | 3    | 3    | 3    |      |      |      |
| CO6     | 3                     | 2   | 1   | 1   | 3   | 3   | 3   | 3   |     |      | 3    | 2    |      | 3    | 3    |
| Average | 2.67                  | 2.5 | 2.2 | 2.2 | 2   | 2.5 | 2.5 | 3   | 1.8 | 3    | 2.67 | 2.25 | 3    | 3    | 1.5  |

**CO Attainment levels:**

Attainment Levels: 3,4, 2, 1, 5, 3

**Future Course Mapping:**

*Cloud Computing and Security, IoT Security, Ethical Hacking & Cyber Forensics*

**Job Mapping:**

*Security Engineer/Network Security Engineer*

*Information Security Analyst*

*Cyber Security Analyst*

*Cyber Security Associate*

*Manager-Information Security Services*

*Security Consultant*

*Penetration Testing Engineer*

**Syllabus Template**

**FF No. : 654**

**CIS3009: Deep Learning**

**Credits: 4**

**Teaching Scheme Theory: 2 Hours/Week**

**Tutorial: 1 Hours/Week**

**Lab: 2 Hours/Week**

**Course Prerequisites:**

Linear algebra, probability theory and statistics, Digital signal processing, Computer vision

**Course Objectives:**

- 1.To explore data preprocessing methods, including feature selection and dimensionality reduction techniques like PCA and LDA.
2. To Apply various supervised learning algorithms, including linear and kernel-based models.
- 3.To evaluate classification methods and their applications in solving problems.
4. To understand the importance of deep learning and its variants
5. To understand the basics of Recurrent neural network models of NN
6. To build deep nets with applications to solve real world problem

**Course Relevance:**

In today's data-driven landscape, machine learning and deep learning are critical for addressing complex industry challenges and driving innovation. As businesses increasingly rely on data analytics for decision-making, machine learning offers essential tools for predictive modeling, customer insights, and operational efficiency. Deep learning, with its ability to analyze vast amounts of unstructured data, is revolutionizing sectors such as healthcare through improved diagnostics, finance through algorithmic trading, and autonomous systems via enhanced perception and decision-making capabilities. This course prepares students to meet the growing demand for expertise in these

technologies, equipping them with the skills to develop intelligent solutions that can transform industries and improve overall productivity.

### **Section 1: Topics/Contents**

#### **Unit-I : Introduction Machine Learning [4 Hours]**

Motivation and role of machine learning in computer science and problem solving, Machine Learning Workflow, Introduce paradigms of Learning, Data Preprocessing and Feature Engineering, Feature Selection and Extraction Techniques, **Dimensionality Reduction Algorithms** : Principal Component Analysis (PCA), Linear Discriminant Analysis (LDA)

#### **Unit-II: Supervised Learning Algorithms [6 Hours]**

**Linear Models:** Linear Regression: Simple Linear Regression, Multiple Linear Regression, Polynomial Linear Regression, **Evaluation Metrics:** MAE, RMSE, R2, MSE, Logistic Regression, Ridge and Lasso Regression.

**Kernel Based Algorithms:** Support Vector Machine (SVM)- Linear Support Vector Machines, Linear Classification, Kernel based classification, **Probability-Based Algorithms** : Naive Bayes' Classifiers, Multinomial Naïve Bayes, and Gaussian Naive Bayes.

#### **Unit-III : Supervised Learning: Classification [4 Hours]**

K-nearest neighbor classifier, Decision Tree, **Ensemble Learning:** Bagging, Boosting, Random Forest, Adaboost, **Evaluation Metrics and Score:** Accuracy, Precision, Recall, Fscore, Cross-validation, Micro-Average Precision and Recall, Micro-Average F-score, Macro-Average Precision and Recall, Macro-Average F-score.

### **Section 2: Topics/Contents**

#### **Unit-IV: Introduction to Deep Learning [5 Hours]**

Introduction, Evolution of AI, Machine Learning vs Deep Learning, Deep Learning types, Stages in ML/DL project, Applications of Deep Learning, Introduction to DL Frameworks Keras, PyTorch, Caffe, Shogun. **Basic Tensor Operations:** Creating, manipulating, and visualizing tensors. **Building a Neural Network:** Step-by-step implementation using Keras and PyTorch.

#### **Unit-V: Convolution Neural Network**

CNN architecture overview, Building blocks of Convolutional Network-Convolution, activation functions (ReLU), pooling, fully connected layers. Padding, Strides, Typical Settings, the Fully Connected Layers **Advanced Architectures:** LeNet-5, AlexNet, VGG-16, ResNet. **Training and Optimization:** Training strategies, regularization, transfer learning. Implementation of neural network for a case study, case study: Real time applications

#### **Unit-VI: Recurrent Neural Networks (RNN) [5 Hours]**

Sequence modeling: Recurrent nets RNN architecture, bidirectional RNNs, Challenges in training RNNs, Long Short-Term Memory (LSTM), Vanishing and exploding gradient problem, Auto encoders, **Applications of RNNs:** Language modeling, speech recognition, machine translation.



**List of Practical :**

1. Perform PCA in dimension reduction of numerical data
  - a. Pre-process the data through standardization.
  - b. Perform PCA to reduce dimension.
  - c. Construct the scree plot.
  - d. Data visualization in lower dimensional representation.
2. Implement Simple and Multiple Linear Regression to predict continuous variables.
  - a. Perform data preprocessing (handle missing values, feature scaling).
  - b. Fit a **Simple Linear Regression** model on a dataset (e.g., predicting house prices).
  - c. Extend to **Multiple Linear Regression** with multiple features.
  - d. Evaluate models using **MSE**, **RMSE**, and **R<sup>2</sup> Score**.
  - e. Visualize the regression line and predictions.
3. Write a program to implement k-Nearest Neighbour algorithm to classify the iris data set. Print both correct and wrong predictions. Python ML library classes can be used for this problem.
4. Write a program to implement the naïve Bayesian classifier for a sample training data set stored as a .CSV file. Compute the accuracy of the classifier, considering few test data sets.
5. Learn Decision trees for regression and classification problem
  - a. Split the data set into training and test sets.
  - b. Build the decision tree
  - c. Check model performances on training and test data sets.
  - d. Apply cost complexity pruning to overcome overfitting problem
  - e. Apply Random Forest algorithm to overcome overfitting problem.
  - f. Apply Ada-boost ensemble method on Decision stumps.
6. Build a Multiclass classifier using the CNN model. Use MNIST or any other suitable dataset.
  - a. Perform Data Pre-processing
  - b. Define Model and perform training
  - c. Evaluate Results using confusion matrix
7. Convolutional neural network (CNN) Use any dataset of plant disease and design a plant disease detection system using CNN.
8. Use MNIST Fashion Dataset and create a classifier to classify fashion clothing into categories. Using CNN
9. Implementation of RNN model for Stock Price Prediction
10. Using LSTM for prediction of future weather of cities in Python
11. Implement a basic **RNN** for handling sequential data.
  - a. Build an **RNN** for a time-series prediction task.
  - b. Train on sequential data (e.g., stock prices).



- c. Evaluate the model using **MSE** or **RMSE** for regression tasks.
  - d. Visualize predictions vs actual values over time.
12. Design an object detection model using deep neural networks for simple objects.
  - a. Select appropriate dataset and perform data pre-processing
  - b. Define architecture in terms of layers
  - c. Evaluate Model performance Label the object with appropriate text.
13. Train a CNN model using data augmentation to improve generalization.
  - a. Apply data augmentation techniques like rotation, zooming, and flipping.
  - b. Train the CNN on augmented data and compare performance with the original model.
  - c. Plot training and validation accuracy for both models.

**List of Seminar Topics:**

1. Explainable AI (XAI): Making Machine Learning Models Transparent
2. Reinforcement Learning in Autonomous Systems
3. Federated Learning: Collaborative Learning Without Centralized Data
4. Classification of skin cancer with deep neural networks
5. Self-Supervised Learning: Closing the Gap Between Supervised and Unsupervised Learning
6. Convolutional Neural Networks (CNNs) in Image Processing
7. Accelerating Deep Network Training by Reducing Internal Covariate Shift
8. Deep learning applications for predicting pharmacological properties of drugs
9. GAN (Generalized Adversarial network)
10. Auto encoders
11. LSTM

**List of Course Group Discussion Topics:**

1. Supervised vs. Unsupervised Learning: When to Use Which Approach?
2. The Role of Data Quality in Machine Learning: Can Good Data Outperform Advanced Algorithms?
3. Reinforcement Learning vs. Traditional Machine Learning
4. Hyperparameter tuning: Is there a rule of thumb?
5. Deep Learning vs. Traditional Machine Learning
- 6 Which cost function: Least squared error or binary cross entropy?
7. Convolutional Neural Networks (CNNs) vs. Recurrent Neural Networks (RNNs): Strengths and Weaknesses
8. Need of hundred classifiers to solve real world classification problem
9. Which optimization: Batch gradient descent or stochastic gradient descent
10. Data Privacy in the Age of AI: Machine Learning and Deep Learning in Sensitive Domains
11. The Role of Data Augmentation in Enhancing Deep Learning Models

**List of Design based Home Assignments:**

**Design:**

1. Design a Machine Learning Model for Predicting Housing Prices
2. Development of control system for fruit classification based on convolutional neural networks
3. Classifying movie review using deep learning

4. Sentiment analysis of the demonetization of economy 2016 India
5. Predicting Students Performance in Final Examination
6. Design an LSTM for Machine Translation

**Case Study:**

1. Credit Risk Scoring for Banking
2. Churn Prediction for Telecom Industry
3. Convolutional Neural Networks for Visual Recognition
4. Deep Learning for Natural Language Processing
5. Scalable object detection using deep neural networks

**Blog**

1. Brain tumor segmentation with deep neural networks
2. Region-based convolutional networks for accurate object detection and segmentation
3. Human pose estimation via deep neural networks
4. Content Based Image Retrieval
5. Visual Perception with Deep Learning
6. Music genre classification system

**Surveys:**

1. Machine translation using deep learning - survey
2. Shaping future of radiology using deep learning
3. Training Recurrent Neural Networks
4. Text generation with LSTM
5. Deep learning applications in Biomedicine

**Assessment Scheme:** Ensures 360 degree assessment and covers all aspects of Bloom's Taxonomy.

**Laboratory Continuous Assessment:** 100 Marks converted to 10 Marks

**Course Project:** End Semester Examination: 100 Marks converted to 20 Marks

**Lab Examination:** End Semester Examination (Written): 100 Marks converted to 50 Marks

**Comprehensive Viva Voce:** End Semester Examination: 100 Marks converted to 20 Marks

**Text Books:**

1. T. Mitchell, "Machine Learning", McGraw-Hill, 1997.
2. Anup Kumar Srivastava, Soft Computing, Alpha Science International limited. 2009.
3. Goodfellow, I., Bengio, Y., and Courville, A., Deep Learning, MIT Press, 2016.
4. C., M., Pattern Recognition and Machine Learning, Springer, 2006.

**Reference Books:**

1. Yegnanarayana, B., Artificial Neural Networks PHI Learning Pvt. Ltd, 2009.
2. Golub, G., H., and Van Loan, C., F., Matrix Computations, JHU Press, 2013.
3. Satish Kumar, Neural Networks: A Classroom Approach, Tata McGraw-Hill Education, 2004.
4. Richard S. Sutton and Andrew G. Barto, "Reinforcement Learning: An Introduction" by Seth Weidman, "Deep Learning from Scratch: Building with Python from First Principles" O'Reilly

5. Francois Duval, “Deep Learning for Beginners, Practical Guide with Python and Tensorflow”

**MOOCs Links and additional reading material:**

1. [www.nptelvideos.in](http://www.nptelvideos.in)
2. <https://nptel.ac.in/noc/courses/noc20/SEM1/noc20-cs11>
3. <https://nptel.ac.in/noc/courses/noc20/SEM1/noc20-cs50>
4. <https://www.my-mooc.com/en/categorie/deep-learning>

**Course Outcomes:**

The student will be able to –

- 1) Understand the role of machine learning, data preprocessing techniques, including feature selection, PCA, and LDA, to improve model performance.
- 2) Apply linear models such as Linear Regression and Logistic Regression, and kernel-based models such as SVM for classification tasks.
- 3) Apply ensemble techniques like Bagging and Boosting, and classification algorithms such as KNN and Decision Trees to solve classification problems.
- 3) Compare different classification models in terms of accuracy and computational efficiency in various real-world applications.
- 4) Apply a basic convolutional neural network using a deep learning framework
- 5) Evaluate the performance of CNN models and analyze the benefits of transfer learning using pretrained models in specific applications.
- 6) Analyze the strengths and weaknesses of RNNs in handling long-term dependencies in sequence modeling tasks.

**CO-PO Map:**

|         | Program Outcomes (PO) |     |     |      |     |      |     |     |     |      |      |      | PSO  |      |      |
|---------|-----------------------|-----|-----|------|-----|------|-----|-----|-----|------|------|------|------|------|------|
| CO/PO   | PO1                   | PO2 | PO3 | PO4  | PO5 | PO6  | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 | PSO1 | PSO2 | PSO3 |
| CO1     | 3                     | 3   | 3   | 2    | 3   | 1    |     | 1   | 1   | 1    |      | 2    |      | 3    | 3    |
| CO2     | 3                     | 3   | 3   | 2    | 3   | 2    |     | 1   | 1   | 1    |      | 2    |      | 3    | 3    |
| CO3     | 2                     | 3   | 3   | 3    | 3   | 2    |     | 1   | 1   | 1    |      | 2    |      | 3    | 3    |
| CO4     | 3                     | 3   | 3   | 3    | 3   | 2    |     | 1   | 1   | 1    |      | 2    |      | 3    | 3    |
| CO5     | 3                     | 3   | 3   | 3    | 3   | 2    |     | 1   | 1   | 1    |      | 2    |      | 3    | 3    |
| CO6     | 3                     | 3   | 3   | 3    | 3   | 2    |     | 1   | 1   | 1    |      | 2    |      | 3    | 3    |
| Average | 3                     | 3   | 3   | 2.66 | 3   | 1.83 |     | 1   | 1   | 1    |      | 2    |      | 3.0  | 3.0  |

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

**CO Attainment levels:**

- CO1 - Level 3
- CO2 - Level 3
- CO3 - Level 4
- CO4 - Level 5

CO5 - Level 5

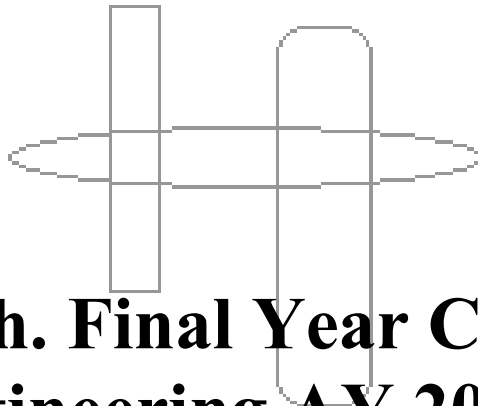
CO6 - Level 2

**Future Course Mapping:**

Advanced course on Deep learning including Auto encoders and Boltzmann machines, Reinforcement Learning etc

**Job Mapping:**

Deep learning engineer, Data Scientist and Algorithm Architect with industries in domains Healthcare, Industrials & Energy, Automobiles, Finance & Insurance, Human Resources, Agriculture, Cybersecurity, Ad & Marketing, Media and Entertainment, Government, Defence, Data Analytics



# **B. Tech. Final Year Computer Engineering AY 2025-26**

## **Module VII and VIII Course Content**

**Syllabus Template**

**FF No. : 654**

**Coursera Specialization Courses**

**Credits: 2**

| <b>Sr. No.</b> | <b>Course code</b> | <b>Specialization Name</b>                  | <b>Link</b>                                                                                                                                                                                                                                                                                           |
|----------------|--------------------|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1              | MD4228             | IBM Full Stack Software Developer           | <a href="https://www.coursera.org/professional-certificates/ibm-full-stack-cloud-developer">https://www.coursera.org/professional-certificates/ibm-full-stack-cloud-developer</a>                                                                                                                     |
| 2              | MD4230             | IBM Back-End Developer                      | <a href="https://www.coursera.org/professional-certificates/ibm-backend-development">https://www.coursera.org/professional-certificates/ibm-backend-development</a>                                                                                                                                   |
| 3              | MD4248             | IBM DevOps and Software Engineering         | <a href="https://www.coursera.org/professional-certificates/devops-and-software-engineering">https://www.coursera.org/professional-certificates/devops-and-software-engineering</a>                                                                                                                   |
| 4              | MD4262             | Salesforce Sales Development Representative | <a href="https://www.coursera.org/professional-certificates/salesforce-sales-operations">https://www.coursera.org/professional-certificates/salesforce-sales-operations</a>                                                                                                                           |
| 5              | MD4238             | Microsoft Cybersecurity Analyst             | <a href="https://www.coursera.org/programs/faculty-development-program-d5iiv/professional-certificates/microsoft-cybersecurity-analyst?source=search">https://www.coursera.org/programs/faculty-development-program-d5iiv/professional-certificates/microsoft-cybersecurity-analyst?source=search</a> |
| 6              | MD4243             | IBM Data Engineering                        | <a href="https://www.coursera.org/professional-certificates/ibm-data-engineer">https://www.coursera.org/professional-certificates/ibm-data-engineer</a>                                                                                                                                               |
| 7              | MD4245             | IBM Data Science                            | <a href="https://www.coursera.org/professional-certificates/ibm-data-science">https://www.coursera.org/professional-certificates/ibm-data-science</a>                                                                                                                                                 |
| 8              | MD4247             | IBM Data Warehouse Engineer                 | <a href="https://www.coursera.org/professional-certificates/data-warehouse-engineering">https://www.coursera.org/professional-certificates/data-warehouse-engineering</a>                                                                                                                             |
| 9              | MD4257             | IBM Mainframe Developer                     | <a href="https://www.coursera.org/professional-certificates/ibm-mainframe-developer">https://www.coursera.org/professional-certificates/ibm-mainframe-developer</a>                                                                                                                                   |

|    |        |                  |                                                                                                                                                       |
|----|--------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10 | MD4269 | Google UX Design | <a href="https://www.coursera.org/professional-certificates/google-ux-design">https://www.coursera.org/professional-certificates/google-ux-design</a> |
|----|--------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|

**Syllabus Template**

**FF No. : 654**

**CI 4001: Generative AI**

**Credits: 2**

**Teaching Scheme Theory: 2 Hours/Week**

**Course Prerequisites:**

Python, Basic ML, Linear Algebra, Neural Networks

**Course Objectives**

1. To introduce the foundational concepts and evolution of Generative AI models such as GANs, VAEs, and Diffusion models.
2. To explain the basics of text processing and natural language understanding using real-world examples.
3. To provide a conceptual understanding of sequence modeling using RNNs, LSTMs, and GRUs.
4. To describe advanced NLP techniques including Attention Mechanism, Transformers, and BERT.
5. To familiarize students with the architecture and functioning of Large Language Models (LLMs) such as GPT and BERT.
6. To analyze practical applications and ethical concerns associated with Generative AI.
7. To promote awareness of responsible AI practices through case-based learning and discussions.

## **Unit-I: Processing Text with Python Essential Training**

Introduction to Text and NLP, Text Cleaning and Tokenization, Stemming and Lemmatization, Stop Words and POS Tagging, Named Entity Recognition (NER), Text Vectorization (BoW, TF-IDF, Embeddings), Sentiment Analysis, Text Classification Basics, Introduction to Text Generation, NLP Libraries: NLTK, spaCy, Hugging Face

## **Unit-II: Hands-On Natural Language Processing**

Text Data Loading and Preprocessing, Tokenization and POS Tagging with spaCy, Named Entity Recognition (NER), Text Classification using Scikit-learn, Word Embeddings (Word2Vec, GloVe), Introduction to Transformers and BERT

## **Unit-III: Advanced NLP with Python for Machine Learning**

Text Feature Engineering, N-grams and Language Models, Topic Modeling with LDA, Sequence Models with RNN and LSTM, Attention Mechanism Basics, Introduction to Transfer Learning in NLP

## **Unit-IV Deep Learning Foundations: Natural Language Processing with Tensor Flow**

Introduction to TensorFlow for NLP, Text Vectorization with TensorFlow, Building Neural Networks for Text Classification, Word Embeddings with Embedding Layer, Implementing RNNs and LSTMs in TensorFlow, Introduction to Transformer Models using TensorFlow

## **Unit-V Recurrent Neural Networks**

Introduction to Sequential Data and RNNs, RNN Architecture and Working, Vanishing Gradient Problem, Long Short-Term Memory (LSTM) Networks, Gated Recurrent Unit (GRU), Text Generation with RNNs

## **Unit VI Generative AI: Working with Large Language Models**

Overview of Large Language Models (LLMs), Pretraining and Fine-Tuning of LLMs, Prompt Engineering Techniques, In-Context and Few-Shot Learning, Introduction to Retrieval-Augmented Generation (RAG), Applications of LLMs in Text Generation and Chatbots

### **1. Reference Print Books**

1. D. Foster; Generative Deep Learning: Teaching Machines to Paint, Write, Compose, and Play; 1st Edition; O'Reilly Media; 2019
2. J. Shane; You Look Like a Thing and I Love You: How AI Works and Why It's Making the World a Weirder Place; 1st Edition; Voracious; 2019

3. T. Mitchell; Machine Learning; 1st Edition; McGraw Hill; 1997
4. S. Russell, P. Norvig; Artificial Intelligence: A Modern Approach; 4th Edition; Pearson; 2020

### **Reference Electronic Books**

1. H. Schmidt-Walter, R. Kories; Electrical Engineering: A Pocket Reference; Artech House, 2007; Accessed: Oct. 16, 2016; [Online]. Available: <https://ebookcentral.proquest.com>
2. F. Chollet; Deep Learning with Python; 2nd Edition; Manning Publications; 2021; Accessed: May30,2024; Available: <https://www.manning.com/books/deep-learning-with-python-second-edition>
3. I. Goodfellow, Y. Bengio, A. Courville; Deep Learning; MIT Press; 2016; Accessed: Apr. 22, 2024; Available: <https://www.deeplearningbook.org>

### **MOOCs and Other Learning Resources**

- 1.I.Goodfellow,Y.Bengio;Introduction to Generative Adversarial Networks; DeepLearning.AI(Coursera); <https://www.coursera.org/learn/build-basic-generative-adversarial-networks-gans>; Accessed: May 28, 2024
2. L. Fei-Fei, J. Ngiam; CS231n: Convolutional Neural Networks for Visual Recognition; Stanford University; <http://cs231n.stanford.edu>; Accessed: May 25, 2024
- 3.C.Olah;TheIllustrated Transformer; Distill.pub; <https://distill.pub/2016/deconv-checkerboard/>; Accessed: May 26, 2024
- 4.OpenAI; ChatGPT Prompt Engineering for Developers; OpenAI; <https://platform.openai.com/education>; Accessed: May 29, 2024

**Syllabus Template**

**FF No.: 654**

### **CI4005: Natural Language Processing**

**Credits: 2**

**Teaching Scheme Theory: 2 Hours/Week**

**Course Prerequisites:** Theory of Computer Science, Compiler Design

### **Course Objectives:**

1. To understand morphology for given natural language
2. To learn how to design lexical analyzer for given natural language
3. To learn how to design Syntactic Analyzer for given natural language



4. To learn how to design type dependency parser using pragmatic approach for given natural language
5. To understand the scientific process for machine transliteration, machine translation and information retrieval for given natural language using statistical approach

**Course Relevance:** Although Natural Language Processing (NLP) has been with us for quite some time, it has only recently gained industry-wide attention, thanks to Deep Learning. Today, NLP is a core competence area in Data Science and IT, with applications spanning across sectors that rely on harnessing language data's potential. Essentially, NLP applications are designed to extract relevant and meaningful information from natural human language data and impart machines with the ability to interact with humans.

## **SECTION 1: TOPICS/CONTENTS**

### **Unit-I Introduction**

**[5 Hours]**

What is natural language processing? Applications of NLP, Origins of NLP, Challenges of NLP, Language and Knowledge, Language and Grammar, Processing Indian Languages. Grammar-based language models, lexical functional Grammar(LFG), Government and Binding (GB), Lexical functional Grammar Model, Generative grammars, Statistical Language Model.

### **Unit-II Regular Expressions and Automata**

**[5 Hours]**

Formal Language Theory: Basic Notions, Basic Regular Expression Patterns, Disjunction, Grouping and Precedence, Advanced Operators, Substitution, Finite State Automata, NFSA. Words and Transducers, Morphology, Inflectional Morphology, Derivational Morphology, Finite State Morphological Parsing, Construction of Finite State Lexicon, Finite State Transducers, FST for Morphological Parsing.

### **Unit-III Theory of parsing / Syntactic Analysis**

**[4 Hours]**

Context Free Grammar, parsing, Top-down Parsing, Bottom-up parsing, Probabilistic parsing, Indian Languages parsing Semantic Analysis: Meaning Representation, Lexical Semantic, Ambiguity, Word Sense Disambiguation, Discourse processing, Natural Language Generation.

## **SECTION2: TOPICS/CONTENTS**

### **Unit-IV Computer Linguistics**

**[5 Hours]**

Machine Transliteration using Statistical Language modeling: N-gram model, Machine Transliteration: Rule-based, Phonology and Stress Analysis based and Statistical based, Support vector

machine, Memory Entropy Model, Hidden Markov Model, Conditional Random Fields, Evaluation Metrics

### **Unit-V Machine Translation**

**[5 Hours]**

Introduction, Problems in MT, Characteristics of Indian Languages, Machine Translation Approaches, Direct Machine Translation, Rule-Based MT, Corpus Based Machine Translation, Semantic/Knowledge based MT Systems, Translation involving Indian Languages, Statistical-Based using MT Tools - GIZA++, SRTLM and Moses, Evaluation Metrics

### **Unit-VI Information Retrieval**

**[4 Hours]**

Designing features for IR Systems, IR Models, Classical IR Models, Non Classical IR Models. Evaluation of IR Systems, NLP in IR, Relation Mapping, and Knowledge based Approaches, Conceptual Graphs in IR, Cross Language Information Retrieval, Evaluation Metrics.

### **List of Course Project Areas:**

1. Biomedical Text Mining.
2. Computer Vision and also NLP.
3. Deep Linguistic Processing.
4. Controlled Natural Language.
5. Language Resources and also Architectures for NLP.
6. Sentiment Analysis and also Opinion Mining.
7. Recognizing Similar Texts
8. Inappropriate Comments Scanner
9. Language Identifier
10. Image-Caption Generator

### **List of Design based Home Assignments:**

#### **Design:**

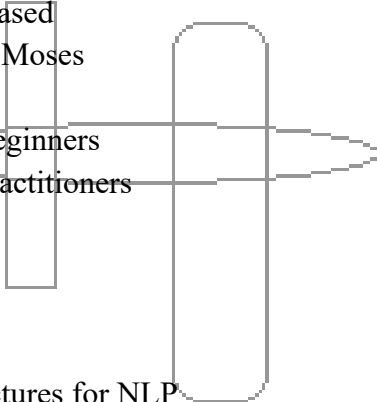
1. Use a simple method to classify positive or negative sentiment in tweets
2. Use a more advanced model for sentiment analysis
3. Use vector space models to discover relationships between words and use principal component analysis (PCA) to reduce the dimensionality of the vector space and visualize those relationships
4. Write a simple English-to-French translation algorithm using pre-computed word embeddings and locality sensitive hashing to relate words via approximate k-nearest neighbors search
5. Create a simple auto-correct algorithm using minimum edit distance and dynamic programming
6. Write a better auto-complete algorithm using an N-gram model (similar models are used for translation, determining the author of a text, and speech recognition)
7. Write your own Word2Vec model that uses a neural network to compute word embeddings using a continuous bag-of-words model
6. Train a neural network with GLoVe word embeddings to perform sentiment analysis of tweets

8. Train a recurrent neural network to perform NER using LSTMs with linear layers
- Translate complete English sentences into French using an encoder/decoder attention model
9. Build a transformer model to summarize text
1. Build a chatbot using a reformer model.

**Case Study:**

1. Clinical Documentation
2. Speech Recognition
3. Computer-Assisted Coding (CAC)
4. Data Mining Research
5. Automated Registry Reporting
6. Clinical Decision Support
7. Clinical Trial Matching
8. Prior Authorization

**Blog:**

1. Machine Translation: Rule-Based
  2. Machine Translation: Statistical-Based
  3. MT Tools - GIZA++, SRTLM and Moses
  4. GIZA++, SRTLM and Moses
  5. Natural Language Resources for Beginners
  6. Natural Language Resources for Practitioners
  7. Biomedical Text Mining
  8. Computer Vision and also NLP
  9. Deep Linguistic Processing
  10. Controlled Natural Language.
  11. Language Resources and Architectures for NLP
  12. Sentiment Analysis and also Opinion Mining
  13. NLP includes Artificial Intelligence
- 

**Survey:**

1. Language Models
2. Top-down Parsing
3. Bottom-up parsing
4. Probabilistic parsing
5. Indian Languages parsing

**Assessment Scheme:** Ensures 360 degree assessment and covers all aspects of Bloom's Taxonomy.

**MCQ Exam – Section I** - Mid Semester 30 Marks converted to 30 equivalent Marks

**Home Assignment** - End of Semester 100 Marks converted to 10 equivalent Marks

**MCQ Exam – Section II** - End of Semester 30 Marks converted to 30 equivalent Marks

**Comprehensive Viva Voce -End of Semester 100Marks converted to 30 equivalents Marks**

**Text Books:**

1. Tanveer Siddiqui and U S Tiwary, "Natural Language Processing and Information Retrieval" Fourth Impression, Oxford, ISBN-13:978-019-569232-7.
2. Daniel Jurafsky and James H Martin., "Speech and Language Processing", 2nd edition, Pearson, Second Impression-2014, ISBN: 978-93-325-1841-43.
3. Christopher D. Manning and Hinrich Schutze., "Foundations of Statistical Natural Language Processing", MIT Press, 1999.

**Reference Books:**

1. Alexander Clark (Editor), Chris Fox (Editor), Shalom Lappin (Editor), "The Handbook of Computational Linguistics and Natural Language Processing".
2. Natural Language Processing with Python – Analyzing Text with the Natural Language Toolkit, Steven Bird, Ewan Klein, and Edward Loper, O'Reilly Publication
3. Natural Language Processing with Python CookBook, Krishna Bhavsar, Naresh Kumar, Pratap Dangeti, Packt Publication.
4. Ralph Grishman, " Computational Linguistics: An Introduction (Studies in Natural Language Processing)", Cambridge University Press, ASIN : B01MQYCTOB.

**MOOCs Links and additional reading material:**

- [www.nptelvideos.in](http://www.nptelvideos.in)
- [www.nfnlp.com](http://www.nfnlp.com)
- <https://www.mooc-list.com/tags/nlp>
- <https://www.my-mooc.com/en/mooc/natural-language-processing-nlp/>
- <https://huggingface.co/learn/nlp-course/>
- <https://www.coursera.org/learn/attention-models-in-nlp>

**Course Outcomes:**

The student will be able to:

1. Interpret morphology for given natural language (2)
2. Construct shallow depth lexical analyzer and syntactic analyzer for given natural language(3)
3. Develop shallow depth type dependency parser for given natural language(3)
4. Develop shallow depth machine transliteration, machine translation, information retrieval for given natural language using either linguistic or statistical approach (3)
5. Correlate shallow depth machine transliteration, machine translation, information for given natural language (4)
6. Evaluate machine transliteration, machine translation and information retrieval results using standard evaluation metrics (5)

**CO-PO Map:**

| CO/PO | Program Outcomes (PO) |     |     |     |     |     |     |     |     |      |      |      | PSO  |      |      |
|-------|-----------------------|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|------|------|------|
|       | PO1                   | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 | PSO1 | PSO2 | PSO3 |
| CO1   | 3                     | 2   |     | 1   | -   | -   | -   | -   | -   | -    | -    | 2    | 2    | 2    | 3    |
| CO2   | 3                     | 2   | 2   | 1   | 3   | -   | -   | 2   | -   | -    | -    | 2    | 2    | 2    | 3    |
| CO3   | 2                     | 2   | 3   | 2   | 3   | -   | -   | -   | 1   | -    | -    | 2    | 2    | 2    | 3    |
| CO4   | 3                     | 3   | 3   | 2   | 3   | 2   | -   | -   | -   | -    | -    | 2    | 2    | 2    | 3    |

|         |     |      |     |      |   |     |     |     |     |     |     |     |     |     |     |
|---------|-----|------|-----|------|---|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| CO5     | 3   | 2    |     | 2    | 3 | 2   | -   | -   | 1   | 1   | 2   | 2   | 2   | 2   | 3   |
| CO6     | 3   | 2    |     | 2    | 3 | 2   | 2   | 2   |     | 1   | 2   | 2   | 2   | 2   | 3   |
| Average | 2.8 | 2.16 | 2.7 | 1.88 | 3 | 2.0 | 2.0 | 2.0 | 1.0 | 1.0 | 2.0 | 2.0 | 2.0 | 2.0 | 3.0 |

### **CO Attainment levels:**

Weights for attainment levels: L1 - Easy-0.75 L2 - Comfortable-0.7 L3 – Medium – 0.65  
L4 – Somewhat difficult – 0.6 L5 – Difficult – 0.55

CO1 – L2, CO2 – L3, CO3 – L3, CO4 – L3, CO5 – L4 and CO6 – L5

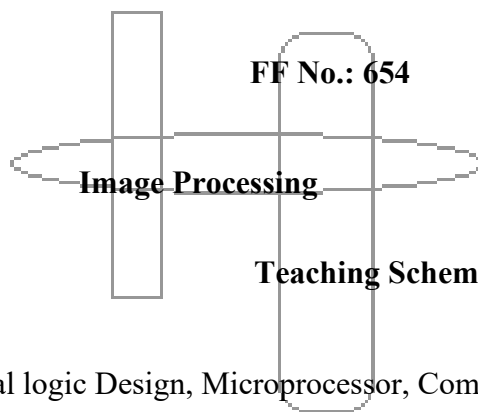
### **Future Course Mapping:**

Computational Linguistics, ANN, RNN, Deep Learning

### **Job Mapping:**

Application Developers, System programmer.

### **Syllabus Template**



**Credits: 2**

**Course Prerequisites:** Digital logic Design, Microprocessor, Computer Organization.

### **Course Objectives:**

- 1.To describe different color models and image processing techniques.
- 2.To analyze image condition and deduce enhancement algorithms.
- 3.To apply image segmentation to identify the region of interest
- 4.To develop an algorithm to recognize the specified objects in the given image.
- 5.To study different image morphological operation.
- 6.To learn different image compression techniques.

### **Course Relevance:**

Vision sense is the most powerful human sense organ. In the world where intelligent automation is taking place, image processing is a vital domain for research and development. In Industry 4.0, image processing systems built around industrial cameras are an essential component in automated production. Throughout all steps of production, from the inspection of raw materials and production

monitoring (i.e., flaw detection) to final inspections and quality assurance, they are an indispensable part of achieving high efficiency and quality standards. In the Entertainment Industry, latest trends such as 4K video streaming requires high quality compression that can provide limited/no loss image quality with high fps. In social networking, sharing images has been a vital part. Creating innovative effects and overall manipulating the images will be explored

### **Section 1: Topics/Contents**

#### **Unit-I Introduction**

**4 Hours**

Elements of image processing system, Scenes and Images, Vector Algebra, Human Visual System, color vision color model: RGB, HVS, YUV, CMYK, YCbCr and some basic relationships between pixels, linear and nonlinear operations, Image sampling and quantization.

#### **Unit-II Image Enhancements**

**5 Hours**

Memory-less operations, Spatial domain image enhancements: Denoising filters, Smoothing Operation, Sharpening Operation, and Contrast stretching /enhancement, histogram and histogram equalization.

#### **Unit-III Image segmentation**

**5 Hours**

Classification of image segmentation techniques: Edge-based Segmentation, Region based techniques. Binarization: Global Thresholding, Adaptive thresholding. Types of Edge detector: derivative filters, Sobel, Canny. Edge linking. Feature Extraction.

### **Section2:Topics/Contents**

#### **Unit-IV Morphological Operation**

**4 Hours**

Binary Morphology, Erosion Dilation, Opening and Closing.

#### **Unit-V Feature Extraction and Object Recognition**

**5 Hours**

Feature points and feature detection (Line, circle and corner). Line detection: RANSAC, Hough Transform. Corner detection: Harris Corner Detector. Feature descriptors, Descriptor matching. SIFT, Boundary representation (Chain code), Boundary detection-based techniques.

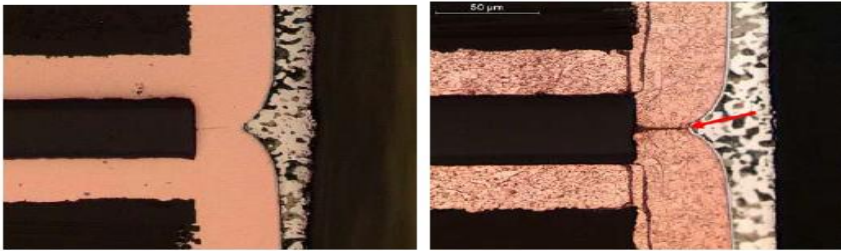
#### **Unit-VI Image Compression**

**5 Hours**

Introduction and need, Coding redundancy, classification of compression techniques (Lossy and lossless- JPEG, Run Length Coding, Huffman Coding, Shannon Fano coding).

**List of Design based Home Assignments:**

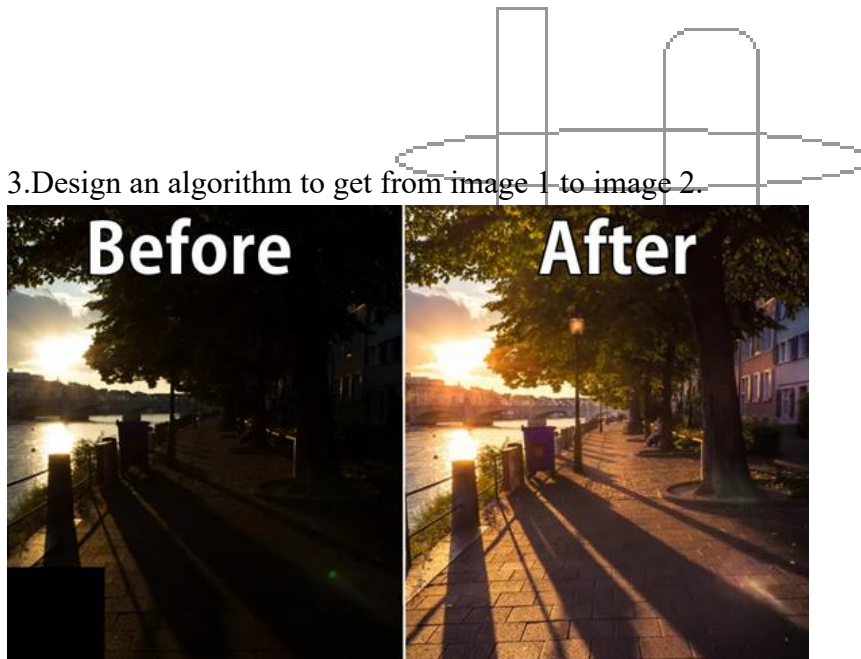
1.Design an algorithm to identify fault in a “PCB inspection system” as shown below



2.Design an algorithm to perform segmentation of the image below to extract the mango from its background.



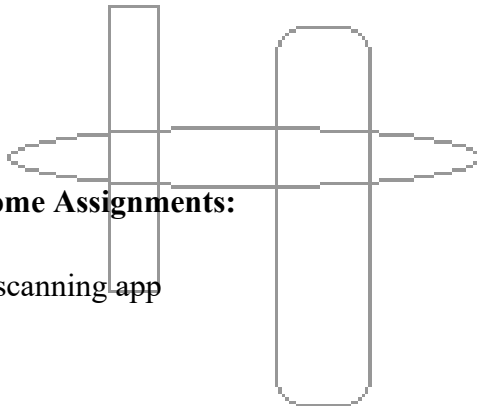
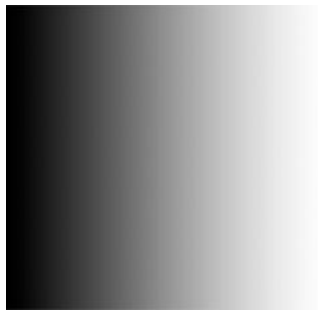
3.Design an algorithm to get from image 1 to image 2.



4.Design an algorithm to recognize character “0” in the image below.



5.Design an algorithm to compress a 300x300 pixel image with horizontal black to white gradient as shown below.



#### **List of Case Study based Home Assignments:**

1. Cam-scanner: Document scanning app
2. Tesseract OCR library
3. Instagram filters
4. OpenCV
5. Google Street View

#### **List of Blog based Home Assignments:**

1. Image processing on Embedded platforms
2. Face recognition system security analysis for authentication
3. Image processing in MSME for effective automation
4. H.264 codec for image streaming
5. Role of mathematics in image processing

#### **List of Survey based Home Assignments:**

1. Image quality metrics
2. Vision based self-driving car safety
3. Compression techniques & codecs
4. State of the art applications such as AR/ XR
5. Human recognition in social networking apps like Facebook



**Assessment Scheme:** Ensures 360 degree assessment and covers all aspects of Bloom's Taxonomy.

**MCQ Exam – Section I** - Mid Semester 30 Marks converted to 30 equivalent Marks

**Home Assignment** - End of Semester 100 Marks converted to 10 equivalent Marks

**MCQ Exam – Section II** - End of Semester 30 Marks converted to 30 equivalent Marks

**Comprehensive Viva Voce** -End of Semester 100Marks converted to 30 equivalents Marks

**Text Books:** (As per IEEE format)

1. Rafael Gonzalez & Richard Woods, "Digital Image Processing," 3<sup>rd</sup> Edition, Pearson publications, ISBN 0132345633.

2. Anil K. Jain, "Fundamental of Digital Image Processing," 5<sup>th</sup> Edition, PHI publication, ISBN 13: 9780133361650.

**Reference Books:** (As per IEEE format)

1. Pratt, "Digital Image Processing," Wiley Publication, 3<sup>rd</sup> Edition, ISBN 0-471- 37407-5.

2. K.R. Castleman, "Digital Image Processing," 3<sup>rd</sup> Edition, Prentice Hall: Upper Saddle River, NJ, 3, ISBN 0-13-211467 -4.

**MOOCs Links and additional reading material:**

1. <https://nptel.ac.in/courses/117/105/117105135/>
2. <https://nptel.ac.in/courses/106/105/106105032/>
3. <https://nptel.ac.in/noc/courses/noc19/SEM2/noc19-cs58/>
4. <https://www.coursera.org/learn/computer-vision-basics>

**Course Outcomes:**

The student will be able to –

1. Recognize different color models and image processing techniques. (1)
2. Select image enhancement algorithm to improve the quality of image. (2)
3. Build image segmentation techniques to identify region of interest. (4)
4. Predict image morphological techniques to resize the image. (3)
5. Construct an algorithm to recognize the specified objects in the given image. (5)
6. Identify different image compression techniques to reduce the size of image. (3)

**CO-PO Map:**

| CO/PO | Program Outcomes (PO) |      |      |      |      |      |      |     |      |      |      |      | PSO   |       |       |       |
|-------|-----------------------|------|------|------|------|------|------|-----|------|------|------|------|-------|-------|-------|-------|
|       | PO 1                  | PO 2 | PO 3 | PO 4 | PO 5 | PO 6 | PO 7 | PO8 | PO 9 | PO10 | PO11 | PO12 | PSO 1 | PSO 2 | PSO 3 | PSO 4 |
| CO1   | 3                     | 2    |      | 1    | -    | -    | -    | -   | -    | -    | -    | 2    | 2     | 2     | 3     | 3     |
| CO2   | 3                     | 2    | 2    | 1    | 3    | -    | -    | 2   | -    | -    | -    | 2    | 2     | 2     | 3     | 3     |
| CO3   | 2                     | 2    | 3    | 2    | 3    | -    | -    | -   | 1    | -    | -    | 2    | 2     | 2     | 3     | 3     |

|         |     |      |     |      |   |     |     |     |     |     |     |     |     |     |     |     |
|---------|-----|------|-----|------|---|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| CO4     | 3   | 3    | 3   | 2    | 3 | 2   | -   | -   | -   | -   | -   | 2   | 2   | 2   | 3   | 3   |
| CO5     | 3   | 2    |     | 2    | 3 | 2   | -   | -   | 1   | 1   | 2   | 2   | 2   | 2   | 3   | 3   |
| CO6     | 3   | 2    |     | 2    | 3 | 2   | 2   | 2   |     | 1   | 2   | 2   | 2   | 2   | 3   | 3   |
| Average | 2.8 | 2.16 | 2.7 | 1.88 | 3 | 2.0 | 2.0 | 2.0 | 1.0 | 1.0 | 2.0 | 2.0 | 2.0 | 2.0 | 3.0 | 3.0 |

**CO Attainment levels:**

Weights for attainment levels: L1 - Easy-0.75 L2 - Comfortable-0.7 L3 – Medium – 0.65

L4 – Somewhat difficult – 0.6 L5 – Difficult – 0.55

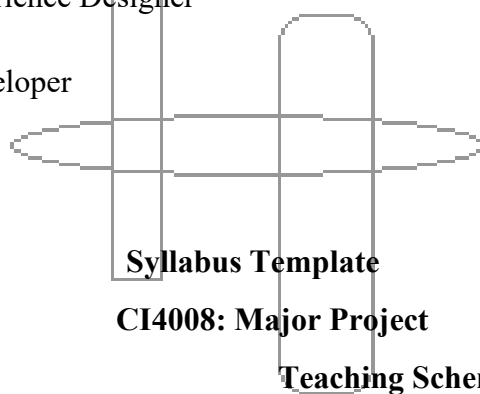
CO1 – L2, CO2 – L2, CO3 – L4, CO4 – L3, CO5 – L5 and CO6 – L3

**Future Course Mapping:**

1. Augmented Reality
2. Multimedia Processing

**Job Mapping:**

1. Augmented Reality Experience Designer
2. Automation Engineer
3. Embedded Software Developer
4. Image Processing Expert



**Course Prerequisites:** Project Based Learning

**Aim**

This course addresses the issues associated with the successful management of a project. The course emphasizes project life cycle phases requirement engineering, system analysis and system design. A further aim is for students to heighten personal awareness of the importance of developing strategies for themselves and working with peers to create desired outcomes. The Project Work can lead to:

- Transform existing Ideas into conceptual models.
- Transform conceptual models into determinable models.
- Use determinable models to obtain system specifications.
- Select optimum specifications and create physical models.

- Apply the results from physical models to create real target systems.

### **Project Group and Topic Selection and Synopsis:**

The project work needs to be undertaken by a group of maximum FOUR and minimum of THREE students. The Project work will be jointly performed by the project team members. The student needs to identify a technological problem in the area of Computer Engineering or Information Technology of their choice and address the problem by formulating a solution for the identified problem. The Project Group will prepare a synopsis of the project work which will be approved by the concerned faculty member. The project should not be a reengineering or reverse engineering project. In some cases, reverse engineering projects will be permissible based on the research component involved in it. The project work aims at solving a real world technical problem. Hence ample literature survey is required to be done by the students. Application-oriented projects will not be acceptable. Low-level custom User Interface development and its allied mapping with a particular technology will not be accepted.

### **Overview of the Course:**

1. The Student Project Group is expected to make a survey of situation for identifying the requirements of selected Technological Problem. The Student Project Group will be monitored by Internal Guides and External Guides (if any).
2. The project requires the students to conceive, design, implement and operate a mechanism (the design problem). The mechanism may be entirely of the student's own design, or it may incorporate off-the-shelf parts. If the mechanism incorporates off-the-shelf parts, the students must perform appropriate analysis to show that the parts are suitable for their intended purpose in the mechanism.
3. The project must be based on a Fresh Idea or Implementation of a Theoretical Problem – meaning that there is not a known Solution to the design problem Or Create a Better Solution.
4. The project must have an experimental component. Students must conceive, design, implement and operate an appropriate experiment as part of the project. The experiment might be to collect data about some aspect of the design (i.e., to verify that the design will work as expected). Alternatively, the experiment could be to verify that the final mechanism performs as expected.
5. Upon receiving the approval, the Student Project Group will prepare a preliminary project report consisting , Feasibility Study Document, System Requirement Specification, System Analysis Document, Preliminary System Design Document. All the documents indicated will have a prescribed format.
6. Upon project completion, the Student Project Group will prepare a detailed Project Report consisting Semester I Preliminary Project document along with Detailed System Design Document, Implementation and Testing Document with conclusion and future scope of the Project Work. All the documents indicated will have a prescribed format. The Project Report ideally should consist of following documents : (Exceptions may be there based on the nature

of the project, especially if some of the following documents are not applicable to a particular project as determined by the project guide, coordinator and head of department).

| Sr . | Project Item                                                                         |
|------|--------------------------------------------------------------------------------------|
| 1    | Project Cover Front Page                                                             |
| 2    | Project Completion Certificate [Institute]                                           |
| 3    | Project Completion Letter<br>[In case of Sponsored Projects]                         |
| 4    | Acknowledgments                                                                      |
| 5    | Table of Contents                                                                    |
| 6    | List of Figures                                                                      |
| 7    | List of Tables                                                                       |
| 8    | Project Synopsis<br>[Problem Background, Existing System Details, Proposed Solution] |
| 9    | Feasibility Study Report                                                             |
| 10   | Project Plan                                                                         |
| 11   | System Requirement Specification                                                     |
| 12   | System Analysis Document: UML Use Case Diagrams                                      |
| 13   | System Analysis Document: UML Sequence Diagrams                                      |
| 14   | System Analysis Document: UML State Diagrams                                         |
| 15   | System Design Document with Module Specifications                                    |
| 16   | System Implementation                                                                |
| 17   | System Testing and Experimental Findings                                             |
| 18   | Conclusion                                                                           |
| 19   | References                                                                           |

7. The Project Work will be assessed jointly by a panel of examiners consisting faculty and industry experts. The Project Groups will deliver the presentation and demonstration of the Project Work which will be assessed by the panel.
8. The Student Project Group needs to actively participate in the presentation. The panel of examiners will evaluate the candidate's performance based on presentation skills, questions based on the Project Work and overall development effort taken by the candidates.

**Note:**

The student needs to design and develop solution for the identified technological problem in the area of Computer Engineering or Information Technology of their choice. The Project Implementation needs to be completed using best possible use of available technologies as applicable to deal with the

complexity of the project. The Project Group will prepare a detailed report of the project work which will be approved by the concerned faculty member. The Project Report need to be submitted both in Hard form and Soft form in CD. The Soft Copy of the Project Report must accompany other project deliverables as well.

Assessment: MSE and ESE

1. Mid Semester Assessment – 50 Marks to be converted to 30 Marks.
2. End Semester Assessment – 100 Marks to be converted to 70 Marks.

**Mid Semester Assessment**

| <b>Sr. No.</b> | <b>Parameter</b>                                                         | <b>Marks</b> |
|----------------|--------------------------------------------------------------------------|--------------|
| 1              | Problem Statement                                                        | 10           |
| 2              | Literature Review                                                        | 10           |
| 3              | Group formation and identification of individual responsibility          | 10           |
| 4              | Objective of Project activity                                            | 10           |
| 5              | Knowledge of domain, latest technology and modern tools used /to be used | 10           |
| <b>TOTAL</b>   |                                                                          | <b>50</b>    |

**End Semester Assessment**

| <b>Sr. No.</b> | <b>Parameter</b>                                         | <b>Marks</b> |
|----------------|----------------------------------------------------------|--------------|
| 1              | Realization of project as per problem statement          | 10           |
| 2              | Design, Testing / Experimentation, Analysis / Validation | 30           |
| 3              | Documentation and Report Writing                         | 20           |
| 4              | Quality of Work                                          | 15           |
| 5              | Performance in Question & Answers Session                | 15           |
| 6              | Regular interaction with guide                           | 10           |
| <b>TOTAL</b>   |                                                          | <b>100</b>   |

**Course Outcomes:**

Upon completion of the course, graduates will be able to -

CO1: Identify the real life problem from societal need point of view

CO2: Prepare the requirement engineering, feasibility analysis documents

CO3: Form the teams and share responsibilities according to individual skill strengths

CO4: Create design documents to build software solutions

CO5: Develop software solutions based on standard engineering specifications

CO6: Perform the verification and validation up to the mark

**CO PO Map**

| CO/PO   | Program Outcomes (PO) |      |      |      |      |      |      |     |      |      |      |      | PSO   |       |       |       |
|---------|-----------------------|------|------|------|------|------|------|-----|------|------|------|------|-------|-------|-------|-------|
|         | PO 1                  | PO 2 | PO 3 | PO 4 | PO 5 | PO 6 | PO 7 | PO8 | PO 9 | PO10 | PO11 | PO12 | PSO 1 | PSO 2 | PSO 3 | PSO 4 |
| CO1     | 2                     | 3    |      | 2    |      |      |      |     | 3    |      |      | 3    |       |       |       | 3     |
| CO2     | 2                     | 3    | 3    | 2    | 2    |      |      |     | 3    | 3    | 2    | 3    | 3     |       | 3     | 3     |
| CO3     | 2                     | -    | -    | -    |      |      |      |     | 3    |      | 2    | 3    | 3     |       |       | 3     |
| CO4     | 2                     | 3    | 3    | 2    | 2    | 3    | 3.0  | 2.0 | 3    | 3    | 2    | 3    | 3     | 3     | 3     | 3     |
| CO5     | 2                     | 3    | 3    | 2    | 2    |      |      |     | 3    | 3    | 2    | 3    | 3     | 3     | 3     | 3     |
| CO6     | 2                     | 2    | 2    | 3    | 2    |      |      |     | 3    | 2    | 2    | 3    | 3     | 2     | 3     | 3     |
| Average | 2.0                   | 2.8  | 2.75 | 2.83 | 2.0  | 3.0  | 3.0  | 2.0 | 3.0  | 2.75 | 2.0  | 3.0  | 3.0   | 1.75  | 3.0   | 3.0   |

**CO attainment levels**

CO1 -4 CO2 -2 CO3-4 CO4-5 CO5 -1 CO6-3

**Syllabus Template**

**FF No. : 654**

**CI4010, CI4011, CI4012, CI4013: Industry Internship, Project Internship , Research Internship, Global Internship, International Internship**

**Credits:..16.....**

**Teaching Scheme Theory: ...32... Hours/Week**

**Guidelines:**

Internships are educational and career development opportunities, providing practical experience in a field or discipline. Internships are far more important as the employers are looking for employees who are properly skilled and having awareness about industry environment, practices and culture. Internship is structured, short-term, supervised training often focused around particular tasks or projects with defined time scales. Core objective is to expose technical students to the industrial environment, which cannot be simulated/experienced in the classroom and hence creating competent professionals in the industry and to understand the social, economic and administrative considerations that influence the working environment of industrial organizations. Engineering internships are intended to provide students with an opportunity to apply theoretical knowledge from academics to the realities of the field work/training.

The following guidelines are proposed to give academic credits for the internship undergone as a part of the B.Tech. Engineering curriculum.

**Duration:**

Industry Internship will be started at the beginning of the semester 7 or semester 8 or yearlong for the duration 6 months or 12 months.

**Identification of Internship work:**

Student may choose to undergo Internship at Industry/Govt./NGO/MSME/Innovation/IPR/Entrepreneurship. Contacting various companies for Internship and Internship work identification process should be initiated at the end of 6<sup>th</sup> semester in coordination with training and placement cell/ industry institute cell/ internship cell. This will help students to start their internship work on time. Students can take internship work in the form of online/onsite work from any of the following but not limited to:

- Working for consultancy/ research project
- Contribution in Incubation/ Innovation/ Entrepreneurship Cell/ Institutional Innovation Council/ start-ups cells of institute
- Industry / Government Organization Internship,
- Internship through Internshala
- In-house product development, intercollegiate, inter department research internship under research lab/group, micro/small/medium enterprise/online internship
- Research internship under professors, IISC, IIT's, Research organizations

**Internship Documents Submission:**

Students must submit internship offer letter, internship completion letter, FF 1029 (Students feedback form), FF 1030 (Industry feedback about interns).

Students must present their internship progress time to time to faculty mentors. Faculty mentors and industry mentors both can evaluate the progress of the intern combiningly.

### **Internship Work Evaluation:**

In-semester and end semester internship evaluation and assessment will be done by internal (Faculty mentor) and external examiners - a supervisor from industry.

After completion of Internship, the student should prepare a comprehensive report to indicate what he has observed and learnt in the internship/training period. The student may contact Industrial Supervisor/ Faculty Mentor/TPO for assigning special topics and problems and should prepare the final report.

If the student remain absent without prior intimation to the department/institute/concern authority/T & P Cell, his entire training can be cancelled and he will fail.

### **Course Outcomes: Industry Internship**

**On the completion of course, students will able to-**

1. Understand real-world applications, workplace environment and operating procedures
2. Adapt skill for learning and applying modern tools and technologies
3. Apply professional values and ethical standards
4. Perform as an individual and as a team member effectively to changing conditions
5. Encompass improved writing, verbal communication and documentation skills
6. Learn about career positions and occupations along with the qualities and training required to obtain those positions

### **CO-PO Map: Industry Internship**

| CO/PO   | Program Outcomes (PO) |      |      |      |      |      |      |     |      |      |      |      | PSO   |       |       |       |
|---------|-----------------------|------|------|------|------|------|------|-----|------|------|------|------|-------|-------|-------|-------|
|         | PO 1                  | PO 2 | PO 3 | PO 4 | PO 5 | PO 6 | PO 7 | PO8 | PO 9 | PO10 | PO11 | PO12 | PSO 1 | PSO 2 | PSO 3 | PSO 4 |
| CO1     | 2                     | 3    |      | 2    |      |      |      |     | 3    |      |      | 3    |       |       |       | 2     |
| CO2     | 2                     | 3    | 3    | 2    | 3    | 2    | 3    | 2   | 3    |      | 2    | 3    | 3     | 3     | 3     | 3     |
| CO3     | 2                     | 3    | 3    | 2    |      |      |      | 2   | 3    |      | 2    | 3    | 3     | 3     | 3     | 3     |
| CO4     | 2                     |      |      |      |      |      |      |     | 3    |      |      | 3    |       |       |       |       |
| CO5     | 2                     |      | 2    |      | 3    |      |      | 2   | 3    | 3    | 2    | 3    |       |       | 3     |       |
| CO6     | 2                     |      |      |      |      | 2    |      |     | 3    |      |      | 3    |       |       |       |       |
| Average | 2.0                   | 3.0  | 2.66 | 2.0  | 3.0  | 2.0  | 3.0  | 2.0 | 3.0  | 3.0  | 2.0  | 3.0  | 3.0   | 3.0   | 3.0   | 2.66  |

### **CO attainment levels:**

CO1 – 3, CO2 –5, CO3 –2, CO4 –2, CO5 – 3, CO6 –2



**Course Outcomes: Research Internship**

**On the completion of course, students will able to-**

1. Develop an ecosystem to promote entrepreneurship and research culture among the students.
2. Learn first-hand to apply techniques, resources, modern engineering tools for prediction modelling to complex engineering activities.
3. Apply reasoning informed by the contextual knowledge to access societal, health, safety, legal and cultural issues and consequent responsibilities.
4. Perform as an individual and as a team member.
5. Understand Engineering and Management Principles.
6. Exercise R & D aptitude focusing on the knowledge creation and dissemination through engineering artifacts creation, construction and presentation.

**CO-PO Map: Research Internship**

| CO/PO   | Program Outcomes (PO) |      |      |      |      |      |      |     |      |      |      |      | PSO   |       |       |       |
|---------|-----------------------|------|------|------|------|------|------|-----|------|------|------|------|-------|-------|-------|-------|
|         | PO 1                  | PO 2 | PO 3 | PO 4 | PO 5 | PO 6 | PO 7 | PO8 | PO 9 | PO10 | PO11 | PO12 | PSO 1 | PSO 2 | PSO 3 | PSO 4 |
| CO1     | 2                     |      |      |      |      |      |      |     |      |      |      | 3    |       |       |       | 3     |
| CO2     | 3                     | 3    | 3    | 2    | 3    | 2    | 2    | 2   |      |      |      | 3    | 3     | 3     | 3     | 3     |
| CO3     | 3                     | 3    | 3    | 2    |      | 3    |      |     |      |      |      | 3    | 3     |       | 3     | 3     |
| CO4     |                       |      |      |      |      |      |      |     | 3    |      |      | 3    |       |       |       | 3     |
| CO5     | 3                     | 2    | 3    | 2    |      |      |      |     |      |      |      | 3    | 3     |       | 3     | 3     |
| CO6     | 3                     | 3    | 2    | 3    |      |      | 3    | 2.0 |      | 3.0  | 2.0  | 3    |       |       | 3     | 3     |
| Average | 2.8                   | 2.75 | 2.75 | 2.33 | 3.0  | 2.5  | 2.5  | 2.0 | 3.0  | 2.75 | 2.0  | 3.0  | 3.0   | 3.0   | 3.0   | 3.0   |

CO1 – 3, CO2 –4, CO3 –5, CO4 –2, CO5 – 3, CO6 –4